

Radiology AIIMS 2017

Question 1:

A 30-year-old female came to OPD with complaints of backache. On examination, Schober test was positive and her eyes and nose were hyperpigmented. Probable diagnosis is



- a) Hypoparathyroidism
- b) Fluorosis
- c) Ochronosis
- d) Ankylosing spondylitis

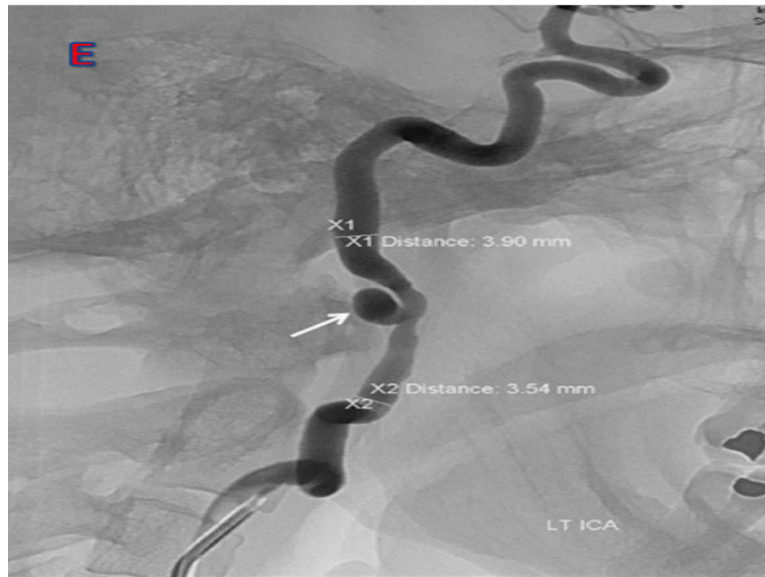
Question 2:

A 30-year-old woman presented with complaints of bone pain and abdominal cramps. Her family says she has a history of abnormal behavior. The consultant doctor arrived at a provisional diagnosis based on the clinical features. Which of the following would be the best investigation to arrive at a definitive diagnosis?

- a) MRI
- b) Ultrasonogram
- c) Sestamibi scan
- d) CT scan

Question 3:

Based on the given image, what is the most likely diagnosis?



- a) Carotico-cavernous fistula
- b) Pseudoaneurysm of internal carotid artery
- c) Tumour blush of angiofibroma
- d) Extensive head and neck hemangioma

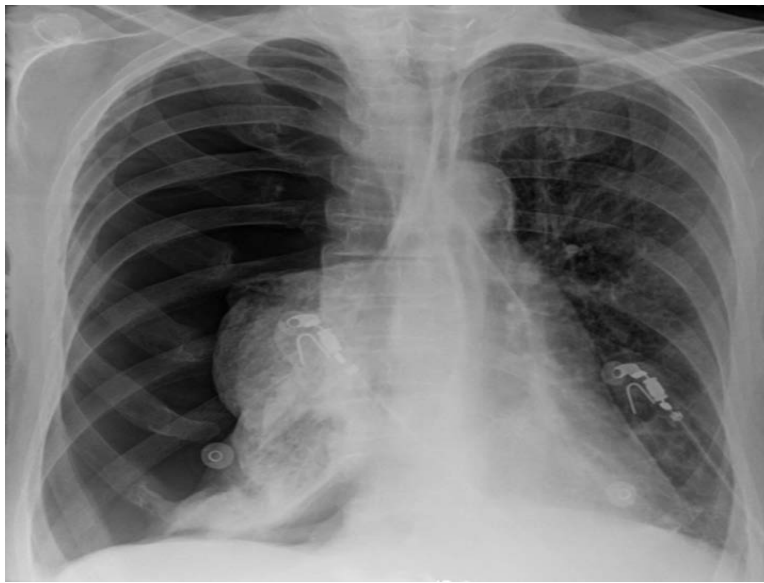
Question 4:

Which of the following will have reduced uptake on FDG PET scan?

- a) Typical carcinoid
- b) Small cell carcinoma
- c) Atypical carcinoid
- d) Large cell neuroendocrine carcinoma

Question 5:

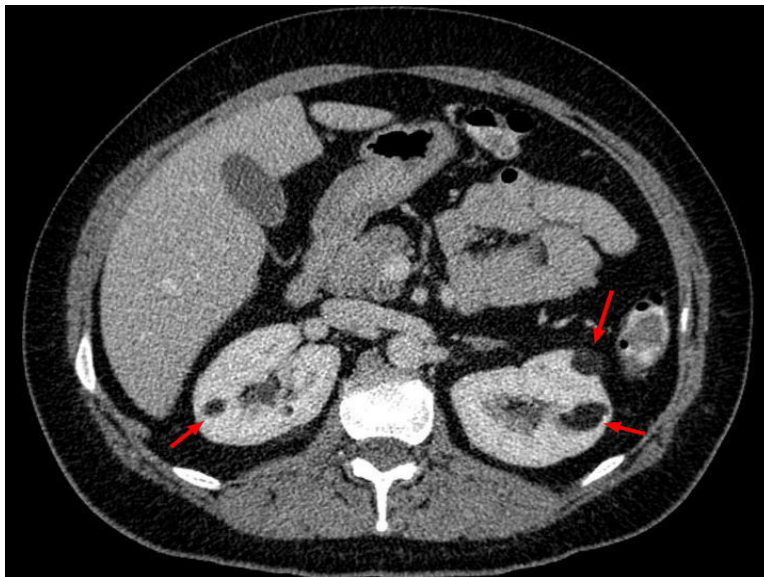
A thin, 34-year-old male, presents with sharp pain and sudden onset dyspnea. Chest X-ray was performed and is shown below. What is the most probable diagnosis?



- a) Pulmonary embolism
- b) Pneumothorax
- c) Hydropneumothorax
- d) Pleural effusion

Question 6:

A 45-year-old woman comes to the emergency department with severe back pain, vomiting, and hypotension. CT findings are given below. What is the diagnosis?



- a) Renal cyst

- b) Renal cell carcinoma
- c) Para pelvic cyst
- d) Renal angiomyolipoma

Question 7:

A 25-year-old male came with complaints of acute abdominal pain. A plain X-ray abdomen erect was taken which is shown below. What is your diagnosis?



- a) Subdiaphragmatic abscess
- b) Pneumoperitoneum
- c) Pneumopericardium
- d) Intestinal obstruction

Answer Key

Question No.	Correct Option
1	c
2	c
3	b
4	a
5	b

6	d
7	b

Detailed Explanations

Solution to Question 1:

Given X-ray shows intervertebral disc calcification and hyperpigmentation of nose and eyes are present. Both features are indicative of ochronosis as diagnosis.

- Ochronosis also known as alkaptonuria is an autosomal recessive disorder.
- It is due to deficiency of homogentisic acid oxidase leading to accumulation of homogentisic acid.
- Clinical features: Triad of
 - Homogentisic aciduria (urine turns dark color on standing due to oxidation or alkalinisation)
 - Eumelanin like pigmentation of skin, sclera, cartilages etc.
 - Degenerative chronic arthropathies especially of spine, hip, knees.

Option a: Hypoparathyroidism doesn't explain hyperpigmentation and other clinical features of hypoparathyroidism aren't mentioned.

Option b: Fluorosis - dental symptoms are not mentioned and osteoporosis will be marked in case of fluorosis. Also, hyperpigmentation is not a typical feature of fluorosis.

Option d: Ankylosing spondylitis – females are unlikely to be affected and it doesn't explain hyperpigmentation.

Solution to Question 2:

The given clinical features are suggestive of primary hyperparathyroidism. Sestamibi scan is the best investigation to arrive at a definitive diagnosis. Parathyroid adenoma is the most common cause of primary hyperparathyroidism.

Clinical features of hyperparathyroidism:

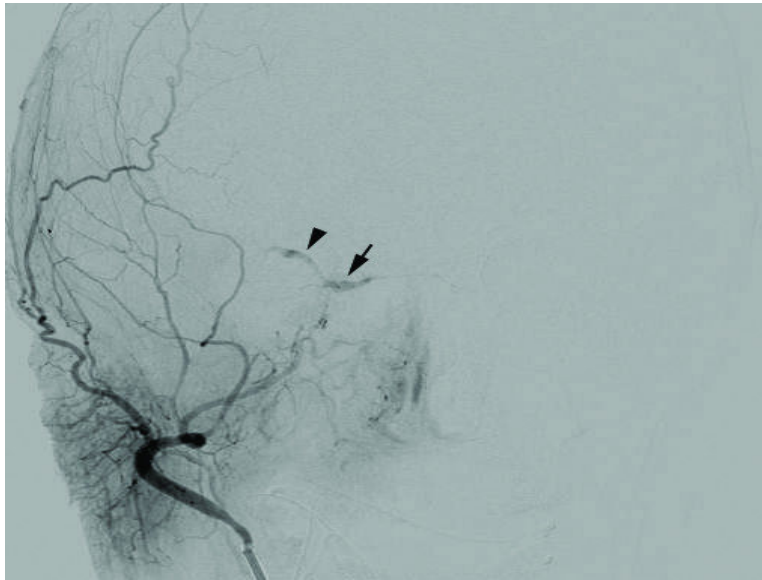
- Bones- bone pain, pathological fractures
- Stones- nephrocalcinosis
- Groans- abdominal pain
- Moans- psychotic symptoms
- Fatigue overtones- fatigue, malaise

Solution to Question 3:

The given image is a digital subtraction angiogram(DSA) and it shows a pseudoaneurysm of the internal carotid artery.

The right ICA injection anteroposterior view shows the neck of the pseudoaneurysm originating from the transverse segment of the cavernous portion of the ICA.

Option A: Carotico-cavernous fistula is characterized by the premature filling of the veins as shown in the image below:



Option C: Angiofibroma is characterised by tumour blush. It is the radiological enhancement of a tumour that reveals opacification of the well-vascularized tumours.

Option D: Hemangioma can be differentiated from other vascular malformations by the presence of well-circumscribed mass demonstrating intense tissue staining, usually in a lobular pattern.

Solution to Question 4:

Typical carcinoid is more likely to have reduced uptake on FDG PET scan than an atypical carcinoid.

The low metabolic rate of tumors makes the tumor negative (show reduced uptake) on an FDG PET scan.

Typical examples are:

- Bronchoalveolar carcinoma (Adenocarcinoma in situ)
- Carcinoid tumour (Typical & Atypical)

Solution to Question 5:

The given chest radiograph clearly depicts the presence of pneumothorax, which represents air in the pleural space.

- The right hemithorax shows blackening (depicts area of low density) due to the presence of air.
- The visceral pleural line has separated from the chest wall by a transradiant zone devoid of vessels.
- The collapsed lung can be seen with a clear border.
- There is a mediastinal shift to the left side (contralateral), which is typically seen with cases of tension pneumothorax.

All the above features are suggestive of pneumothorax.

Note: Pleural effusions signify the presence of fluid accumulation in the pleural space. The fluid may have varying degrees of mobility depending on the nature of effusion (free-flowing or loculated).

While a small amount of free fluid is usually hard to detect on an erect PA chest x-ray, a significant pleural effusion (200-500 mL) is represented by obscured lateral and posterior costophrenic angles and the hemidiaphragm affected. Subsequently, homogenous opacification of the lower lung field develops, associated with concave superior margin relative to the lung. This margin is higher laterally than medially.

A massive pleural effusion cause dense opacification usually associated with a contralateral mediastinal shift.



Here is an image of a left-sided pleural effusion representing the homogenous opacification of the left lower and middle lung fields. There is associated blunting of the left sided costophrenic angle. The superior border of the effusion is concave and is higher laterally than medially.

Solution to Question 6:

The given CT image shows well-circumscribed lesions with low attenuation (black in colour) that indicate fatty tissue. This is suggestive of renal angiomyolipomas.

Angiomyolipomas (AMLs) are benign tumours that occur sporadically or in association with tuberous sclerosis. It is commonly seen in adults between 30 and 50 years of age, with a female predilection. AMLs predominantly affect the kidneys, though they can occur in other sites such as the liver (a common extra-renal location) and less frequently in the ovary, fallopian tube, spermatic cord, palate, and colon.

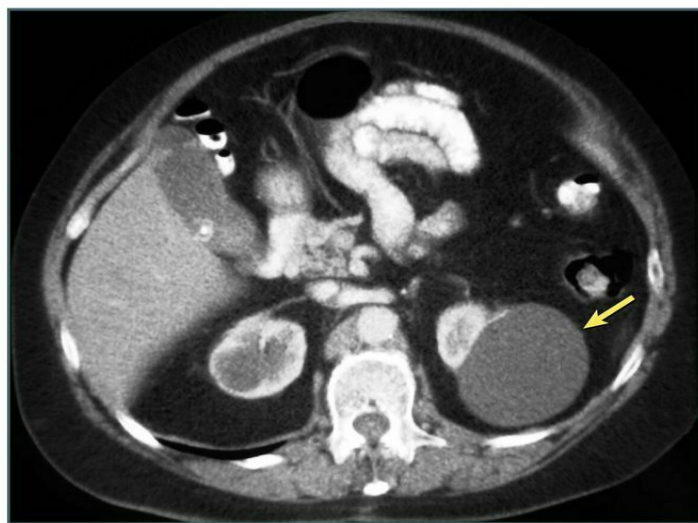
Histologically, angiomyolipomas consist of perivascular epithelioid cells with varying proportions of vascular, immature smooth muscle cells, and fat cells. Patients with AMLs present with flank pain, a palpable mass, hematuria, recurrent urinary tract infections, hypertension, or renal failure. Larger tumours increase the risk of complications, such as spontaneous retroperitoneal hemorrhage, which can manifest as sudden-onset abdominal or flank pain, along with symptoms like nausea, vomiting, and hypotension.

Everolimus is used for the treatment of angiomyolipomas. Treatment should be considered for asymptomatic, growing AMLs measuring ≥ 3 cm in diameter. Angiomyolipomas do not normally require surgery unless life-threatening bleeding is present.

Other options:

Renal cyst is a broader term that includes the cystic lesions of the kidney. They can be simple cysts that are benign. Complex or complicated cystic lesions need evaluation and can be malignant. Bosniak classification is used for renal cysts. Cysts appear as hypodense lesions (greyish) with a thin wall, with or without septations.

CT image showing a simple renal cyst:



Renal cell carcinoma (RCC) is the most common malignant tumour of the kidney. It usually occurs in the age group 50-70 years and macroscopic hematuria is seen in about 60% of the cases. It is a differential for angiomyolipoma but it is important to note that macroscopic fat is almost always associated with ossification or calcification.



Solution to Question 7:

The above X-ray clearly shows the presence of subdiaphragmatic free gas in the right upper quadrant. This indicates pneumoperitoneum.

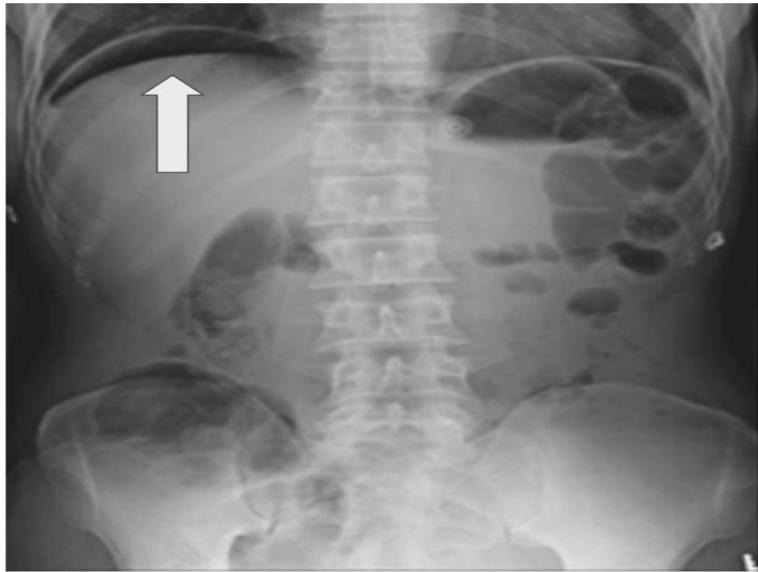
Causes of pneumoperitoneum include:

- Bowel perforation
- Peptic ulcer
- Ischemic bowel
- Necrotizing enterocolitis.
- Small bowel perforation
- Trauma
- Penetrating injuries
- During surgery / laparoscopy

Signs of pneumoperitoneum:

- Bowel-related signs
- Double wall sign (also known as Rigler's sign or gas-relief sign)
- Telltale triangle sign (also known as triangle sign)
- Peritoneal ligament-related signs
- Football sign
- Falciform ligament sign
- Lateral umbilical ligament sign (also known as inverted "V" sign)
- Urachus sign

- Right upper quadrant signs
- Cupola sign
- Fissure for ligamentum teres sign
- Hepatic edge sign
- Lucent liver sign
- Morison's pouch sign (doge cap sign)
- Periportal free gas sign



Radiology AIIMS 2018

Question 1:

A child presents with history of having ingested a button battery. What is your next course of action?

- a) Give oral fluids
- b) Remove endoscopically
- c) Reassure and to check whether passed in stools
- d) Induce vomiting

Question 2:

Which of the following statements are false regarding X Rays?

- a) Loss of soft tissue planes is the earliest X-ray change of infection
- b) Glass pieces can be seen in X-ray images.
- c) Joint space is a misnomer as there is no real space in the joint.
- d) Live bone is more radio-opaque than dead bone on X- Ray images.

Question 3:

A child presented with a history of road traffic accident. The axial CT image of the child, shown below, is suggestive of _____



- a) Extradural Hemorrhage
- b) Subdural Hemorrhage
- c) Subarachnoid Hemorrhage
- d) Intraventricular Hemorrhage

Question 4:

A 39-year-old male patient presented with complaints of fever and cough. His chest radiograph shows air bronchogram sign. The probable diagnosis is :

- a) Pleural effusion
- b) Consolidation
- c) Hilar lymphadenopathy
- d) Bronchiectasis

Question 5:

The Doppler waveform of the venous blood flow in lower limb DVT is _____.

- a) Monophasic
- b) Biphasic
- c) Triphasic
- d) Multiphasic

Question 6:

A middle-aged female patient with a history of cholelithiasis, presented with vomiting and abdominal pain. The abdominal radiograph showed Rigler's triad. Which of these is not a component of Rigler's triad?

- a) Pneumobilia
- b) Small bowel obstruction
- c) Ectopic gall stone
- d) Large bowel obstruction

Question 7:

All among the following are ionizing radiations except:

- a) Alpha rays
- b) Beta rays
- c) X-rays
- d) MRI

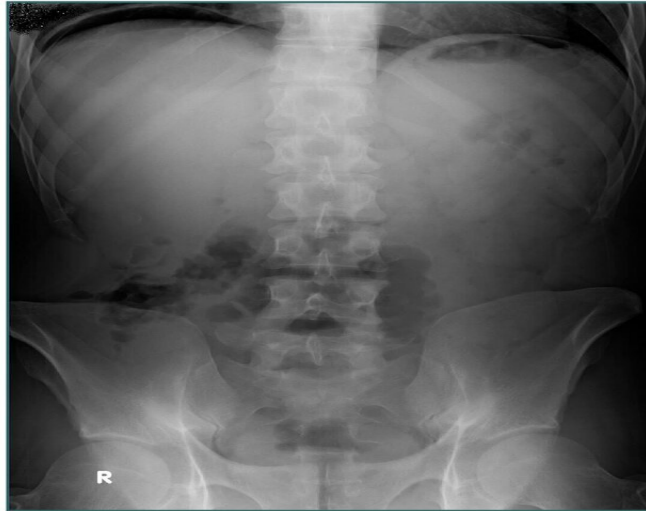
Question 8:

Empty thecal sac sign on MRI is seen in:

- a) Arachnoiditis
- b) Discitis
- c) Vertebral osteomyelitis
- d) Tethered cord syndrome

Question 9:

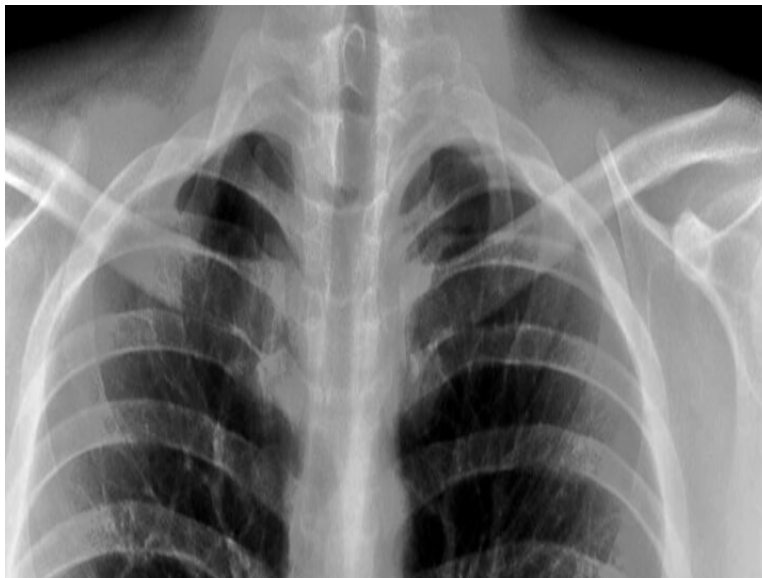
From the X-ray given, pick out the probable diagnosis:



- a) Subdiaphragmatic abscess
- b) Pneumothorax
- c) Pneumoperitoneum
- d) Intestinal obstruction

Question 10:

What is the diagnosis shown in the image below?

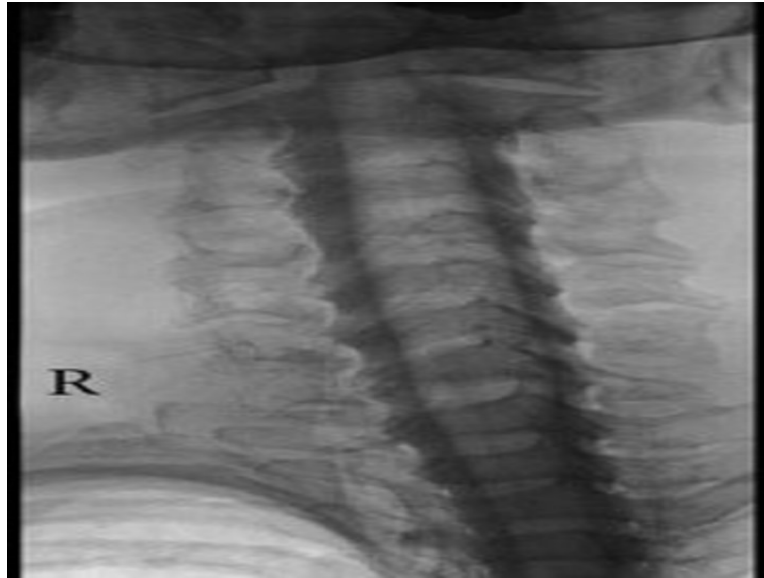


- a) Costochondritis
- b) Spondylolisthesis

- c) Cervical rib
- d) Rib fracture

Question 11:

Identify the procedure.



- a) Myelography
- b) Angiography
- c) DEXA
- d) Projectional radiography

Answer Key

Question No.	Correct Option
1	b
2	d
3	a
4	b
5	a
6	d
7	d

8	a
9	c
10	c
11	a

Detailed Explanations

Solution to Question 1:

This case should be managed by removing the button battery endoscopically after confirming its position on X-Ray.

Button battery ingestion causes alkaline caustic injury, more in the areas surrounding the negative pole.

Management :

- The child is to be kept NPO until the oesophageal position of the battery is ruled out by X-ray scan. Vomiting should never be induced since the battery, which has passed through the esophagus, might lodge back again into it. Reassurance can be done only if the position of the battery is confirmed and if there are no symptoms.
- Since the question does not mention that the child is asymptomatic, the management of choice would be removing the battery endoscopically.
- Management also depends on battery life. Fresh batteries need to be removed as soon as possible as they have a higher concentration of injurious substances.

Solution to Question 2:

Live bone is not more radio-opaque than dead bone on X-ray images.

In fact, dead bone looks more white (more radio-opaque) than live bone on X-Ray as there's no bone resorption activity.

Loss of soft tissue planes is the earliest X-ray change in infections and is seen 24-48 hours after the onset of disease.

Joint space in X-rays is the radiolucency of cartilage being apparently perceived as space.

Glass pieces can be seen on X-ray images as they contain lead.

Solution to Question 3:

The given CT image shows the typical bi-convex hyperdense lesion not crossing cranial sutures, characteristic of an extradural hemorrhage.

As the dura mater tends to adhere to the skull, the hematoma collecting between the skull and this dura is seen on CT sections as a biconvex dense area immediately beneath the skull vault, convex towards both the brain and the vault. They tend not to cross the suture lines.

The temporoparietal convexity is the most common site. The hematoma usually lies beneath a fracture of the squamous part of the temporal bone. The underlying brain is displaced but often appears intrinsically normal.

Monitoring an extradural hemorrhage and immediate intervention if necessary is an important step in management since the brain can herniate with increased bleeding that is causing raised ICP.

Solution to Question 4:

The radiograph shows air bronchogram sign in a lung with consolidation, seen in pneumonia, which is supported by the clinical history of this patient.

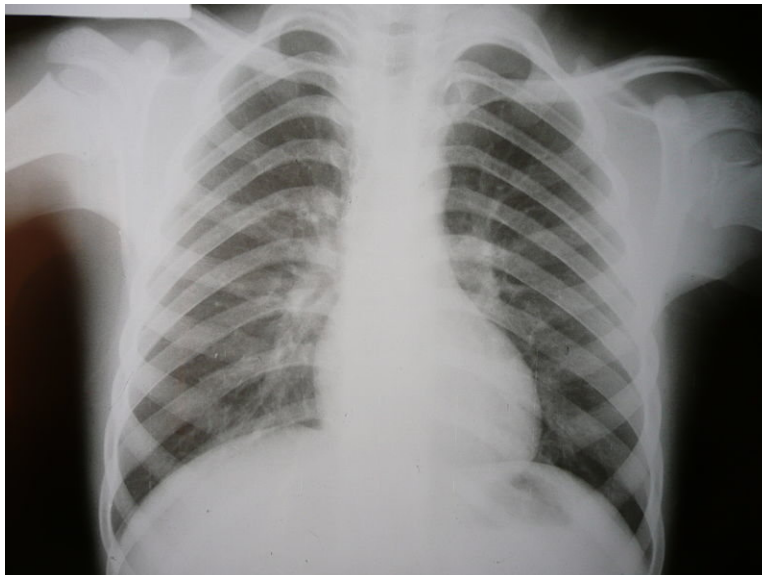
An air bronchogram is a tubular outline of an airway made visible when there is filling of the surrounding alveoli by fluid or inflammatory exudates.

The causes of air bronchogram are as follows:

- Lung consolidation
- Pulmonary edema
- Nonobstructive pulmonary atelectasis
- Severe interstitial disease
- Neoplasm
- Normal expiration

Pleural effusion is seen as a homogenous opacity on chest radiograph.

The given radiograph shows peribronchial thickening and air bronchogram sign:





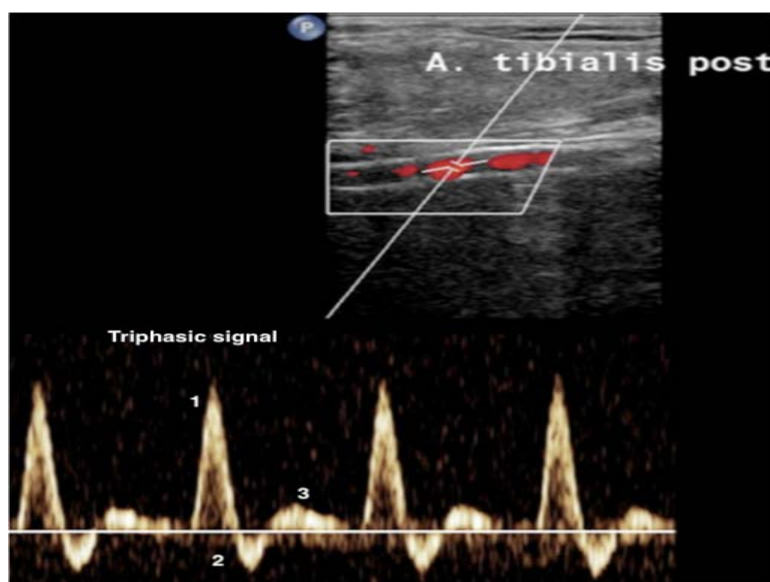
Solution to Question 5:

The Doppler waveform of the venous blood flow in lower limb DVT is monophasic proximal to the level of obstruction.

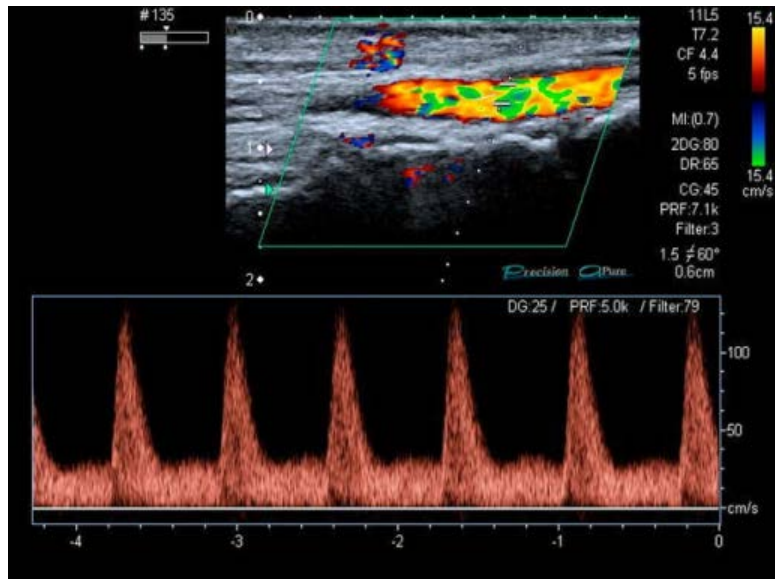
The normal venous waveform shows a variation (phasicity) secondary to respiration referred to as respirophasic waveform. Loss of this phasic variation results in a monophasic waveform suggestive of venous obstruction proximal to the level of interrogation.

A normal artery has a triphasic signal. The waveform becomes biphasic in mild-moderate stenosis and monophasic in severe stenosis.

The image given below shows the normal triphasic signal from an artery.



Monophasic waveform:



Solution to Question 6:

Rigler's triad is seen in gallstone ileus and consists of

- Pneumobilia
- Small bowel obstruction
- Ectopic gall stone (usually in the right iliac fossa)

This should not be confused with Rigler's sign and Rigler's notch sign.

Rigler's Sign:

Seen in pneumoperitoneum. Here, the air is seen on both sides of the bowel wall unlike normally, where the air is seen only in the intraluminal side of the bowel.

Rigler's notch sign:

Refers to indentation seen in the border of a solid lung mass, which is thought to represent a feeding vessel, thus suggesting the presence of bronchial carcinoma. But, this is not specific and can also be seen in granulomatous infections.

Solution to Question 7:

MRI is a technique that produces topographic images by means of magnetic fields and radiofrequency waves.

The ionizing radiations

- Alpha
- Beta

- Gamma
- X-Rays

The non-Ionizing radiations are:

- Radiofrequency waves
- UV rays
- Microwaves

Alpha and beta rays are particulate radiation (particles) whereas gamma and x-rays are electromagnetic rays. All electromagnetic waves have a constant velocity of 3×10^8 m/s and no mass. The electromagnetic spectrum, from low frequency to high frequency consists of:

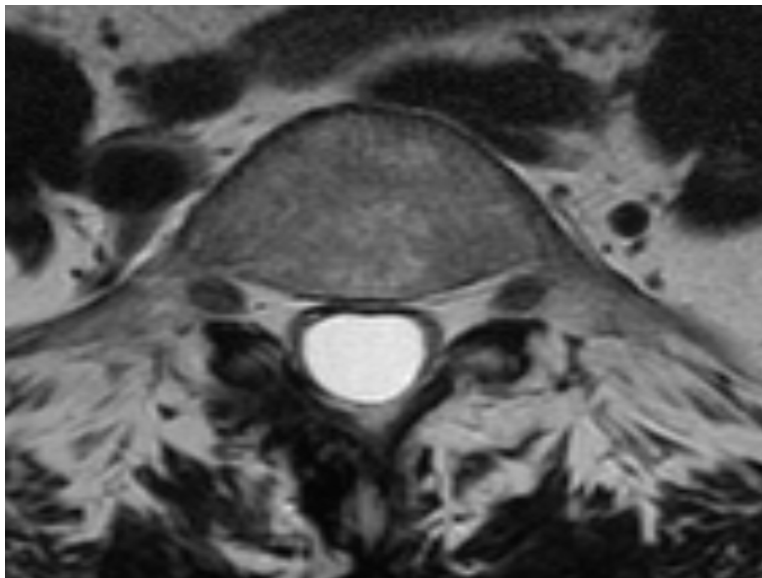
Radiowaves < Microwaves < Infrared < Visible light < Ultraviolet < X-rays < Gamma rays

Note: W. C. Roentgen is the father of X-ray and the first medical X-ray by Roentgen was of his wife Anna Bertha's left hand. Physicists Peter Mansfield and Paul Lauterbur developed MRI-related imaging techniques in medicine and they were awarded the Nobel prize in 2003.

Solution to Question 8:

An empty thecal sac sign is seen in arachnoiditis.

Adhesive arachnoiditis can be diagnosed if the empty thecal sac indication is observed. This occurs due to inflammation, which leads to clumping of nerve roots. This leads to an empty sac appearance. It is best observed on T2-weighted MRI scans of the lumbar spine when the thecal sac appears empty.



Solution to Question 9:

The above X-ray clearly shows the presence of subdiaphragmatic free gas in the right upper quadrant. This indicates pneumoperitoneum.

Pneumoperitoneum is the presence of free gas within the peritoneal cavity but outside the viscera. The most common cause of pneumoperitoneum is bowel perforation.

- Gastric and posterior duodenal perforations are seen in long-standing cases of peptic ulcers.
- Intestinal perforations are seen with ischemia, inflammation, diverticula, or obstruction.
- Other causes also include iatrogenic, penetrating injuries, prolonged mechanical ventilation, peritoneal dialysis, etc.

Signs of pneumoperitoneum:

- Bowel-related signs
 - Double wall sign/Rigler's sign/gas-relief sign- Air is seen on both sides of the bowel wall.
 - Telltale triangle sign/triangle sign - Triangle of gas formed between three loops of the bowel or between the abdominal wall and two loops of the bowel.
- Peritoneal ligament-related signs
 - Football sign- Air outlines the confines of the peritoneal cavity
 - Falciform ligament sign- Air outlines the falciform ligament
 - Lateral umbilical ligament sign/inverted "V" sign- Air outlines the lateral umbilical ligament
 - Urachus sign - Air outlines the urachus
- Right upper quadrant signs
 - Cupola sign- Air outlines the median subphrenic space
 - Fissure for ligamentum teres sign- Air outlines the ligamentum teres
 - Hepatic edge sign- Cigar-shaped free-air pocket in the subhepatic region. It tracks superomedial following the contours of the liver
 - Lucent liver sign - Reduction of hepatic radiodensity due to accumulation of free gas anterior to the liver
 - Morison's pouch sign / Doge cap sign - Triangular/ crescentic gas lucency bound by the 11th rib
- Periportal free gas sign.

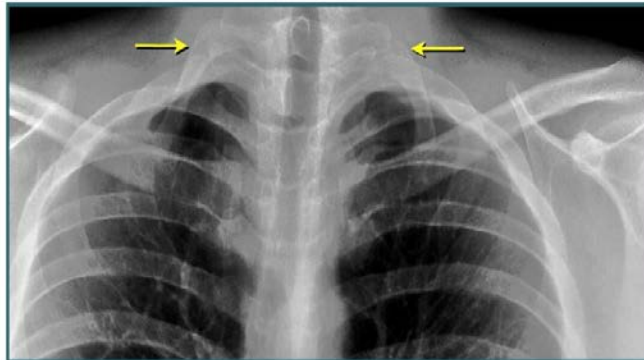
Solution to Question 10:

The given X-ray shows cervical rib.

Patients with a cervical rib usually show a lump and tenderness in the supraclavicular fossa. The pallor and paresthesias are due to vascular and neural compressions respectively. A radiograph of the neck is useful in detecting a cervical rib.

However, the absence of a rib on the x-ray does not rule out the possibility of a cervical rib. CT and MRI are more sensitive.

Bilateral Cervical Ribs



Solution to Question 11:

The procedure depicted in the image represents myelography.

Myelography is a procedure used to evaluate spinal conditions. X-ray myelography can be used to evaluate the spinal canal in patients with spondylolisthesis. While MRI has largely replaced X-ray myelography it is done when there is a contraindication to MRI (Eg; non-compatible pacemaker, severe claustrophobia).

Radiology AIIMS 2019 (May & Nov)

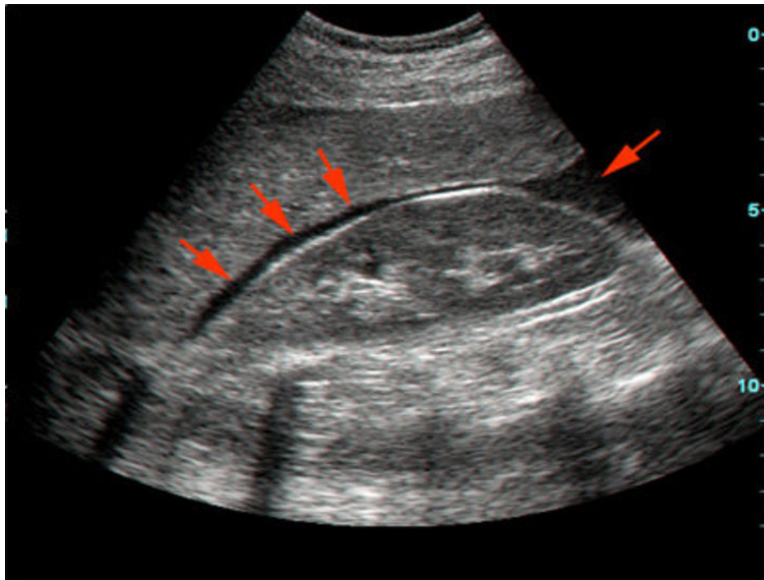
Question 1:

All lung carcinomas for diagnosis requires CECT thorax but MRI is useful in which of the following:

- a) Small cell lung carcinoma
- b) Adenocarcinoma
- c) Carcinoid
- d) Pancoast tumor

Question 2:

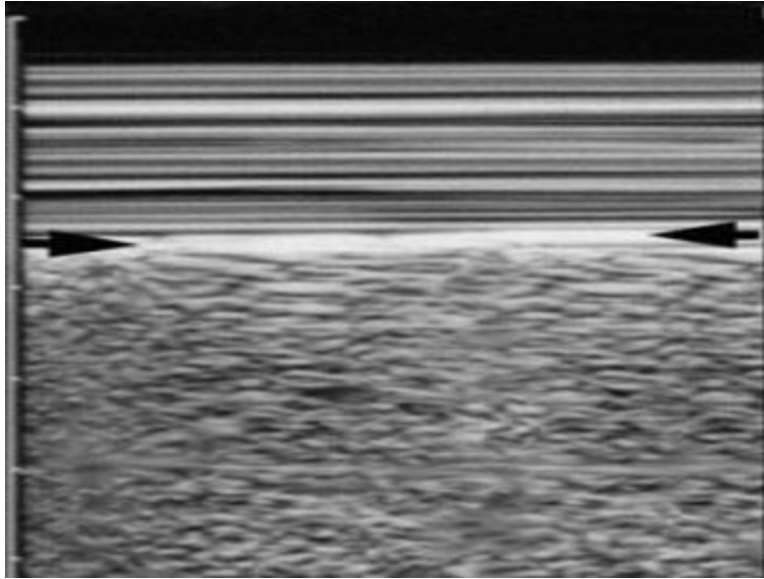
The following image of a FAST scan in a trauma patient suggests:



- a) Positive fast
- b) Negative fast
- c) Equivocal fast
- d) Pneumoperitoneum

Question 3:

The given USG of lung shows:



- a) Stratosphere sign
- b) Sea shore sign
- c) Troposphere sign
- d) Mirror image artifact

Question 4:

An 8-year old child presents with the history of cough for 7 months. He also gives a history of haemoptysis for 1 month and fever for 4 days. The X-ray is given below. What is the diagnosis?



- a) Lung abscess
- b) Consolidation
- c) Hydatid cyst
- d) Thoracic duct malformation

Answer Key

Question No.	Correct Option
1	d
2	a
3	b
4	b

Detailed Explanations

Solution to Question 1:

MRI is useful in Pancoast tumor because of the associated soft tissue infiltration, extent of which can be best visualised using an MRI

Pancoast syndrome/superior sulcus tumor results from a local extension of a tumor growing in the apex of the lung, with the involvement of C8, T1, and T2 nerve roots.

It presents with pain that radiates to the ulnar distribution of the arm, often with the destruction of the 1st and 2nd ribs. Patients with the central or endobronchial growth of the primary tumor may present with a cough, hemoptysis, wheeze, stridor, dyspnea, or post-obstructive pneumonitis.

Regional spread can cause

- Tracheal obstruction,
- Esophageal compression with dysphagia
- Recurrent laryngeal paralysis with hoarseness
- Phrenic nerve palsy with the elevation of the hemidiaphragm and dyspnea
- Sympathetic nerve paralysis with Horner's syndrome (enophthalmos, ptosis, miosis, and anhydrosis).

Solution to Question 2:

The image of FAST in a trauma patient suggests Positive FAST as there is a fluid collection in the abdominal cavity. The thin anechoic area located between the liver(on the left) and the Kidney(on the right) is a collection of fluid visualized in this RUQ Sonogram image.

FAST stands for Focused assessment with sonography for trauma. It is a rapid diagnostic examination to assess patients with potential thoracoabdominal injuries. It is performed within 2-4 minutes.

Technique:

- Patient in the supine position
- 3.5-5.0 MHz convex transducer
- Following regions are scanned :
- Subxiphoid: assess for pericardial effusion and left lobe liver injuries
- Right upper quadrant: assess for right liver injuries, right kidney injury, and Morison pouch
- Left upper quadrant: assess for splenic injury and left kidney injury
- Suprapubic region: assess the bladder and pouch of Douglas

An "extended FAST or eFAST" scan may be performed: left and right thoracic views to assess for pneumothorax and haemothorax.

Sequentially order of FAST is:- pericardial sac, the Right Upper Quadrant (RUQ), Left Upper Quadrant (LUQ), and pelvis.

4 'Ps' are evaluated in sequence: Pericardial sac → Perihepatic region → Perisplenic region → Pelvis.

Note:-

In Pneumoperitoneum, air is collected in the peritoneal cavity which is usually hyperechoic not anechoic and would appear white in the ultrasound image.

Solution to Question 3:

The given image shows the seashore sign seen in M-mode imaging of the normal lung.

M-mode is defined as the time-motion display of the ultrasound wave along a chosen ultrasound line. With M-Mode, even rapid motions can be displayed, measured, and recorded.

In the M-mode of USG, the motionless portion of the chest above the pleural line creates horizontal 'waves', and the sliding below the pleural line creates a granular pattern, the 'sand'. The resultant picture is one that resembles waves crashing in onto the sand and is therefore called the 'seashore sign' and is present in a normal lung.

Option A: Stratosphere sign or barcode sign is a finding in the M-mode seen as parallel horizontal lines above and below the pleural line. This is seen in pneumothorax.

Option C: There is no named sign called troposphere sign.

Option D: Mirror image artifact is seen when there is a highly reflective surface (eg: Diaphragm) in the path of the primary beam.

Solution to Question 4:

The given X-ray shows opacities in the right middle lobe and history of fever, cough and haemoptysis suggesting consolidation due to lobar pneumonia.

Radiology AIIMS 2020 (May and Nov INI-CET)

Question 1:

Which of the following statements is false regarding acute radiation syndrome?

- a) GI manifestations occurs at a lower radiation dose than bone marrow depression
- b) Bone marrow suppression appear at doses greater than 1 Gy
- c) Bone marrow suppression occurs earlier than neurovascular effects
- d) GI manifestations occur at a lower radiation dose than neurovascular effects

Question 2:

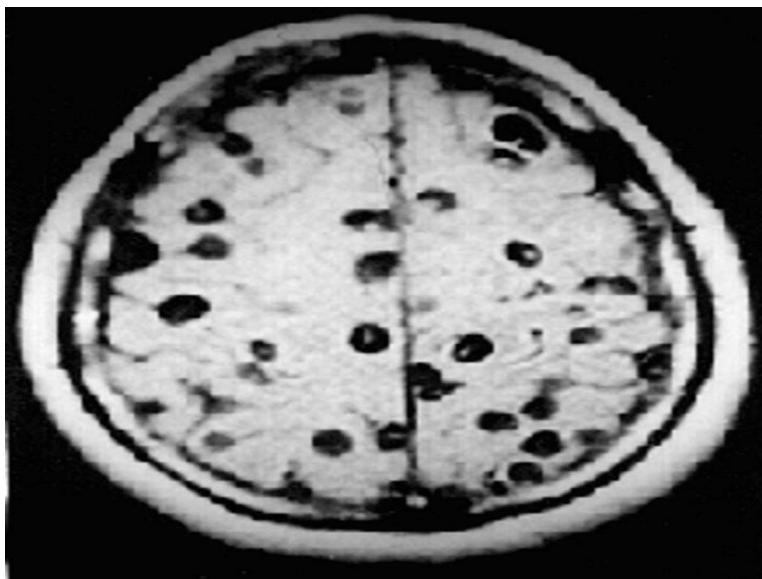
A 46-year-old male presented with chronic back pain and neck stiffness. X-ray cervical spine is given below. What is the diagnosis?



- a) Ankylosing spondylitis
- b) Diffuse idiopathic skeletal hyperostosis
- c) Klippel feil syndrome
- d) Sprengel deformity

Question 3:

An 18-year-old girl presented with seizures and her radiological image is as given below. Knowing that the MR spectroscopy of the lesion showed elevated choline, alanine, and lactate, what is the likely diagnosis?



- a) Cerebral toxoplasmosis
- b) Neurocysticercosis
- c) Cryptococcosis
- d) Multiple tuberculomas

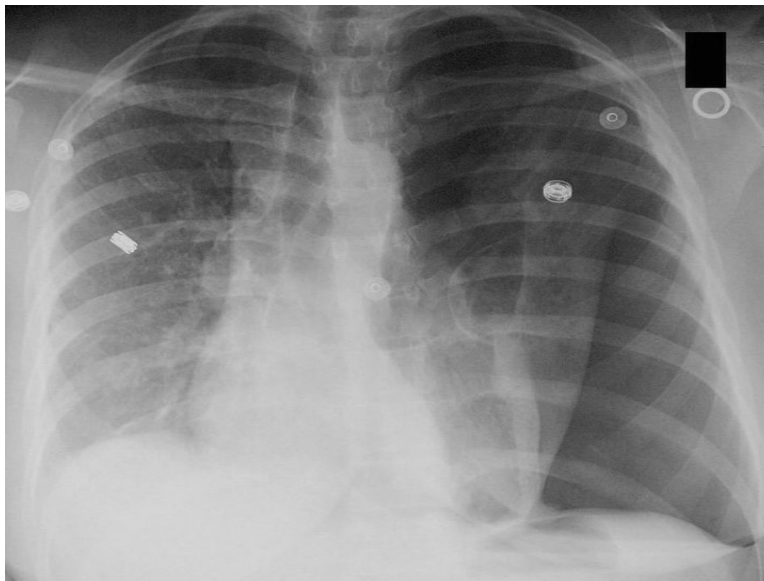
Question 4:

Which of the following is a finding of large bowel obstruction?

- a) Dilated loops of intestine with plica circularis
- b) Dilated loops with haustrations
- c) String of pearls sign
- d) Central dilated bowel loops

Question 5:

A 25-year-old male with Marfanoid features presented with sudden onset breathlessness. The chest X-ray is given below. What is the likely diagnosis?



- a) Pneumothorax
- b) Pleural effusion
- c) Consolidation
- d) Fibrosis

Question 6:

A 22-year-old male was brought to the emergency room following a road traffic accident. On examination, his GCS was E2M2V3. He had no fractures and no hydrocephalus or raised intracranial pressure. Non-contrast CT was performed and was found to be normal. The likely diagnosis is

- a) Diffuse axonal injury
- b) Post concussion syndrome
- c) Extra dural hemorrhage
- d) Herniation of brain

Answer Key

Question No.	Correct Option
1	a
2	b

3	b
4	b
5	a
6	a

Detailed Explanations

Solution to Question 1:

GI manifestations do not occur at a lower radiation dose than bone marrow depression.

Stages of Acute Radiation Syndrome (ARS)

The prodromal phase manifests with non-specific symptoms like nausea, vomiting, anorexia. Early and persistent symptoms predict severity.

Latent phase: The patient looks and feels generally healthy for a few hours or even up to a few weeks. In severe cases, this phase may be absent.

Manifest illness phase: Symptoms are seen as per the radiation dose. Tissues with greater rates of cellular division are most radiosensitive, and hence the hematopoietic and gastrointestinal systems are the most affected.

Recovery phase or Death: The recovery phase will be seen after the manifest stage when the patient recovers from the acute exposure. The time taken for recovery will depend on the severity of the injury and the organ system affected.

Radiation dose	Sub-syndrome	Features
~1-2 Gy	Hematopoietic	Bone marrow suppression
~6-10 Gy	Gastrointestinal	Nausea, vomiting, gastrointestinal bleeding, malabsorption,
~20Gy	Neurovascular	irritability, altered mental status, seizures, coma

Solution to Question 2:

The given clinical scenario and the radiograph showing ossification of the anterior longitudinal ligament, in front of the C2 to C7 cervical vertebral bodies, point towards a diagnosis of Diffuse Idiopathic Skeletal Hyperostosis (DISH). It is also referred to as Forestier disease.

DISH is a common disorder characterized by bone and calcium formation at the sites of ligamentous insertions. It occurs in the middle-aged or elderly. The radiograph shows calcification of anterior and sometimes posterior longitudinal ligaments appearing as flowing or dripping-like

candle wax. It is usually asymptomatic but the calcification might cause compression of the spinal cord and can be better visualized on CT and MRI. In contrast to ankylosing spondylitis, the sacroiliac joints are usually spared.

Solution to Question 3:

The given radiological image and history of seizures suggests a diagnosis of neurocysticercosis.

Neurocysticercosis presents with seizures, hydrocephalus, raised ICP, and may even cause signs of chronic meningitis. The given image shows multilobulated cystic lesions (with scolex) in the subarachnoid space, typical parenchymal brain calcifications are suggestive of neurocysticercosis caused by *T. solium* (Cysticercosis cellulose).

- Cyst with a dot sign: seen in both MRI and CT and the dot represents the eccentrically placed scolex of the parasite.

MR Spectroscopy features of neurocysticercosis include elevated lactate, alanine, succinate, and choline levels and reduced levels of N -acetyl aspartate and creatine. Escobar's pathological stages describe 4 stages - vesicular, colloidal vesicular, granular nodular, and nodular calcified.

Other options

- Solid or ring enhancement is typical in the post-contrast images of tuberculoma. A central focus of calcification with a ring of peripheral enhancement (the "target sign") is also described in tuberculosis but is not specific to it.
- Cerebral toxoplasmosis appears as multiple hypodense regions predominantly in the basal ganglia and at the corticomedullary junction. It can be associated with ring enhancing lesions as well.
- Cryptococcosis is usually associated with normal scans but may show cerebral atrophy, hydrocephalus or mass lesions as well.

Solution to Question 4:

Dilated bowel loops with haustrations is a feature of large bowel obstruction.

In large bowel obstruction, the colon is distended proximally with the distal portion collapsed. The colon is considered dilated if it is more than 6 cm in diameter while the cut off for caecum is 9cm.

Other options:

Dilated loops of the intestine with plica circularis (valvulae conniventes) and predominantly central loops are features of the small bowel obstruction. The small bowel is considered dilated if it is more than 3 cm in diameter.

The string of pearls sign is also a feature of small bowel obstruction seen in plain radiographs or CT. It represents a row of gas bubbles trapped between the valvulae conniventes.

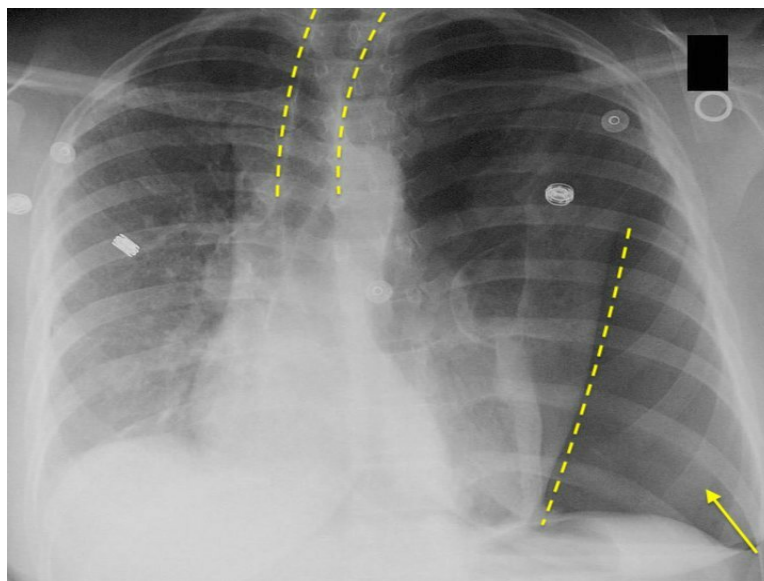
Differences in small and large bowel obstruction on radiography are listed below:

	Small bowel	Large bowel
Valvulae conniventes	Present in jejunum	Absent
No of Loops	Many	Few
Distribution of loops	Central	Peripheral
Haustra	Absent	Present
Diameter	3-5 cm	>5 cm

Solution to Question 5:

The chest X-ray demonstrating tracheal deviation and displacement of the heart away from the affected side along with the partial collapse of the left lung in the above condition points to a diagnosis of left-sided pneumothorax.

Marfan's syndrome is one of the causes for secondary spontaneous pneumothorax in adults.



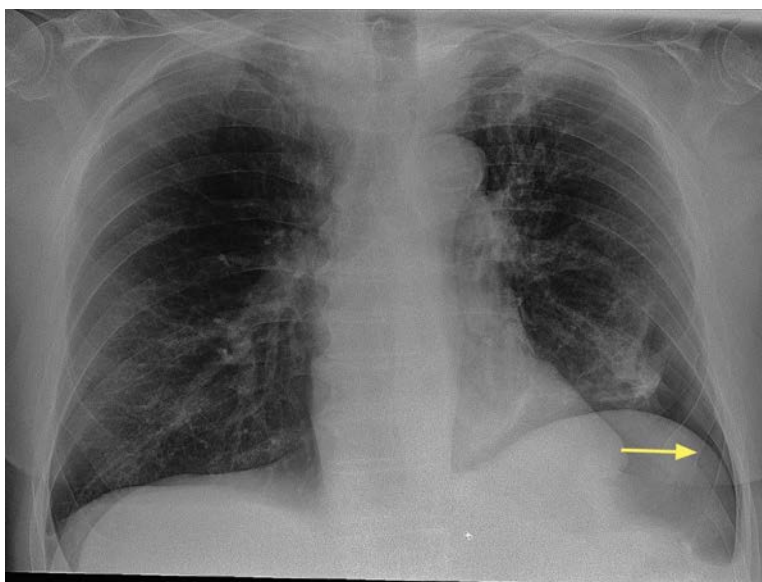
Pneumothorax refers to the collection of air in the pleural space.

Chest radiograph in spontaneous pneumothorax demonstrates:

- Thin, sharp white line, demarcating the visceral pleural edge
- Absence of lung markings seen peripheral to this line
- Peripheral space radiolucent compared to adjacent lung
- Complete lung collapse, as well as subcutaneous emphysema and pneumomediastinum, may be seen
- Deep sulcus sign (on supine chest radiograph)

Deep sulcus sign: On a supine position (common in the ICU), air collects anteriorly and basally, within the non-dependent portions of the pleural space, abnormally deepening the costophrenic angle. This sign is indicative of pneumothorax in such patients and may be the only visible clue, especially when patient can not be imaged upright.

The image below is plain chest x-ray showing pneumothorax with deep sulcus sign (shown by arrow):



When there is a progressive accumulation of air intrapleurally in the thoracic cavity (caused by a valve effect during inspiration and expiration), it is called Tension pneumothorax. It can result in compression of mediastinal structures and compromise cardiopulmonary function. It is life-threatening and requires emergency intervention.

Solution to Question 6:

The given clinical scenario of an RTA with no imaging abnormality along with a disproportionate clinical manifestation suggests diffuse axonal injury.

It is caused due to stretching of axons in subcortical white matter due to sudden deceleration injuries. It is usually associated with mild nonspecific changes in NCCT. It is visualized on T2 MRI characterized by multiple lesions showing microbleeds or linear hypointensities. It may also show disruption of white matter tracts on tractography.

Other options

Post-concussion syndrome usually follows mild traumatic brain injury. It does not have a severe clinical status as seen in this patient.

Extradural hemorrhage is produced usually results secondary to trauma to head. The bleeding is usually arterial and blood collects between outer dura and inner table of the skull. It can very well be visualized on CT.

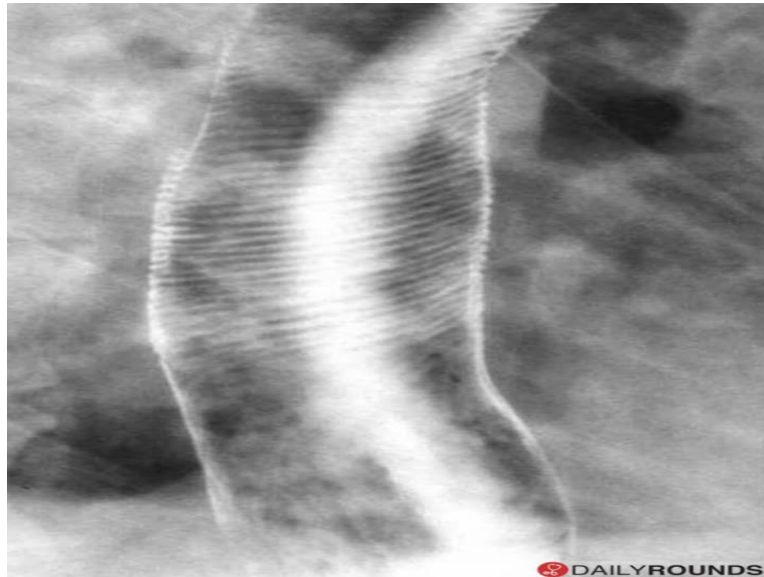
A cerebral herniation is defined as a shift of cerebral tissue in an adjacent space. Intracranial brain herniation can be classified into 3 types - subfalcine, transtentorial and tonsillar. A distinct

clinical picture can be produced as a result of compression of neurovascular structures by the herniating part. CT is the imaging modality of choice because it can be done rapidly.

Radiology NEET 2018

Question 1:

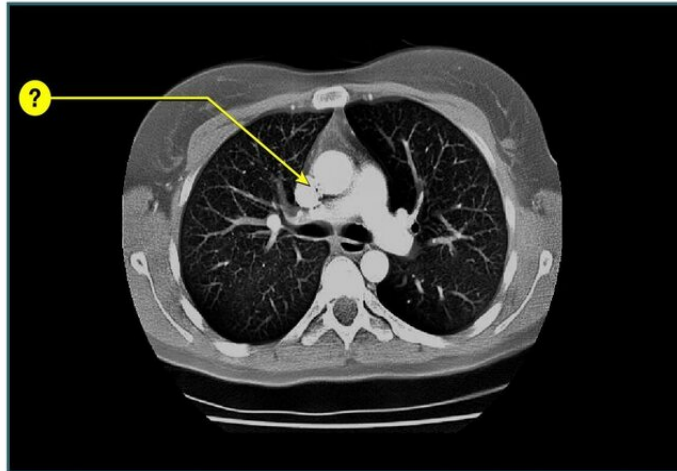
In which of the following conditions is the given X-ray finding seen?



- a) Esophageal atresia
- b) Esophageal stenosis
- c) Reflux esophagitis
- d) Barrett's esophagus

Question 2:

Identify the structure marked in the CT scan shown below:



- a) Descending aorta
- b) Ascending aorta
- c) Superior vena cava
- d) Right pulmonary artery

Question 3:

What condition causes the skull appearance as in the given radiograph?



- a) Hyperparathyroidism
- b) Multiple myeloma

- c) Hyperthyroidism
- d) Paget's disease

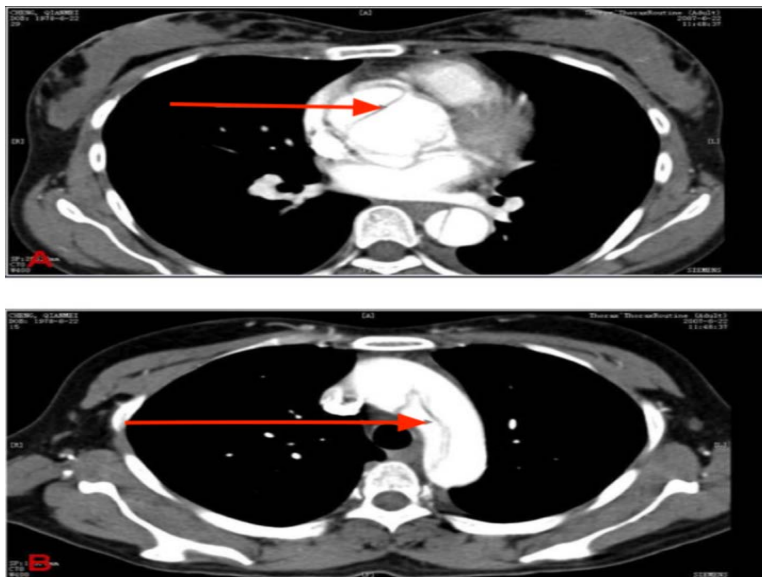
Question 4:

What dye is used for the diagnosis of esophageal perforation?

- a) Gastrograffin
- b) Iohexol
- c) Gadolinium
- d) Barium sulphate

Question 5:

A 50-year-old patient has come with sudden onset of chest pain and his CT scan shows the following picture. What is the diagnosis?



- a) Pulmonary embolism
- b) Aortic dissection
- c) Aortic aneurysm
- d) Myocardial infarction

Question 6:

The following scan is of a young woman on oral contraceptives. The most likely diagnosis is:



- a) Hepatocellular carcinoma
- b) Metastatic infiltrates
- c) Hepatic adenoma
- d) Hemangioma

Question 7:

A child presents with a left-to-right shunt and high-output cardiac failure. The following is his MRA. What is the most likely diagnosis?



- a) Vein of Galen malformation
- b) Dandy Walker syndrome
- c) Pneumocephalus
- d) Venous sinus thrombosis

Question 8:

What is the diagnosis based on the given image?



- a) Uterus didelphys
- b) Bicornuate uterus
- c) Unicornuate uterus
- d) Septate uterus

Question 9:

The Bragg peak effect is pronounced in:

- a) X ray
- b) Proton
- c) Neutron
- d) Electron

Question 10:

The image seen below shows_____.



- a) Magnetic Resonance Imaging
- b) Positron Emission Tomography
- c) Hysterosalpingography
- d) Laparoscopic chromopertubation

Question 11:

Identify the artery marked by blue arrow.



- a) Superior mesenteric artery
- b) Inferior mesenteric
- c) Celiac artery
- d) Intestinal artery

Question 12:

What is the probable diagnosis based on the image of the scan given below?



- a) Pituitary apoplexy
- b) Pituitary macroadenoma
- c) Rathke cleft cyst
- d) Craniopharyngioma

Answer Key

Question No.	Correct Option
1	c
2	c
3	a
4	b

5	b
6	c
7	a
8	c
9	b
10	c
11	a
12	d

Detailed Explanations

Solution to Question 1:

The given image is a barium swallow X-ray showing a feline esophagus suggestive of reflux esophagitis.

The feline esophagus is also known as esophageal shiver. These folds are 1-2 mm thick and run horizontally around the entire circumference of the esophageal lumen. It is thought to be due to the contraction of the muscularis mucosae of the esophagus. This results in the shortening of the esophagus and the bunching up of the mucosa in the lumen.

The appearance is almost always associated with active gastroesophageal reflux. The findings are transient, seen following reflux and not during swallowing. The appearance is confined to the distal two-thirds of the thoracic esophagus.

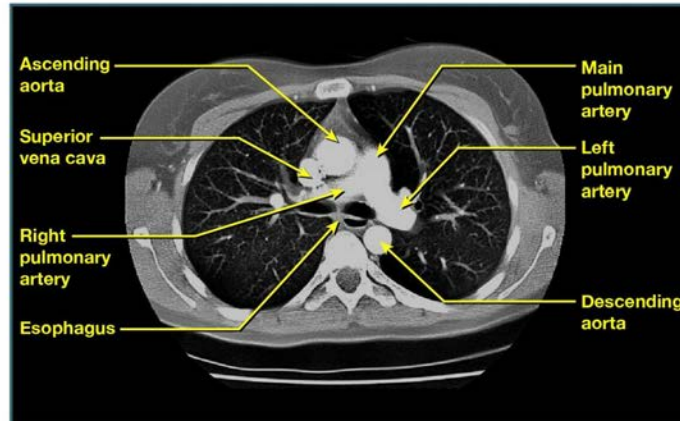
It can also be seen in hiatus hernia and eosinophilic esophagitis. Here, the folds are permanent and associated with strictures.

Solution to Question 2:

The structure marked in the given CT thorax image is the superior vena cava.

The image below depicts all the major structures labelled in the CT thorax image:

CT Thorax



Solution to Question 3:

The given X-ray of the skull shows multiple radiolucent lytic lesions, typical of the salt and pepper appearance (pepperpot appearance). This is seen in hyperparathyroidism.

In hyperparathyroidism, there is resorption of the trabecular bone of the skull (which causes the radiolucent areas) in combination with cystic areas of osteopenia. There is also a loss of definition between the inner and outer tables of the skull.

Other options:

Option B: In multiple myeloma, multiple well-defined "punched out" lytic lesions exist.

Option C: Untreated hyperthyroidism for several years can lead to osteoporosis of bones.

Option D: In Paget's disease, the pelvis, spine, skull, and proximal long bones are most frequently affected. Radiological skull changes include:

- Osteoporosis circumscripta: Large, well-defined lytic lesion
- Cotton wool appearance: Mixed lytic and sclerotic lesions of the skull
- Diploic widening: Both inner and outer calvarial tables are involved, with the former usually more extensively affected
- Tam o'Shanter sign: Frontal bone enlargement, with the appearance of the skull falling over the facial bones, like a Tam o'Shanter hat

Solution to Question 4:

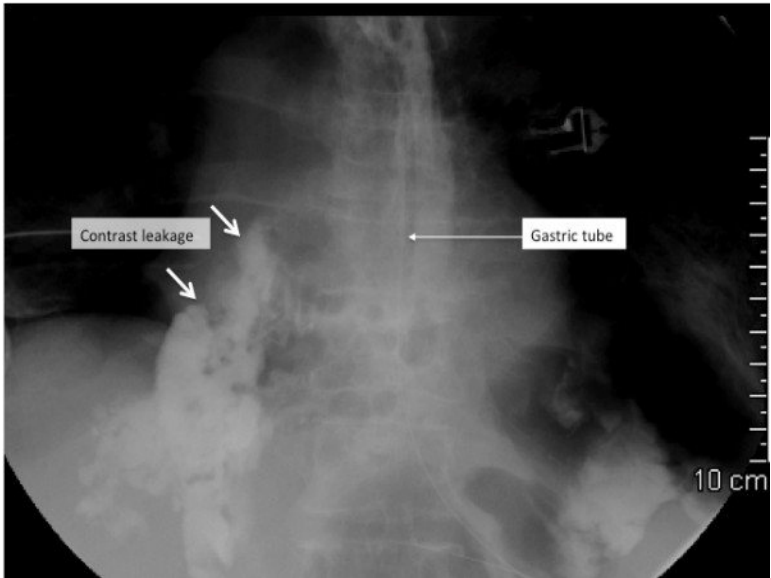
Iohexol is the contrast medium used for the diagnostic work-up for a suspected esophageal perforation.

Although gastrograffin (water-soluble) was used previously, it has a high chance of causing aspiration pneumonitis and pulmonary edema consequently. Hence it is not preferred nowadays. (Option A)

Barium contrast is not used in esophageal perforation as it causes barium-related inflammation of the mediastinum. (Option D)

Given below is a plain chest radiography with a water-soluble contrast swallow, showing contrast leakage in a patient with spontaneous rupture of the esophagus.

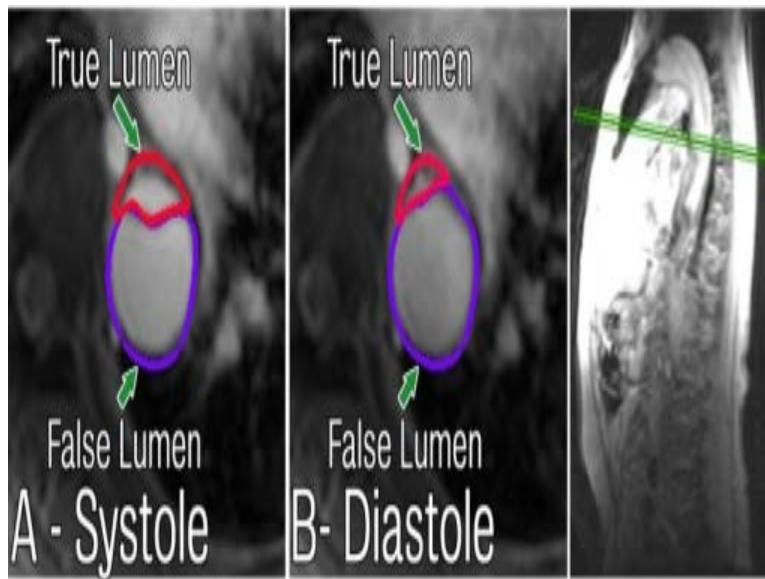
Contraindications to the use of barium contrast	Alternative dyes used
Tracheoesophageal fistula	Dianosil- water-soluble, non-ionic agent
Esophageal perforation	Iohexol
Perforation peritonitis	Gastrograffin / Iohexol



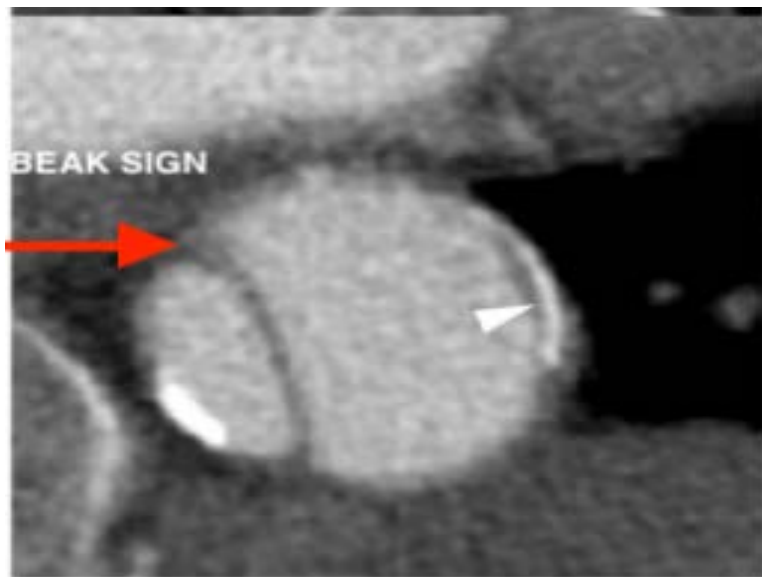
Solution to Question 5:

The given CT scan showing a double lumen and dilatation of the aorta is suggestive of an aortic dissection.

In this condition, blood enters the space below the intima of the aorta. This results in the formation of a true lumen and false lumen. The true lumen is often the smaller of the two because of being compressed by the false lumen, as shown below.



The beak sign in aortic dissection is a reliable sign to identify the false lumen. It refers to the acute angle formed at the edge of the false lumen in aortic dissection in axial cross-section. It is formed by the borders of the outer aortic wall and the intimal flap, and maybe partially thrombosed (blunted beak).



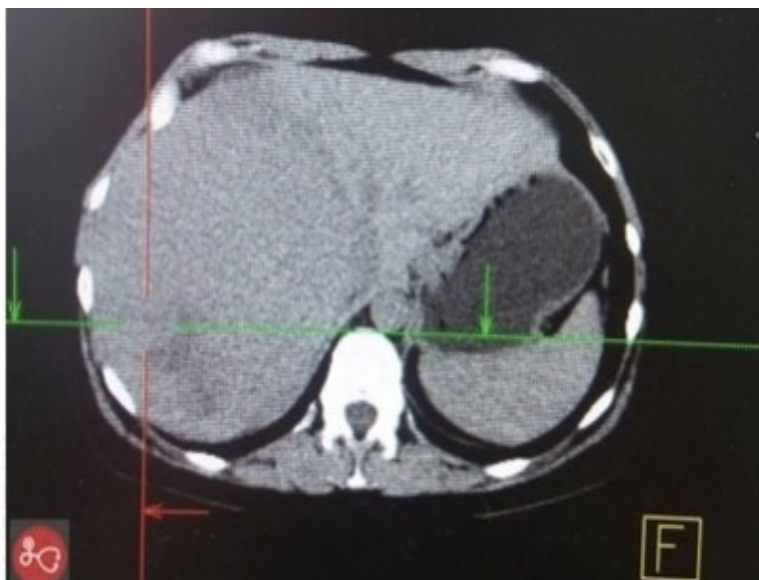
Solution to Question 6:

The image given is of hepatic adenoma. It is the most frequent hepatic tumour in young women on oral contraceptive pills.

Hepatic adenomas, also referred to as hepatocellular adenomas, are benign liver tumours. They are well-defined vascular lesions, which are generally solitary. They are almost always seen in women between the age of 25 and 50 years and are associated with the use of the oral contraceptive pill.

On CT, uncomplicated adenomas are usually homogeneous, with a well-defined margin. They enhance markedly and uniformly during the arterial phase and then merge with the surrounding liver during the portal phase.

Hepatic adenoma on the CT scan is shown below.

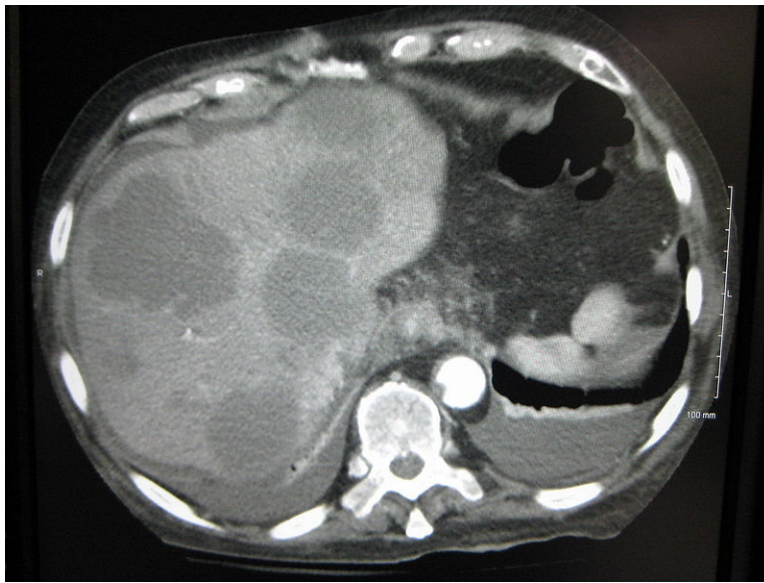


Other options:

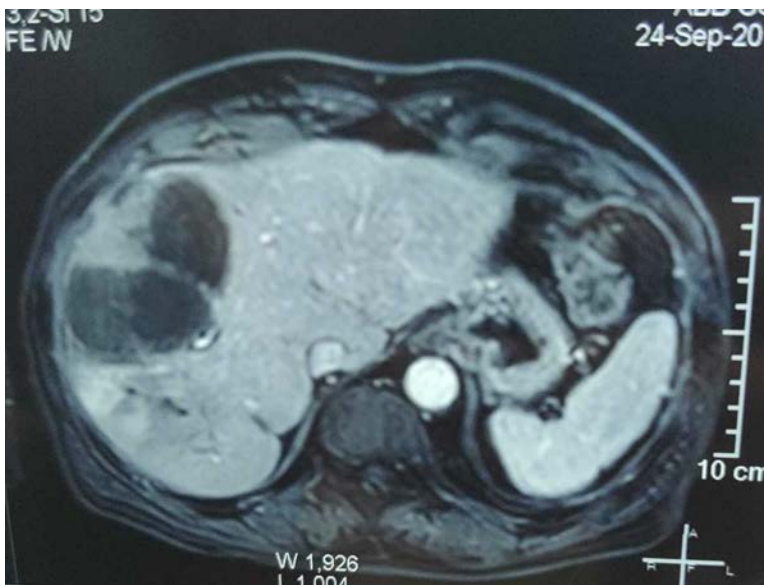
Option D: On contrast-enhanced CT scan, hemangiomas show asymmetrical, nodular peripheral enhancement that is isodense with large vessels and progressive centripetal enhancement. Liver hemangioma on CT scan is shown below.



Option B: Most metastatic deposits appear as areas of low attenuation on unenhanced CT and portal-phase CT. Hypervascular tumours are often visible as low-attenuation lesions on unenhanced images and may enhance transiently in the arterial phase, some becoming invisible in the portal phase. Metastatic infiltrates in the liver in the portal phase on CT scan are shown below.



Option A: Hepatocellular carcinoma (HCC) typically shows a decreased signal on T1-weighted images and a moderately increased signal on T2-weighted images, with internal heterogeneity. MRI is regarded as the most sensitive imaging modality for detecting HCC. Given below is the MRI image of HCC:



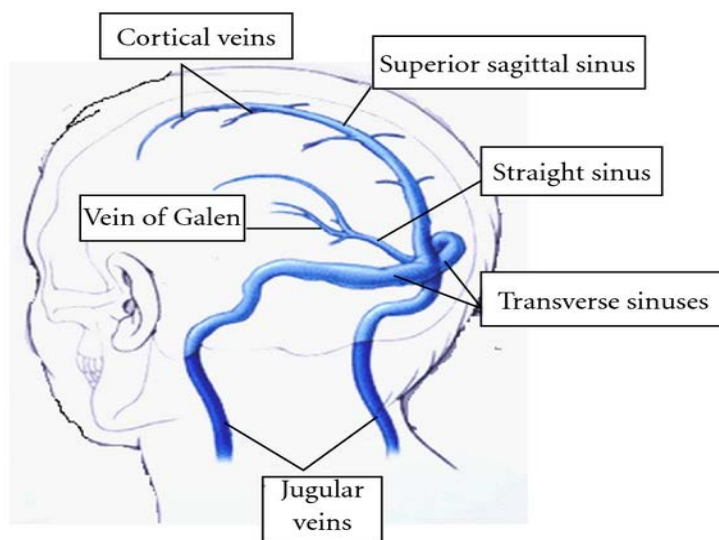
Solution to Question 7:

The MR-Angiography shown is of Vein of Galen Malformation.

Vein of Galen aneurysmal malformations (VGAM's), also known as median prosencephalic arteriovenous fistula, are uncommon intracranial anomalies that tend to present in early childhood with features of a left-to-right shunt and high-output cardiac failure.

The vein of Galen, also known as the great cerebral vein or great vein of Galen, is a short trunk formed by the union of the two internal cerebral veins and basal veins of Rosenthal. It lies in the quadrigeminal cistern. It curves backwards and upward around the posterior border of the splenium of the corpus callosum and drains the anterior and central regions of the brain into the sinuses of the posterior cerebral fossa.

The following image shows the Vein of Galen:



Solution to Question 8:

The given image shows the hysterosalpingogram of a unicornuate uterus.

The endometrial cavity usually assumes a fusiform (banana type) shape, tapering at the apex and draining into a single fallopian tube. The uterus is generally shifted off the midline.

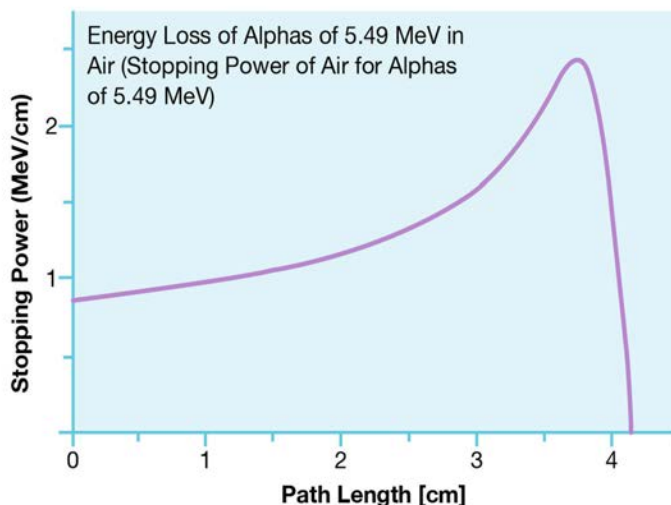
Solution to Question 9:

The Bragg peak effect is pronounced in protons.

The proton is a heavy and charged particle that gradually loses its speed as it interacts with human tissue. It is easily controlled and delivers its maximum dose at a precise depth and can go as far as 32 cm. The absorbed dose increases very gradually with greater depth and lower speed, suddenly rising to a peak when the proton is ultimately stopped. This is known as the Bragg peak.

The behavior of the proton can be precisely determined, and the beam can be directed, allowing the Bragg peak to occur within the tumor site. Immediately following this burst of energy, the protons cease radiating and come to a complete stop. This property of proton therapy enables targeting tumors within the body, delivering radiation directly to the tumor while sparing the surrounding healthy cells, offering a much less invasive alternative to treat cancer.

The Bragg curve shown below plots the energy loss of ionizing radiation during its travel through matter.



Solution to Question 10:

The image given is of the findings seen in hysterosalpingography. The given image shows bilateral retort-shaped fallopian tubes, indicating bilateral hydrosalpinx.

Hysterosalpingography (HSG) is a technique used for visualization of the uterine cavity and fallopian tubes by the injection of a radioopaque dye, and visualization of the dye by fluoroscopy as it passes through the uterine cavity and fallopian tubes.

It is best performed between the end of the menstrual period and ovulation (i.e. 9th or 10th day of the cycle). 10-15ml of the dye is injected using a Leech-Wilkinson cannula, Rubin's cannula, or Foley's catheter. If the tubes are patent, the dye will spill out of the tubal Ostia and smear the adjacent bowel.

It is used to detect:

- Tubal block: This can be due to strictures (fibrotic block), inspissated amorphous material, or spasms. The tubal spasm can occur in an apprehensive woman, and this is prevented by premedication with atropine and analgesia.
- Uterine anomalies
- Uterine polyps
- Uterine synechiae
- The feasibility of tuboplasty, by studying the location and extent of tubal pathology.

Contradictions:

- Presence of genital tract infection and bleeding
- Genital tuberculosis, due to the risk of spreading infection.

- Suspected pregnancy
- Allergy to the dye
- Premenstrual phase (there is a risk of embolism and endometriosis if HSG is done during this phase).

The image given below shows the appearance of a normal hysterosalpingogram.



Laparoscopic chromopertubation is a laparoscopic procedure for establishing the patency of fallopian tubes or verifying the findings of blocked tubes on HSG. Methylene blue is injected through the cervix to visualize the free spill. The absence of spill indicates blocked tubes.

Solution to Question 11:

The artery marked by the blue arrow is the superior mesenteric artery (SMA), which gives rise to various branches that supply the small intestines, cecum, ascending, and part of the transverse colon.

The CT also depicts nutcracker syndrome, a left renal vein entrapment syndrome characterized by extrinsic compression of the left renal vein (marked by a red arrow), most commonly between the abdominal aorta and superior mesenteric artery. This syndrome is one of the postulated reasons for left-sided varicoceles being more common than the right side.

Solution to Question 12:

The given MRI of the brain demonstrates a craniopharyngioma, which is a suprasellar tumour that arises from remnants of Rathke's pouch.

The two variants of craniopharyngioma are:

- Adamantinomatous type (common in children): Predominantly solid with cystic components. Diagnostic features include compact lamellar keratin ("wet keratin") and areas of dystrophic calcification.
- Papillary type (common in adults): Contains both solid sheets and papillae lined by well-differentiated non-keratinized squamous epithelium.

These tumours are usually large and heterogeneous, displaying solid and cystic components. The cysts contain a cholesterol-rich, thick brownish-yellow fluid that is compared to 'machine oil'.

Clinical features include headaches (most common), visual abnormalities - decreased acuity and bitemporal hemianopia, GH deficiency, growth failure, delayed sexual maturation due to pituitary hypofunction, and polyuria (central diabetes insipidus).

Radiologically:

- CT may show calcifications associated with the solid and cystic wall components
- MRI is the imaging modality of choice which shows:

Craniopharyngiomas tend not to expand the pituitary fossa unless they become huge, which is a differentiating feature from pituitary macroadenomas.

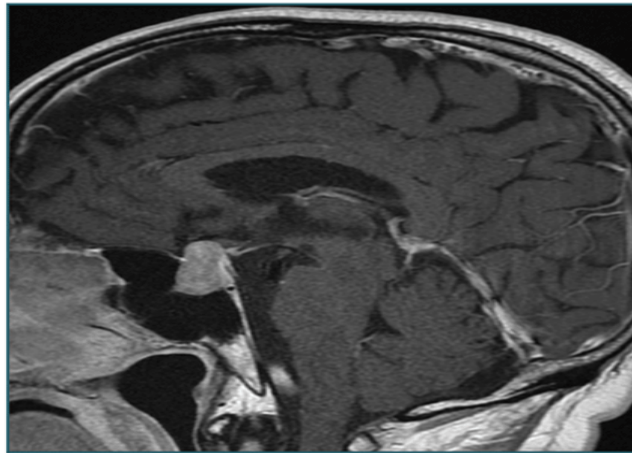
Other options:

Option A: Pituitary apoplexy is an acute clinical condition associated with the rapid expansion of the pituitary gland, which is evident on an MRI as enhancement around a non-enhancing infarcted centre.

Option B: Pituitary macroadenoma (with cystic degeneration or necrosis) shows indentation at the diaphragma sellae, giving rise to a snowman/figure of eight appearance/cottage loaf. Calcification is often absent. The image below shows the snowman appearance:

Cysts	
T1	Iso to hyperintense to the brain (due to high protein content "machinery oil cysts")
T2	Mostly hyperintense
Solid component	
T1	Vivid enhancement
T2	Variable or mixed

Snowman appearance – Pituitary adenoma

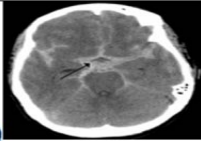
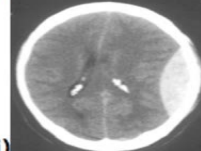
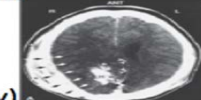


Option C: Rathke cleft cyst shows no solid or enhancing component. Calcification is rare. It is unilocular and the majority are completely or mostly intrasellar.

Radiology NEET 2019

Question 1:

Match the following intracranial bleed:

A)Epidural hematoma	 i)
B)Subdural hematoma	 ii)
C)Subarchnoid haemorrhage	 iii)
D)Intracerebral haemorrhage	 iv)

- a) A -i B-ii C- iii D- iv
- b) A -ii B-iv C-i D-iii
- c) A-iv B-iii C -ii D-i
- d) A -iv B-ii C-iii D-i

Question 2:

The following appearance in IVP is caused by:



- a) Horse shoe kidney
- b) Ureterocoele
- c) Bladder tumour
- d) Posterior urethral valve

Question 3:

Which of the following investigations is shown in the radiograph?



- a) Barium swallow
- b) Barium meal

- c) Barium meal follow through
- d) Barium enema

Question 4:

What is the best imaging study for earliest diagnosis of cerebral infarct?

- a) FLAIR MRI
- b) Non-contrast CT
- c) CECT
- d) Diffusion weighted MRI

Question 5:

During evaluation of PUO, a patient's chest X-ray is as seen below. What is your diagnosis?



- a) Miliary tuberculosis
- b) Pulmonary edema
- c) Necrobiotic nodules
- d) Tuberculoma

Question 6:

Identify the structure marked in the following MRI scan:



- a) Midbrain
- b) Cerebellum
- c) Brainstem
- d) Cerebrum

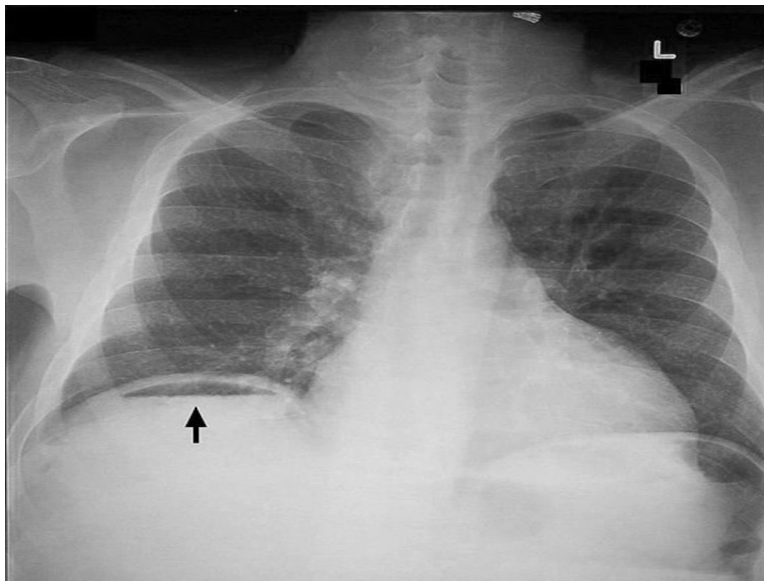
Question 7:

Which of the following is a water soluble contrast?

- a) Iohexol
- b) Barium
- c) Bromium
- d) Calcium

Question 8:

All of the following could be the cause of the following X-ray finding except:



- a) Duodenal ulcer perforation
- b) Post laparoscopy
- c) Chilaiditi syndrome
- d) Acute intestinal obstruction

Answer Key

Question No.	Correct Option
1	b
2	b
3	b
4	d
5	a
6	b
7	a
8	d

Detailed Explanations

Solution to Question 1:

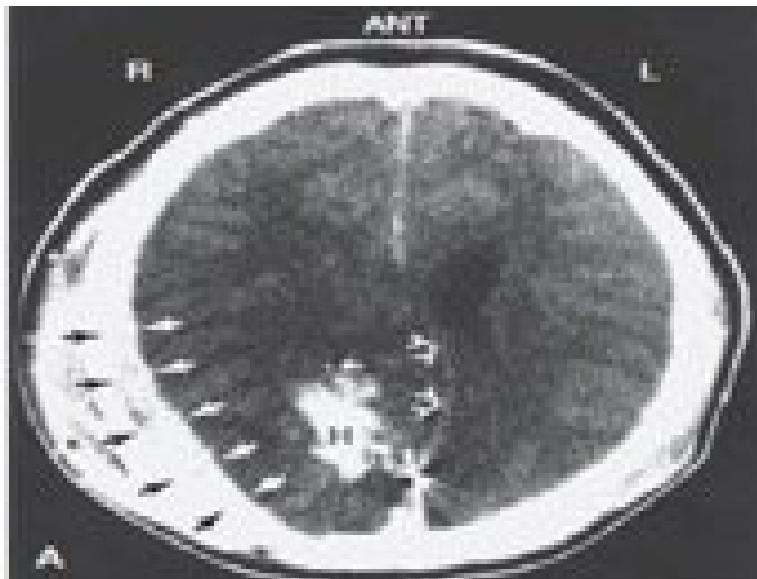
Epidural hematoma:



Noncontrasted axial computed tomography scan showing a lenticular area of increased density in left parietal region.

- Typically occur in the groove of the middle meningeal artery associated with temporal bone fractures that have resulted in a tear of middle meningeal artery.
- They tend to not to cross suture lines of the skull.

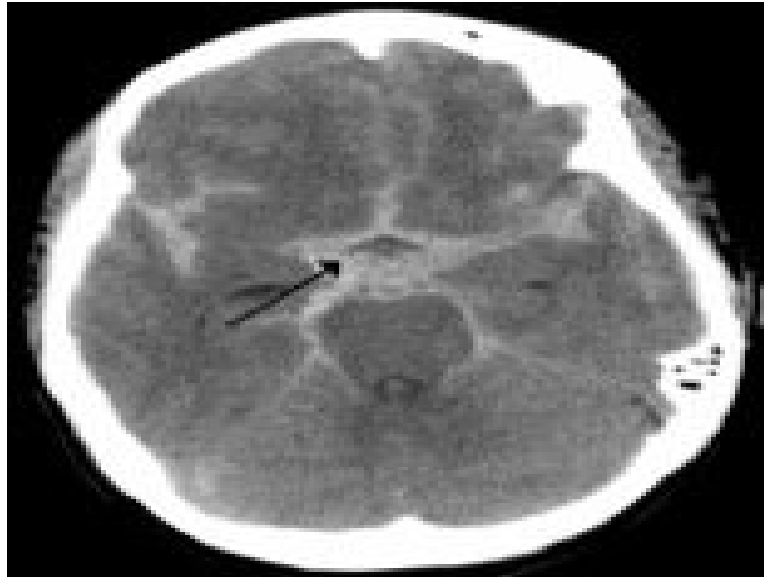
Subdural hematoma:



A non contrasted computed tomography scan of acute subdural hematoma shows a crescentic area of increased density in the right posterior parietal region between brain and the skull. An area of intraparenchymal haemorrhage also seen. In addition, mass effect causes a midline shift to left.

- They can cross suture lines but they do not cross the tentorium or falx.
- Can be acute or chronic

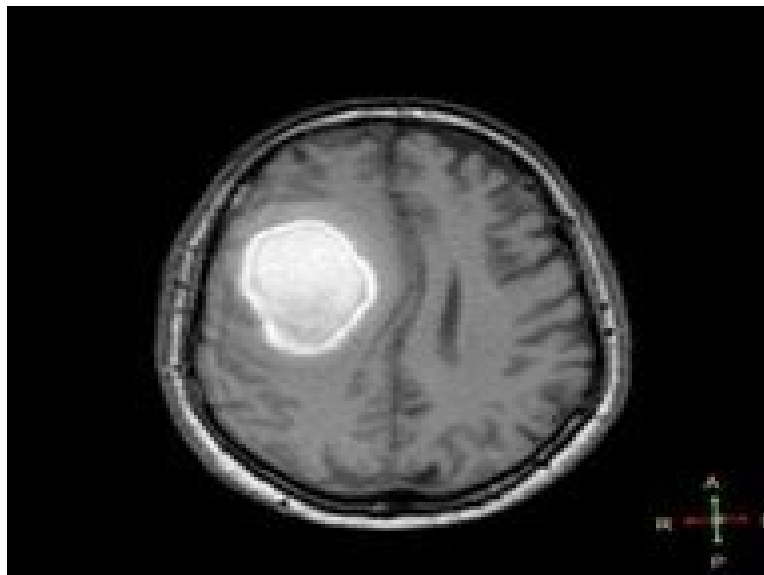
Subarachnoid haemorrhage:



A non contrasted axial computed tomography scan shows the blood as areas of increased density near the base of the brain shows blood in Texaco star pattern formed blood radiating from suprasellar cistern into the sylvian fissures and anterior interhemispheric fissure.

- Most often accompanied by severe sudden onset headache.
- Can really be visualised in acute stage, when blood is radiographically dense (whiter) than the CSF.

Intracerebral haemorrhage:



Non contrast computed tomography shows a large area of fresh blood appearing white in the region of right thalamus. Blood is denser than CSF so settles dependently.

This settling is not seen with subarachnoid or interparenchymal blood.

Solution to Question 2:

The given intravenous pyelography (IVP) depicts the cobra head or adder head deformity or spring onion sign seen in bilateral ureterocele. These signs refer to a dilated, contrast-filled distal ureter surrounded by a thin rim of the ureteric wall and bladder mucosa.

Ureterocele is a cystic enlargement of the intramural portion of the ureter, more commonly seen in females than males. It is thought to be caused due to ureteric orifice atresia. They are similar to ectopic ureters, which typically drain the upper pole of the kidney and are associated with dysplastic or non-functional renal tissue.

If they present clinically, ureteroceles often manifest with urinary tract infections (UTIs) in children, while in adults, they are more likely to present with stones in the lower ureter. Ureteroceles can enlarge and obstruct the bladder neck or the contralateral ureteric orifice.

The classic sign of a ureterocele on intravenous urography is a 'cobra head', 'adder head', or 'spring onion sign'. On ultrasonography, it can be seen as a fluid-filled cystic mass in the bladder.

The treatment of simple ureteroceles is surgical excision with re-implantation of the ureter.

Solution to Question 3:

The given image is a double-contrast radiograph of a barium meal. It is used to examine the lower half of the esophagus, stomach, and entire duodenum. It is used in identifying ulcers, small fleshy lumps (polyps), tumours.

Barium meal follow-through is useful in visualising small bowel pathology and evaluating the function of the small bowel. It is performed following a barium meal examination where X-rays are taken 10-15 minutes after the intake of barium. This allows time for the barium to reach the small intestine. It is inferior to enteroclysis enema, where contrast is administered through a nasogastric tube.

GI Imaging	Preparations of barium	Structures best visualized
Barium swallow	Barium paste	Esophagus
Barium meal	95% BaSO ₄	Stomach and duodenum
Barium meal follow-through	50% BaSO ₄	Small bowel
Barium enema	25% BaSO ₄	Large bowel

The below radiograph is a double-contrast barium enema with a normal large bowel structure. The contrast is administered through a rectal tube and a series of radiographs are obtained.



Solution to Question 4:

The best imaging modality for the early diagnosis of cerebral infarction is diffusion-weighted MRI (DWI), which can detect infarcts as early as 30 minutes of onset and identify cytotoxic oedema within a few hours of ischemic stroke.

However, the initial investigation in suspected stroke cases is non-contrast CT due to its rapid acquisition time, widespread availability, and cost-effectiveness. It helps quickly rule out hemorrhagic stroke and facilitates the timely administration of intravenous tPA in eligible patients.

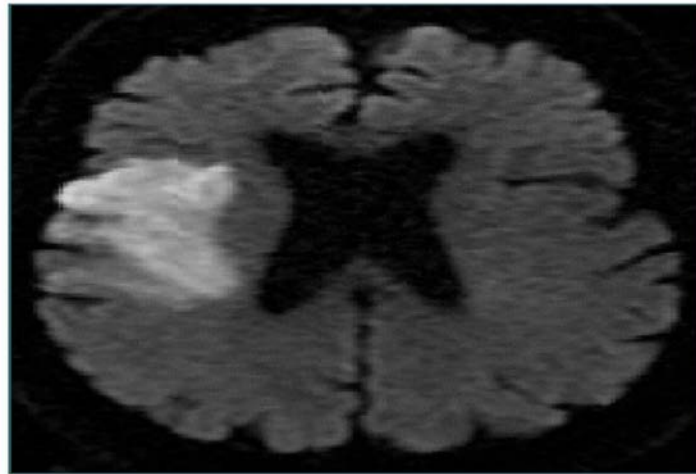
Acute MCA territory infarct on CT is shown below:

CT of Middle Cerebral Artery Territory Infarct



The image below is of a DWI MRI of a patient clearly showing the extent of the infarct.

Diffusion-Weighted MRI of
Middle cerebral Artery Infarct



Solution to Question 5:

The given chest radiography image shows miliary tuberculosis.

It is the widespread dissemination of *Mycobacterium tuberculosis*, via hematogenous spread.

Classic miliary TB is defined as millet-like (mean, 2 mm; range, 1-5 mm) seeding of TB bacilli with uniform size and distribution in the lung, as seen on chest radiography. This particular sign is called the miliary shadow which is also seen in sarcoidosis and histoplasmosis.

Sputum-smear microscopy is negative in most cases.

Physical findings include hepatomegaly, splenomegaly, and lymphadenopathy

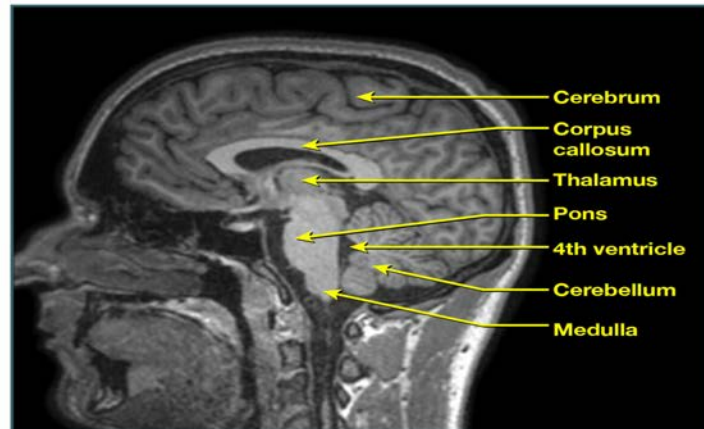
Bronchoalveolar lavage and transbronchial biopsy are used to provide bacteriologic confirmation

Solution to Question 6:

The given image shows an MRI of the sagittal section of the brain. The area marked with a red arrow is the cerebellum.

The other areas also visualized in the image are :

MRI Brain- Sagittal T1



Solution to Question 7:

Iohexol is a nonionic, water-soluble radiographic contrast medium.

It is used to visualize the following structures:

- Arteries
- Veins
- Ventricles of the brain
- The urinary system
- Joints

It can be used during computer tomography (CT scan) as well. It can be given orally, intravenously, or can be injected into the body cavity.

Solution to Question 8:

The given X-ray shows air under the diaphragm, indicative of pneumoperitoneum, which is not caused by acute intestinal obstruction.

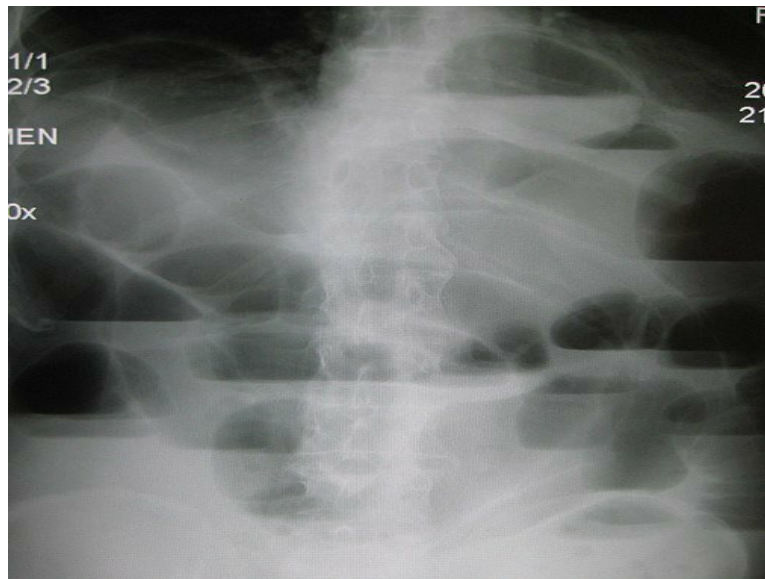
Causes of pneumoperitoneum include:

- Bowel perforation: Gastric and posterior duodenal perforations in long-standing cases of peptic ulcers., ischemic bowel, necrotizing enterocolitis
- Small bowel perforation secondary to trauma or penetrating injuries
- Post-surgery/laparoscopy
- Causes other than perforation are:

- Chilaiditi syndrome - the intestine is seen between the liver and diaphragm giving a false impression of free air
- Sub-phrenic abscess
- Postoperatively, air may be present in the abdomen for up to 7 days

An intestinal obstruction would show multiple dilated loops on X-ray with possible air-fluid levels. Only when the obstruction becomes severe and there is a perforation, pneumoperitoneum occurs.

The image given below is an abdominal radiograph showing multiple dilated loops with air-fluid levels suggestive of intestinal obstruction.



Radiology NEET 2020

Question 1:

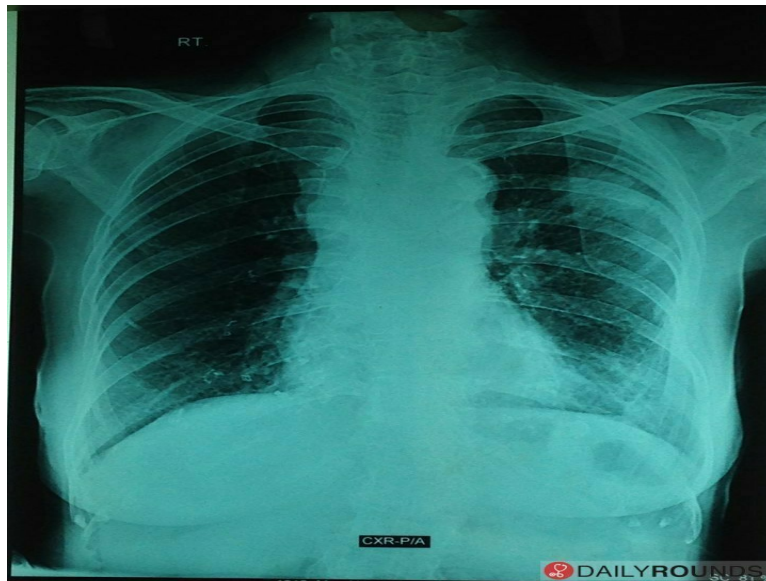
A 56-year-old female patient presented with progressive dysphagia for solids, loss of weight, and loss of appetite. A barium swallow done is given below. What is the most probable diagnosis?



- a) Carcinoma esophagus
- b) Diffuse esophageal spasm
- c) Esophageal leiomyoma
- d) Esophageal web

Question 2:

A 30-year-old woman presents with persistent hypertension, chest pain and cramping in her legs, particularly during exertion. A chest radiograph was taken, which reveals the condition, as shown in the image below. What is the likely underlying cause of her hypertension?



- a) Coarctation of aorta
- b) Essential hypertension
- c) Aortic aneurysm
- d) Aortaarteritis

Question 3:

A 51-year-old man complains of abdominal pain and blood-stained stools. The barium study of his intestines shows the following finding. What is the most likely diagnosis?

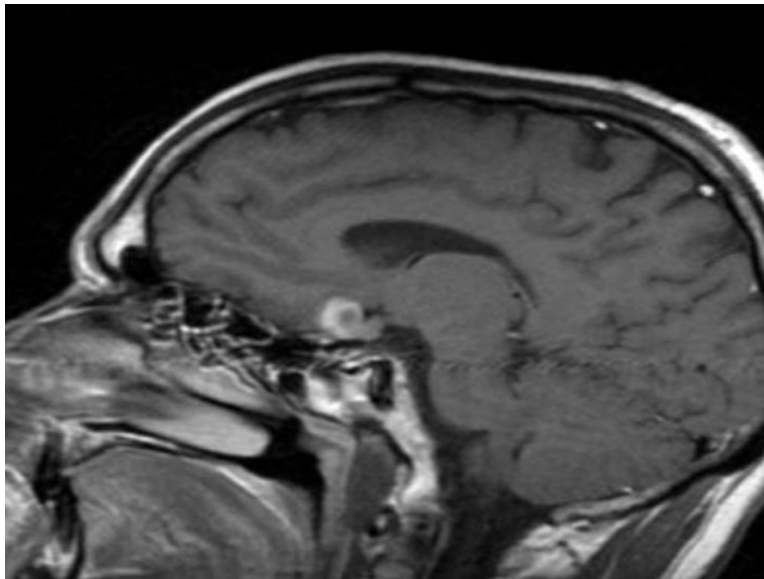


- a) Intussusception

- b) Volvulus
- c) Diverticulitis
- d) Ulcerative colitis

Question 4:

A cystic lesion in the suprasellar region with calcification is seen on the MRI. What is the most likely diagnosis?



- a) Craniopharyngioma
- b) Pituitary adenoma
- c) Meningioma
- d) Oligodendroglioma

Question 5:

Personal monitoring of radiation is done by:

- a) TLD badge
- b) Collimators
- c) Linear accelerator
- d) Grid

Question 6:

The appearance of USG at saphenofemoral junction, showing common femoral vein, common femoral artery and the great saphenous vein is called as:

- a) Mickey mouse sign
- b) String sign
- c) Tillaux sign
- d) Stemmer sign

Question 7:

Plethoric lung fields are seen in:

- a) Total Anomalous Pulmonary Venous Connection
- b) Tricuspid atresia
- c) Ebstein anomaly
- d) Tetralogy of Fallot

Question 8:

A patient complains of neck stiffness and severe headache. The CT image shown below is suggestive of which of the following conditions?



- a) Meningitis

- b) Subarachnoid hemorrhage
- c) Intraparenchymal hemorrhage
- d) Hemorrhagic stroke

Answer Key

Question No.	Correct Option
1	a
2	a
3	a
4	a
5	a
6	a
7	a
8	b

Detailed Explanations

Solution to Question 1:

A patient with the given history and barium swallow findings of irregular narrowing, mucosal destruction, and shouldered margins of esophagus, are suggestive of carcinoma esophagus.

Esophageal carcinoma is more common in men, with tobacco use and alcohol being the strongest risk factors. The squamous cell carcinoma variant usually affects the upper 2/3rd, and the adenocarcinoma variant usually affects the lower 1/3rd of the esophagus. Clinical features include progressive dysphagia (most common), regurgitation of food, vomiting, odynophagia and weight loss.

On barium swallow, carcinoma esophagus presents with irregular narrowing and ulceration of esophagus. Flexible endoscopy with biopsy is preferred for confirmation of diagnosis. Contrast-enhanced CT and PET can be used to assess distant metastasis. Endoscopic ultrasonography (EUS) is the investigation of choice to assess tumor invasion and lymph node metastasis.

Other options:

Option B: Diffuse esophageal spasm results in the corkscrew appearance of the esophagus on barium study.

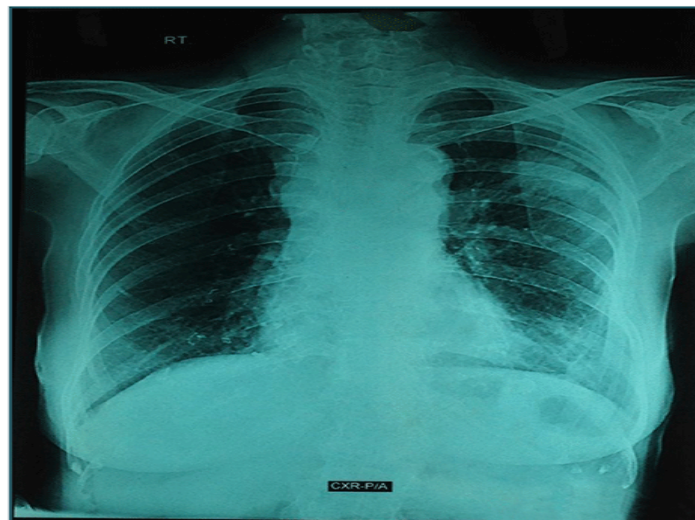
Option C: Esophageal leiomyoma appears as a smooth, semilunar filling defect on barium swallow. it is seen as a discrete ovoid mass that is well outlined by barium.

Option D: Esophageal webs occur in the upper and middle esophagus as a thin membrane that can be identified by the translucent part on contrast radiology.

Solution to Question 2:

This clinical scenario and the image showing the figure of 3 sign point towards coarctation of aorta as the cause of hypertension in this patient.

The figure of 3 sign refers to the indentation of the aorta at the coarctation site. There is pre and post-stenotic dilatation along with left paramediastinal shadow, as shown below.



Other chest X-ray features in coarctation of the aorta are dilated left subclavian artery, dilated ascending aorta and notching of ribs 4-8 due to rib erosion by dilated collateral vessels.

Solution to Question 3:

The given clinical scenario and the radiological images are suggestive of intussusception.

In intussusception, the occluding mass prolapsing into the lumen has been termed the coiled spring appearance (barium in the lumen of the intussusceptum and in the intraluminal space). In adults, it develops due to a tumor of the bowel.

The intussusception appears as a sausage-shaped mass in a longitudinal plane or a target mass in the transverse plane on CT and USG.

Ultrasound signs include:

- Target sign (also known as the doughnut sign)

- Pseudokidney sign- this is seen in the longitudinal section of the bowel
- the edematous bowel resembles renal parenchyma and
- the mesentery which is dragged into the intussusception resembles renal hilum
- Crescent in a doughnut sign.

Note: The colographic appearance of intussusception is also described as a claw sign, pincer defect, shouldering effect, and coiled-spring pattern.

Option B: Volvulus presents with constipation, abdominal bloating, nausea, and/or vomiting. It shows coffee bean sign

Option C: Diverticulitis presents with pain in the left iliac fossa

Option D: Ulcerative colitis causes backwash ileitis

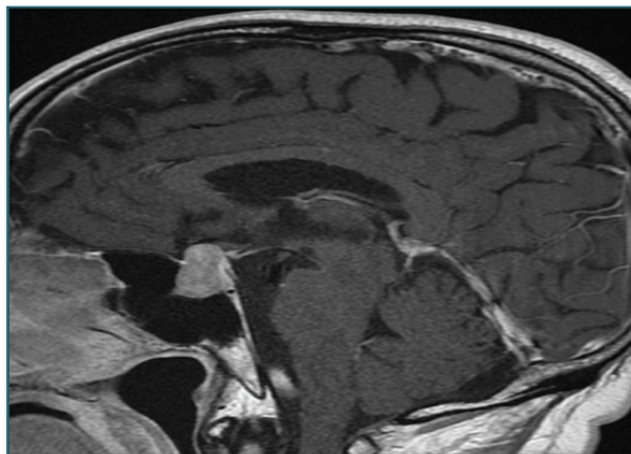
Solution to Question 4:

A cystic lesion in the suprasellar region with calcification is suggestive of craniopharyngioma. It is derived from Rathke's pouch and arises near the pituitary stalk, commonly extending into the suprasellar cistern.

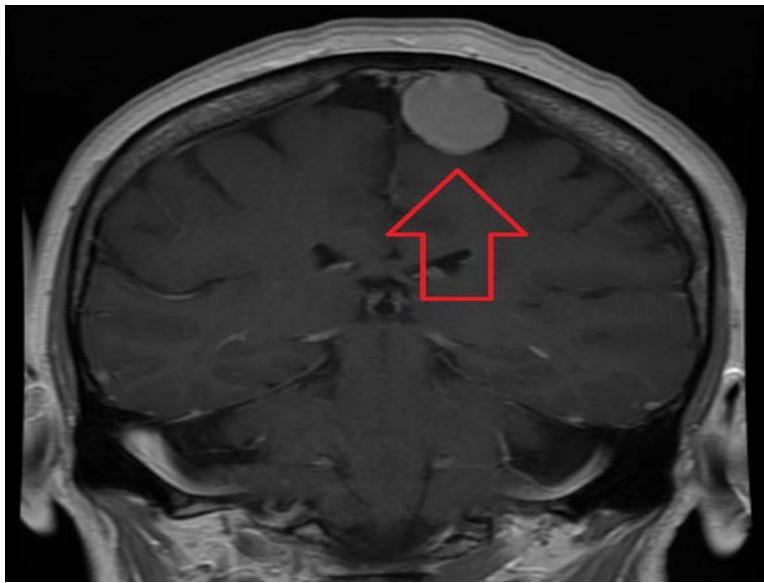
Other options:

Option B: Pituitary adenomas are seen in the sellar region. Microadenomas are best shown on contrast-enhanced images. Macroadenomas are seen as snowman or figure of eight appearance.

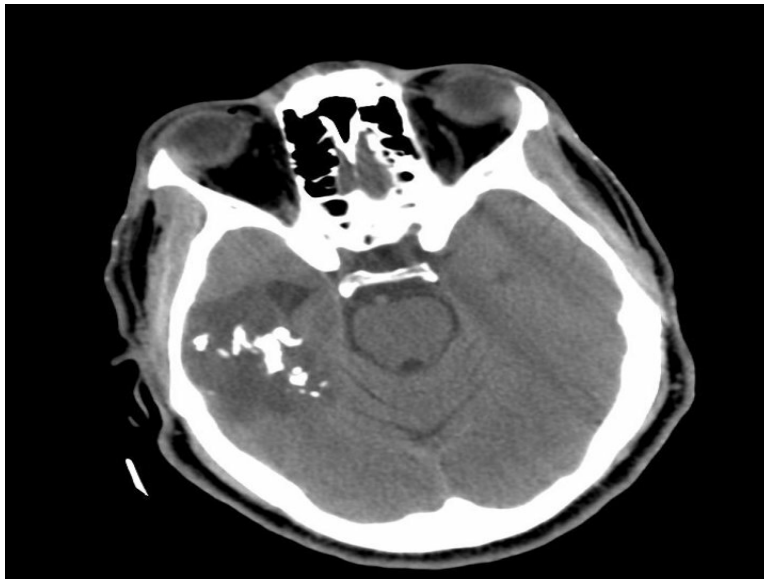
Snowman appearance – Pituitary adenoma



Option C: Meningioma is a supratentorial tumor, mainly in the parasagittal region. A classical dural tail may be seen in intracranial meningiomas but is not pathognomic. The MRI image below shows a well-defined, hyper-intense, and homogeneously enhanced mass close to the midline suggestive of meningioma.



Option D: Oligodendroglioma is exclusively found in the cerebral hemispheres, most commonly in the frontal lobes. Upto 90% of oligodendrogliomas contain visible calcification on CT. The CT image below shows a calcified cortical mass that suggests oligodendroglioma.



Solution to Question 5:

Personal monitoring of radiation is done by a TLD badge.

The TLD badge (thermoluminescent dosimeter) should be positioned according to local or national practices. In India, one TLD badge has to be worn at the chest level inside the lead apron if the apron is used. An additional wrist badge has to be worn for procedures requiring hands close to the primary beam.

The thermoluminescent material used is calcium sulfate doped with dysprosium or lithium fluoride. Radiation exposure is monitored every three months. The maximum dose limit for occupational radiation workers is 30 mSv in a year, provided the cumulative dose in a block of 5 years does not exceed 100 mSv.

The given image shows a TLD badge.



Solution to Question 6:

The normal anatomy of the femoral vessels and great saphenous vein at the saphenofemoral junction on USG is called the mickey mouse sign.

The given images below show the mickey mouse sign (FV- femoral vein; FA- femoral artery; GSV- great saphenous vein).



Other options:

Option B: String sign (of Kantor) on barium studies appears as a thin, thread-like stream of contrast passing through the stenosed bowel segment, particularly in Crohn's disease.

Option C: Tillaux sign is said to be positive if the mass/swelling is mobile perpendicular to the mesentery attachment line. Tillaux's sign is seen in a mesenteric cyst.

Option D: Stemmer's sign is the inability to tent the skin at the base of the toes due to skin thickening, as seen in lymphedema.

Solution to Question 7:

Plethoric lung fields are seen in Total Anomalous Pulmonary Venous Connection (TAPVC). In this condition, pulmonary venous obstruction results in pulmonary arterial hypertension, resulting in plethoric lung fields.

	Increasedpulmonary blood flow (pulmonary plethora)	Decreasedpulmonary blood flow
Acyano tic	Ventricular septal defect, Atrial septal defect, Patent ductus arteriosus	—
Cyanotic	TAPVC	Ebstein anomaly, Tetralogy of Fallot, Tricuspid atresia

Solution to Question 8:

The clinical manifestations of neck stiffness and severe headache with CT revealing blood in the basal cistern points out to the diagnosis of subarachnoid hemorrhage (SAH).

Subarachnoid hemorrhage (SAH) is the bleeding into the subarachnoid space, which is the area between the arachnoid membrane and the pia mater surrounding the brain. The diagnosis is suspected when a hyperattenuating material is seen filling the subarachnoid space. Most commonly this is apparent around the circle of Willis, on account of the majority of berry aneurysms occurring in this region (65%), or in the Sylvian fissure (30%).

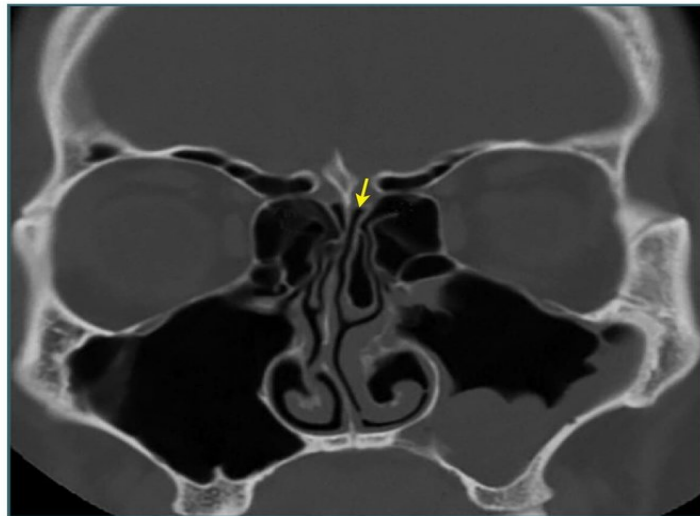
The modified Fisher scale is a method for grading SAH as seen on non-contrast CT (NCCT). The Hunt and Hess scale describes the severity of SAH and is used as a predictor of survival.

Xanthochromia on CSF is indicative of SAH in case of a negative CT scan.

Radiology INI-CET 2021

Question 1:

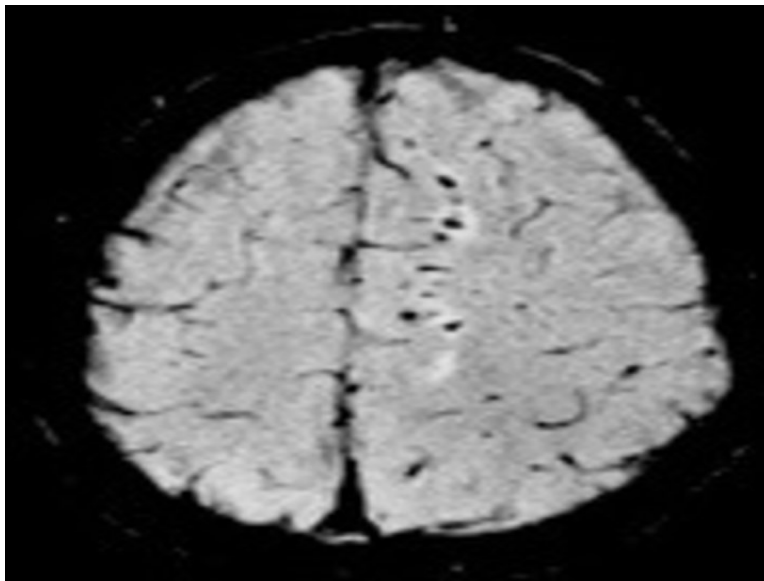
What is the structure marked in the CT scan ?



- a) Anterior ethmoidal canal
- b) Posterior ethmoidal canal
- c) Optic nerve
- d) Cribriform plate

Question 2:

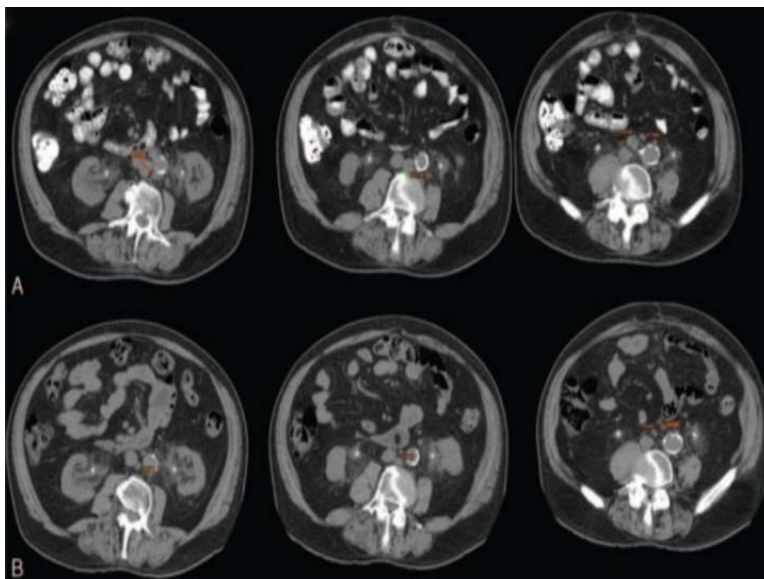
What type of MRI image is given below?



- a) Proton density image
- b) Susceptibility weighted image
- c) Gradient echo image
- d) Inversion recovery image

Question 3:

During rounds, your senior was discussing the given image. The following is an image of which of the following investigations?



- a) Contrast Dye study

- b) CT scan
- c) X-ray
- d) Angiography

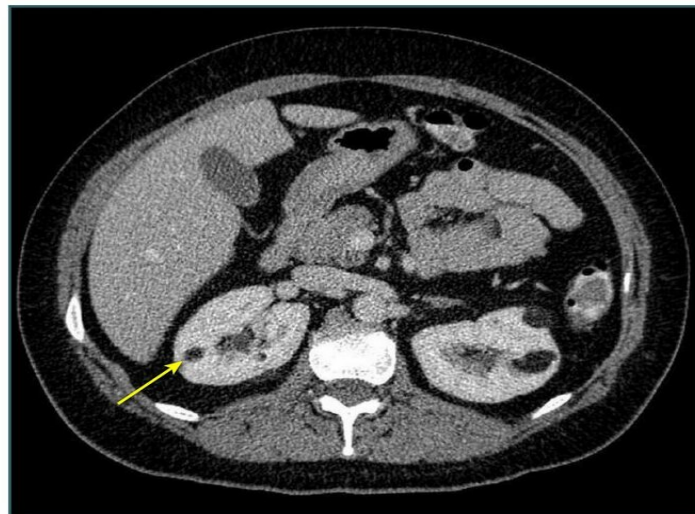
Question 4:

A 22-year-old male was brought to the emergency room following a road traffic accident. On examination, his GCS was E2M2V3. He had no fractures and no hydrocephalus or raised intracranial pressure. Non-contrast CT was performed and was found to be normal. The likely diagnosis is

- a) Diffuse axonal injury
- b) Post concussion syndrome
- c) Extra dural hemorrhage
- d) Herniation of brain

Question 5:

A 45-year-old patient presented with vague abdominal pain. On USG, he was found to have a renal cyst of Bosniak class III. CECT was done, as shown below. What is the diagnosis?



- a) RCC
- b) Renal oncocytoma
- c) Renal angiomyolipoma
- d) Renal cyst

Question 6:

A young child presented with mild intermittent upper abdominal pain. X-ray is given below. What is the diagnosis?



- a) Bochdalek hernia
- b) Morgagni hernia
- c) Gastric volvulus
- d) Eventration of diaphragm

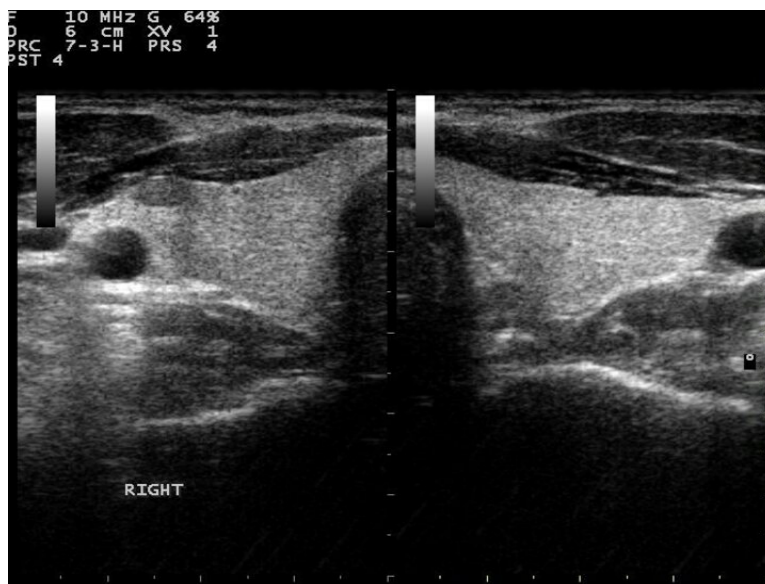
Question 7:

A woman presented with confusion, constipation, and headache. Serum calcium was 13.1mg/dL and ALP was elevated. She was diagnosed with primary hyperparathyroidism. Which of the following is the imaging of choice?

- a) MRI
- b) FDG PET
- c) CT
- d) Sestamibi scan

Question 8:

Identify the imaging modality given below.



- a) USG
- b) Fluroscopy
- c) X-Ray
- d) MRI

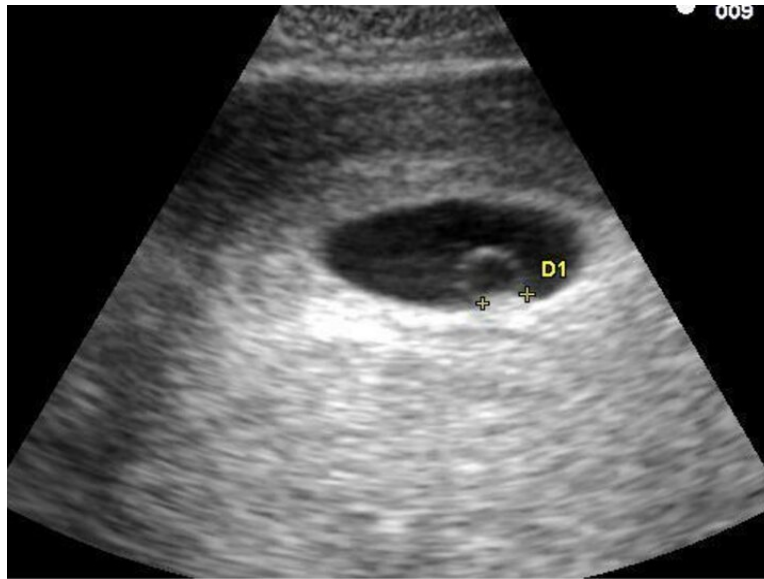
Question 9:

Which of the following is true regarding CORADS?

- a) CORADS- 0 indicates normal lung
- b) CORADS-5 is typical of COVID-19
- c) Pleural thickening is typical of COVID-19
- d) Lobar consolidation is typical of COVID-19

Question 10:

Double bleb sign seen in early pregnancy is due to _____.



- a) Twin pregnancy
- b) Amnion and chorion
- c) Yolk sac and amniotic sac
- d) Heterotopic pregnancy

Question 11:

Which one of the following is the most radioresistant phase of the cell cycle?

- a) S phase
- b) M phase
- c) G₁ phase
- d) G₂ phase

Answer Key

Question No.	Correct Option
1	d
2	b
3	b
4	a
5	c

6	b
7	d
8	a
9	b
10	c
11	a

Detailed Explanations

Solution to Question 1:

The structure marked in the image is the cribriform plate of the ethmoid bone. The image shown is a coronal section of paranasal sinuses at the level of anterior ethmoidal cells.

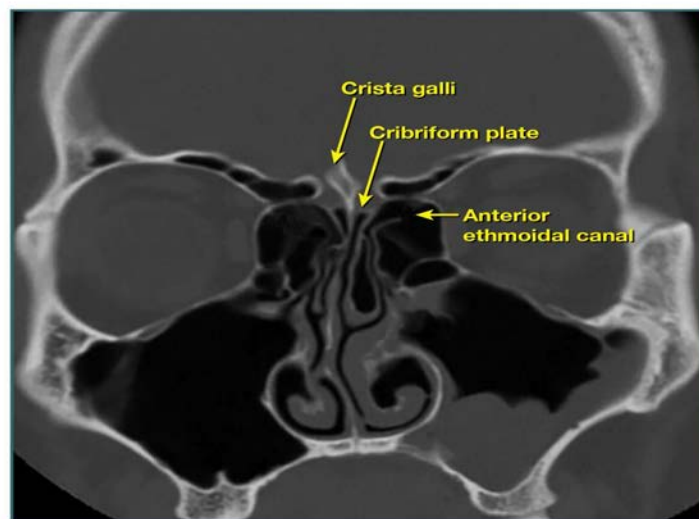
The cribriform plate is identified as a bony structure that separates the nasal cavity from the anterior cranial fossa. It is part of the ethmoid bone and is a sieve-like structure as it is pierced by the olfactory nerve fibers.

Crista Galli is also a part of ethmoid bone that projects superiorly from the cribriform plate. It provides attachment for the falx cerebri.

The anterior ethmoidal canal is identified by its location below the roof of ethmoidal air cells. The anterior ethmoidal artery passes through this canal and it is important to identify this to avoid bleeding from this vessel during sinus surgeries.

The posterior ethmoid canal/foramen is an opening in the ethmoid bone through which the posterior ethmoidal artery, vein, and nerve traverse. It is not visualized in this image as it is a coronal section at the level of anterior ethmoidal cells.

The optic nerve can be identified as an oval intraorbital structure in the coronal sections.



Solution to Question 2:

The image shown is a susceptibility-weighted image(SWI) as it clearly demonstrates microhemorrhages as hypointense foci.

SWI is a form of MRI sequence that is especially sensitive to substances that distort the local magnetic field. Compounds with diamagnetic, paramagnetic, and ferromagnetic properties will interact with the local magnetic field and produce distraction. This would lead to a loss of signal in these areas.

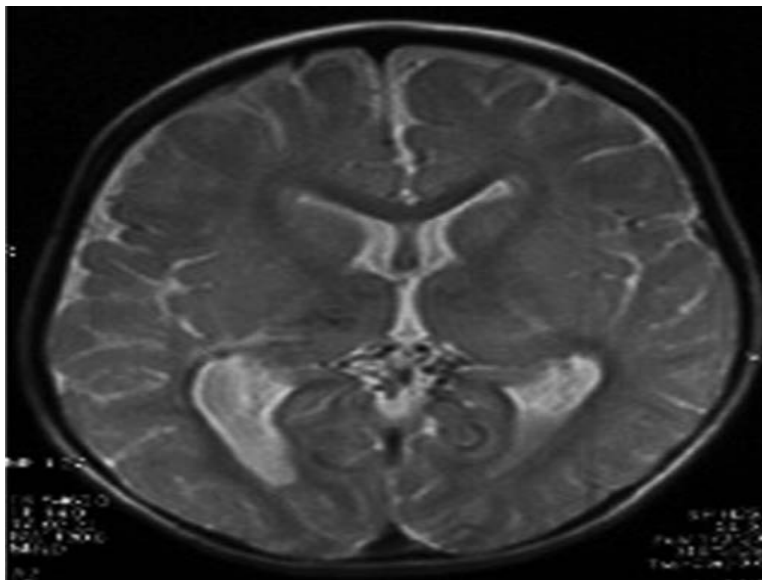
Deoxyhemoglobin, hemosiderin, and ferritin are paramagnetic. Bone minerals and dystrophic calcifications are diamagnetic in nature.

The main clinical use is the identification of microhemorrhages/blood products or calcium that are not detected in other MRI sequences.

Other options:

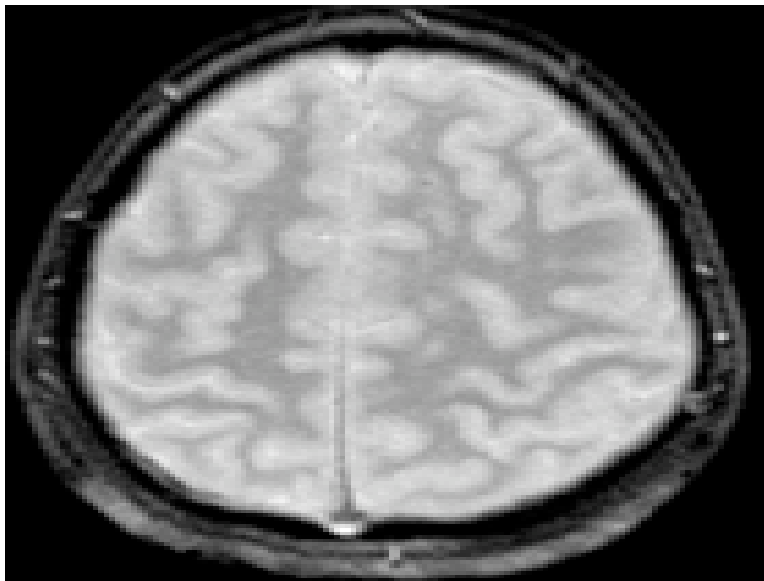
Proton density image:

It is based on the number of nuclei in the area that is imaged. In PD-weighted image tissues with a higher density of protons produce stronger signals and appear brighter. PD weighted image can be identified by comparing fluid with the fat signal. Fluids in these images appear greyish white similar or the appearance of fat in the body.



Gradient echo sequences(GRE):

It uses gradient fields to generate transverse magnetization. In areas of hemorrhage, the iron that is present in the hemoglobin produces its own local magnetic field and dephases the spinning nuclei. This is useful to identify hemorrhagic contusions in the brain and pigmented villonodular synovitis.



Inversion recovery sequences:

This MRI sequence is used to selectively null certain tissues. It can be:

- Short tau inversion recovery(STIR) where fat is nulled. Fluids appear bright and fat appears dark in these images.
- Fluid attenuated inversion recovery(FLAIR) or double inversion recovery(DIR) where fluid is nulled.- The fluid(CSF) normally appears dark and lesions appear to be brighter on these images.

FLAIR image-



Solution to Question 3:

The image in the question is a CT scan as it shows the bone as a high-intensity structure (white appearance).

In CT study X-rays are applied in a circular motion and body tissue slices are recreated mathematically. Different tissues are seen with different intensities depending on the proportion of attenuation of the X-rays that pass through them.

- Air and water are displayed as low intensities (dark appearance) because they produce only little attenuation.
- Bone is displayed as a high-intensity structure (bright appearance) because of its significant attenuation.
- High density is seen with hyperemia, freshly clotted blood, and when contrast is used.
- Edema and necrosis are seen as low-intensity lesions.

Other options:

Contrast dye study: In this imaging study, a special dye or contrast is given to evaluate structures that are not identified by conventional imaging. The contrast added would be visible in the image depending on the type of contrast and tissues targeted. Intravenous pyelogram, barium studies, angiography, etc are all examples of contrast studies.

X-ray: It shows different parts of the body in varying shades of black and white. The difference in the appearance of various tissues is due to the differences in the amount of radiation absorbed by them.

- Bone appears white because calcium has high x-ray absorption.
- Fat and soft tissues appear grey due to less x-ray absorption.
- The least absorption is seen with air which appears black.

Angiography: Here a special dye or contrast is injected to highlight the blood vessels as they are not clearly identified in normal imaging. Hence vessels would be highlighted in an angiogram.

Note: A contrast study is different from a contrast-enhanced CT scan, Intravenous pyelogram, barium studies, angiography, etc are all examples of contrast studies.

Solution to Question 4:

The given clinical scenario of an RTA with no imaging abnormality along with a disproportionate clinical manifestation suggests diffuse axonal injury.

It is caused due to stretching of axons in subcortical white matter due to sudden deceleration injuries. It is usually associated with mild nonspecific changes in NCCT. It is visualized on T2 MRI characterized by multiple lesions showing microbleeds or linear hypointensities. It may also show disruption of white matter tracts on tractography.

Other options

Post-concussion syndrome usually follows mild traumatic brain injury. It does not have a severe clinical status as seen in this patient.

Extradural hemorrhage is produced usually results secondary to trauma to head. The bleeding is usually arterial and blood collects between outer dura and inner table of the skull. It can very well be visualized on CT.

A cerebral herniation is defined as a shift of cerebral tissue in an adjacent space. Intracranial brain herniation can be classified into 3 types - subfalcine, transtentorial and tonsillar. A distinct clinical picture can be produced as a result of compression of neurovascular structures by the herniating part. CT is the imaging modality of choice because it can be done rapidly.

Solution to Question 5:

The CT image showing renal lesions that has hypoattenuating components consistent with macroscopic fat is highly suggestive of renal angiomyolipoma.

Imaging in renal angiomyolipoma: Demonstration of macroscopic fat is the main part of making a diagnosis. USG may show hyperechoic lesions in the cortex with posterior acoustic shadowing. In patients with tuberous sclerosis, the entire kidney is affected, producing an echogenic appearance with loss of normal corticomedullary junction.

CT findings:

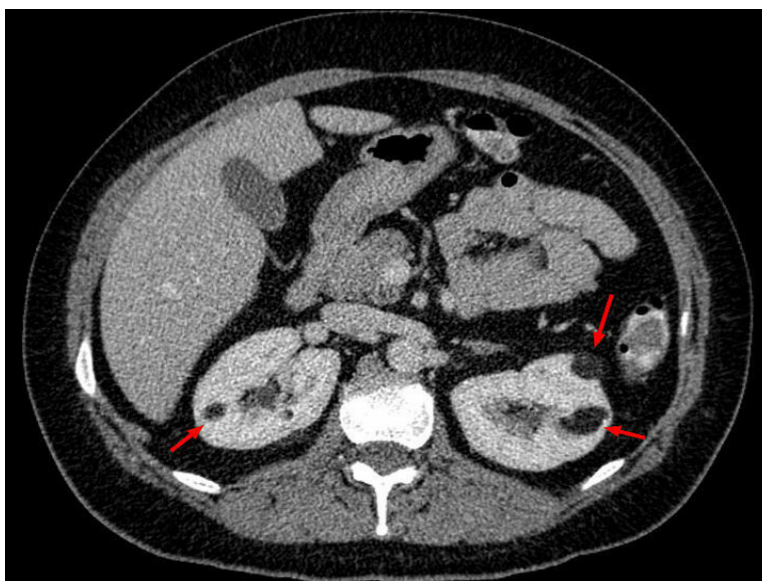
- Cortical lesions that demonstrate macroscopic fat (less than -20 HU)
- Absence of ossification or calcification
- The presence of macroscopic fat is strongly suggestive of this condition, but not pathognomonic as RCC may rarely demonstrate macroscopic fat.
- Approximately 5% may be fat poor.

MRI: It is an excellent tool to demonstrate intramural fat components. Two sequences are used:

- Fat-saturated techniques
- Phase-in and phase-out imaging that shows India ink artifact at the junction between fat and non-fat components.

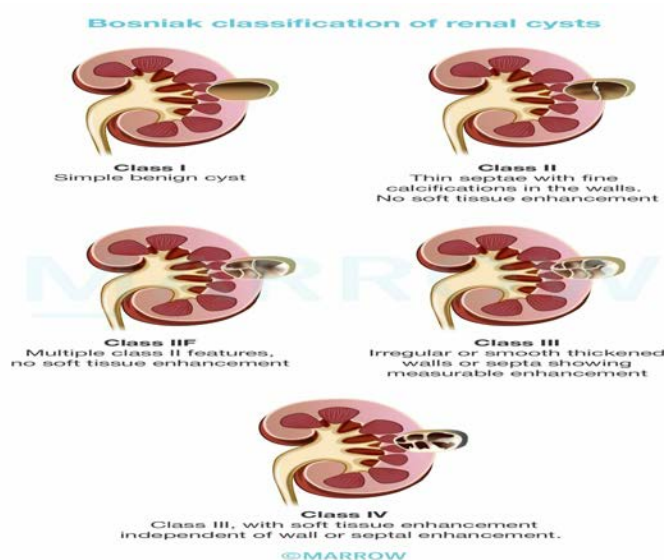
Digital subtraction angiography: They are hypervascular lesions showing 'sunburst appearance' in the arterial phase and 'onion peel appearance' in the venous phase.

Angiomyolipomas are benign tumors that occur sporadically or in association with tuberous sclerosis. The mean age of presentation is 43 years and there is a female predilection. It can present as spontaneous retroperitoneal hemorrhage, especially with larger tumors. The patient can present with flank pain, palpable mass, hematuria, urinary tract infections, hypertension, or renal failure. They usually do not require any intervention. mTOR inhibitors like everolimus can be used to decrease the size of the lesion.



Bosniak classification:

It classifies the renal cystic lesions mainly into four classes based on the CT characteristics.



Class I and II are benign and do not require follow-up imaging or surgery.

Class IIF requires follow-up for about 5 years to exclude malignancy.

Class III needs to be biopsied or surgically explored due to a higher risk of malignancy.

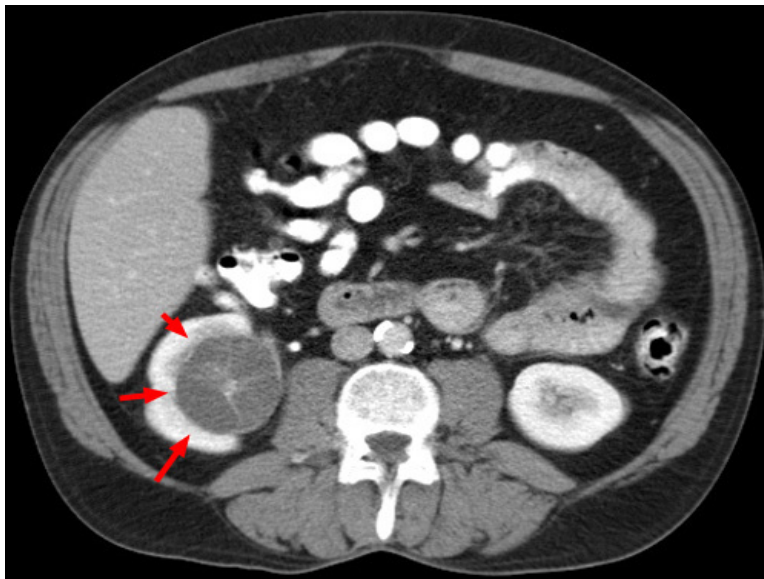
Class IV lesions are clearly malignant and need appropriate treatment.

Other options:

Renal cell carcinoma (RCC): It is the most common malignant tumor of the kidney. It usually occurs in the age group 50-70 years and macroscopic hematuria is seen in about 60% of the cases. It is a differential for angiomyolipoma but it is important to note that macroscopic fat is almost always associated with ossification or calcification.

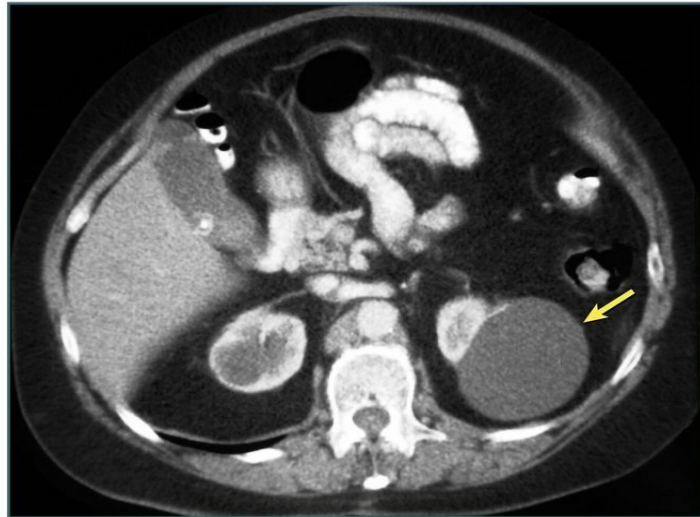


Renal oncocytoma: They are relatively benign in nature and usually present in the 6th to 7th decade. An important point regarding this tumor is its similarity with RCC and hence it is very difficult to differentiate between them preoperatively. A sharp central stellate scar may be seen in one-third of the cases. But RCC may also show this stellate scar appearance in some cases.



Renal cyst: It is a broader term that includes the cystic lesions of the kidney. They can be simple cysts that are benign. Complex or complicated cystic lesions need evaluation and can be malignant. Bosniak classification of renal cysts is given above.

Image showing a simple renal cyst:



Solution to Question 6:

The clinical features and the chest X-ray showing an opacity in the right cardiophrenic angle are suggestive of Morgagni hernia.

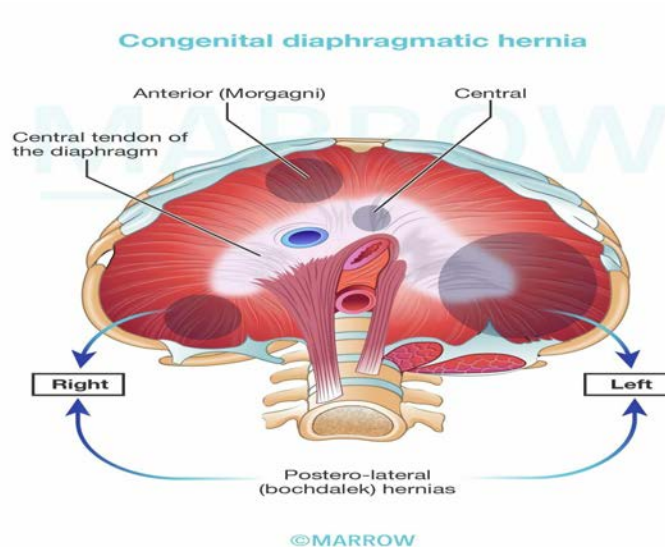
Congenital diaphragmatic hernia (CDH) of anteromedial location is the Morgagni hernia. Here the defect is anterior and retrosternal. It is rarer than bochdalek hernia and accounts for only 2-5% of CDH. About 90% of the cases occur on the right side.

It is less symptomatic than bochdalek hernia and the diagnosis is usually delayed until childhood. The patient presents with respiratory or upper GI symptoms. Respiratory symptoms like distress, tachypnea, and recurrent pulmonary infections are common in children. Other symptoms are abdominal pain, dyspnea, flatulence, and indigestion.

Radiographic findings:

- Radiolucent paracardiac shadow may be seen retrosternally on lateral chest x-ray if there is bowel herniation.
- Rounded opacity in the right cardiophrenic angle may be seen if there is herniation of omentum or solid organs.
- A CT scan may be used to confirm the diagnosis.

Surgical repair is indicated in all cases either by a thoracic or abdominal approach.



Other options:

Bochdalek hernia: It is more common than Morgagni hernia. Here the defect is posterolateral in location. Respiratory distress is experienced by most infants. In the pediatric age group, symptoms are predominantly respiratory due to pulmonary hypoplasia. In adults, it is usually asymptomatic and is incidentally found. About 85% of the cases are left-sided.

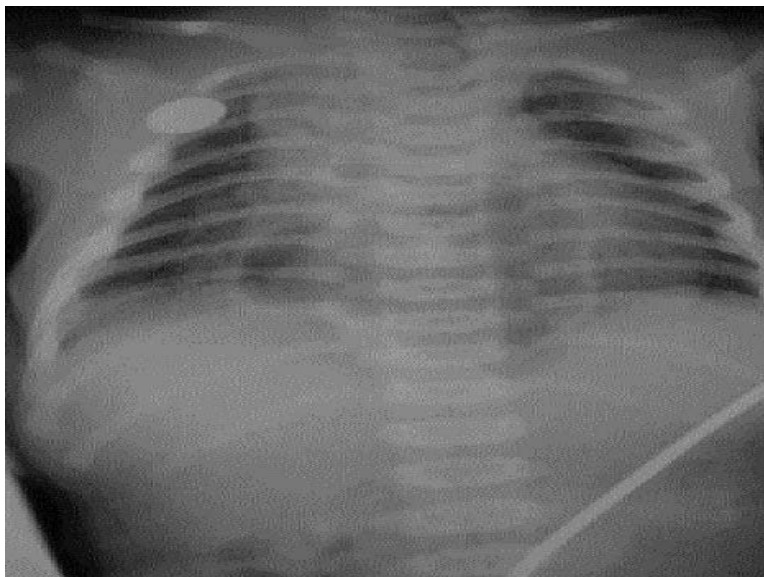


Gastric volvulus: This is an uncommon condition and presents with the characteristic Borchardt triad which includes:

- Sudden-onset severe upper abdominal pain
- Recurrent retching with the production of little vomitus
- Inability to pass a nasogastric (NG) tube

It can be organoaxial or mesenteroaxial. Chest X-ray may show intrathoracic upside-down stomach and retrosternal or mediastinal air-fluid level.

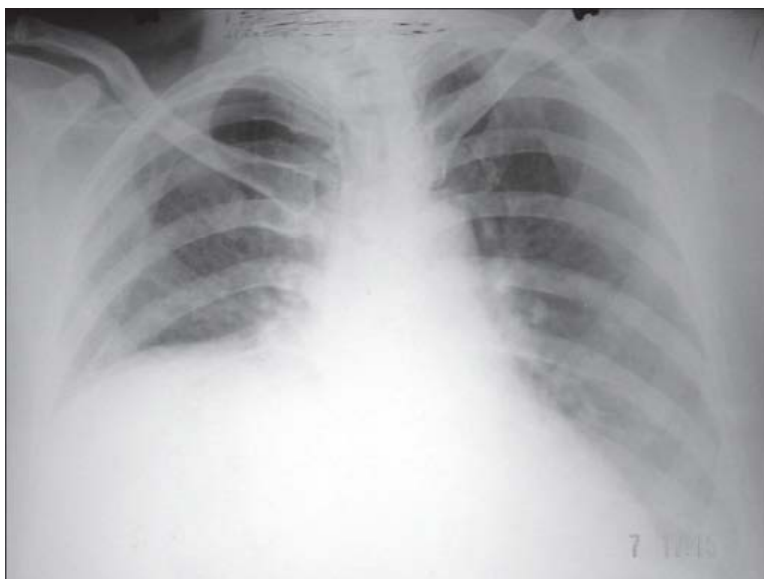
Image showing intrathoracic gastric volvulus with mediastinal gastric bubble-



Eventration of the diaphragm: It refers to the abnormal contour of the dome of the diaphragm without any disruption to the diaphragmatic continuity. It usually involves a segment of the hemidiaphragm. It can cause respiratory distress in newborns and is usually asymptomatic in adults.

Radiographic findings: The affected portion of the hemidiaphragm is elevated and is seen as a smooth hump, while the normal contour is seen in the remaining part. 'Double diaphragmatic contour' may be seen in frontal view and is confirmed with a lateral X-ray.

Image showing eventration of the right dome of the diaphragm-



Solution to Question 7:

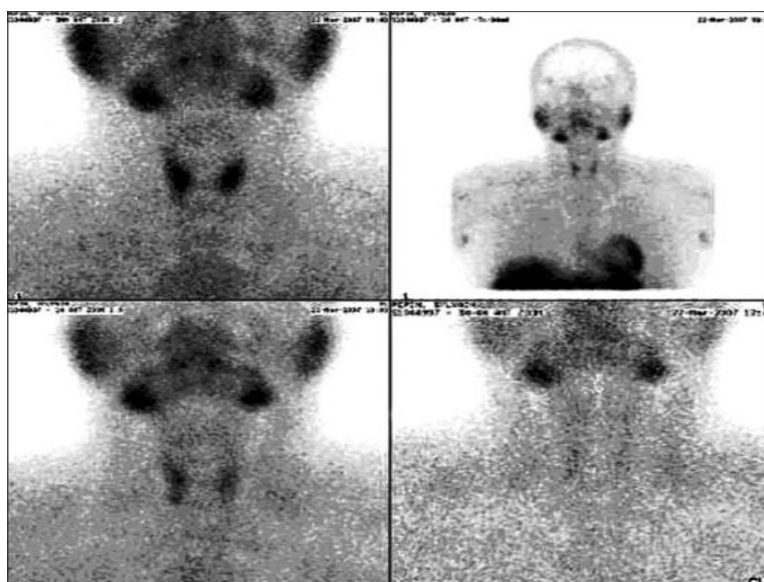
Sestamibi scan is the imaging of choice to delineate parathyroid abnormalities.

Sestamibi is a monovalent lipophilic cation. It shows passive diffusion across cell membranes and accumulates in mitochondria. Because of its accumulation in the mitochondria, there is a difference in the rate at which it gets washed out depending upon the number of mitochondria in each tissue.

In parathyroid adenomas, there is a high concentration of oxyphilic cells with high mitochondrial content. The three protocols for sestamibi scanning are:

- Single-isotope dual-phase scan
- Dual-isotope subtraction imaging
- Single-photon emission computed tomography (SPECT).

SPECT is commonly used because of its improved spatial resolution. Sestamibi combined with CT can be used to detect smaller parathyroid tissues.



Other imaging modalities for parathyroid localization:

- Ultrasonography:
 - It is inexpensive and noninvasive
 - Mediastinal parathyroid tissue cannot be detected because USG cannot penetrate the sternum or clavicles.
- Four-dimensional CT:
 - It has a very high sensitivity
 - Increased radiation exposure than other routine modalities is a potential disadvantage
 - Selective venous sampling for PTH can be used for the exact localization of adenomas

Solution to Question 8:

The image shown in the question is the USG of the neck.

Sonographic examination of the neck shows the thyroid gland, muscles, blood vessels, bony structures or cartilages, and nerves. The thyroid of a healthy patient appears homogeneously echogenic with a uniform echotexture on USG. Lateral to the thyroid, the large blood vessels (the common carotid artery and more laterally, the internal jugular vein) can be identified. In the above image, hypoechoic areas are suggestive of the lumen of vessels seen in the neck i.e common carotid artery and internal jugular vein.

Neck ultrasonography is the preferred modality for routine radiographic evaluation of the thyroid gland, as it offers the most accurate visualization of thyroid nodular disease. It also allows for the concomitant performance of percutaneous fine-needle aspiration biopsy of thyroid lesions.

Neck ultrasound can also be used to localize parathyroid disease.

Other options:

Option B: Fluoroscopy is an imaging modality that allows real-time X-ray viewing of a patient with high resolution. During this procedure, radiologists usually use contrast material (dye) to highlight the area being examined.

Below image shows the fluoroscopy of neck.



Option C: The below image shows the X-ray appearance of neck in AP view.



Option D: Relative to the neck musculature, the thyroid is hyperintense on MRI images.

Solution to Question 9:

Among the given options, CORADS-5 is typical of COVID-19 is the true statement.

COVID-19 Reporting and Data System (CO-RADS) is a categorical assessment for pulmonary involvement of COVID-19 on chest CT.

CO-RADS Category	Summary
CO-RADS 0	Scans are incomplete or of inferior quality
CO-RADS 1	Normal or non-infectious
CO-RADS 2	Typical features of other infections but not of COVID-19
CO-RADS 3	Features compatible with COVID-19 but also with other diseases
CO-RADS 4	Suspicious for COVID-19
CO-RADS 5	Typical for COVID-19
CO-RADS 6	Proven COVID-19 by positive RT-PCR test

CO-RADS categories:

- CO-RADS 0: Not interpretable - Technically insufficient to assign a score: Scans are incomplete or of inferior quality.

- CO-RADS 1: Very low level of suspicion for pulmonary involvement by COVID-19. Mild or severe emphysema, perifissural nodules, lung tumors, or fibrosis.
- CO-RADS 2: Low level of suspicion for pulmonary involvement by COVID-19. Includes infectious etiology that is not compatible with COVID-19 such as bronchitis, bronchiolitis, bronchopneumonia, lobar pneumonia, abscess.
- CO-RADS 3: Equivocal findings for pulmonary involvement by COVID-19. Findings overlap with other viral or infectious etiologies. Could include perihilar ground-glass, homogenous extensive ground-glass with or without sparing of secondary pulmonary lobules.
- CO-RADS 4: High level of suspicion for pulmonary involvement by COVID-19. Findings are similar to CO-RADS 5 but are not present in the typical locations. Findings may also be superimposed on severe pre-existing pulmonary abnormalities.
- CO-RADS 5 based on typical CT findings. It requires the presence of at least one confirmatory pattern (mentioned below).
- CO-RADS 6: Proven COVID-19 by positive RT-PCR test.

CT features typical of pulmonary involvement in COVID-19 include:

- Obligatory features
- Ground glass opacities, with or without consolidation in areas close to the visceral pleural surface, including the fissures.
- Confirmatory features
- Ground glass regions
- Crazy paving pattern
- Organizing pneumonia compatible patterns
- Thickened vessels with parenchymal abnormalities

Solution to Question 10:

The double bleb sign is due to the yolk sac and amniotic sac.

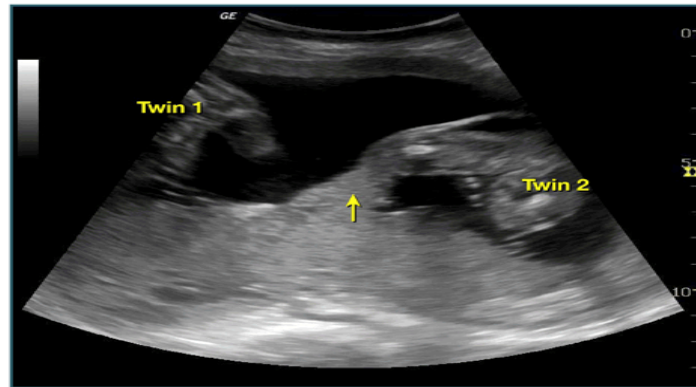
On ultrasonographic examination, a gestational sac may contain the yolk sac and amniotic sac, appearing like two small bubbles. This is known as the double bleb sign. The embryonic disc is located between the yolk sac and the amniotic sac. It is an important feature of an intrauterine pregnancy.

Other options:

In a twin pregnancy, the chorionicity of the placenta is best diagnosed with the help of an ultrasound examination.

- In dichorionic gestation, the twins are separated by amniotic membranes and an intervening chorion. So, a thicker twin peak sign or lambda (λ) sign is formed at the point at which the amnions meet the placenta.

Twin peak/ Lambda sign seen in dichorionic pregnancy



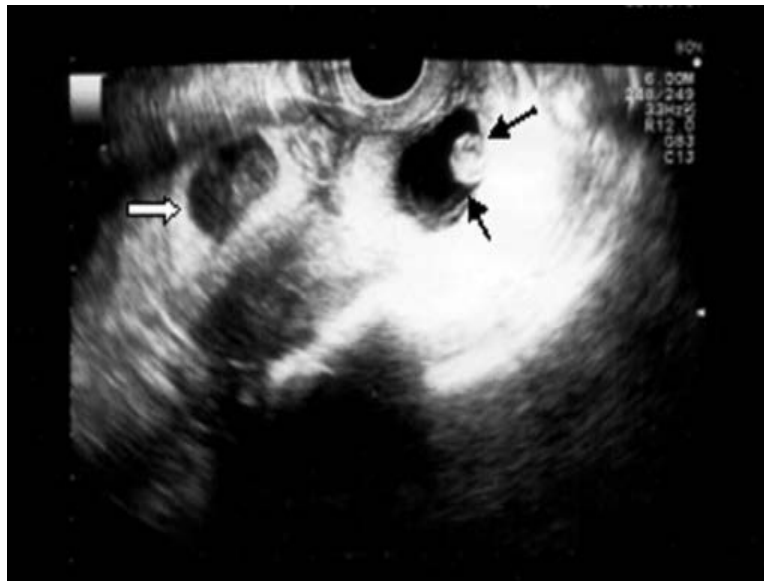
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- In monochorionic diamniotic gestation, the twins are separated only by the two amniotic membranes with no intervening chorion. So, a thin T sign is formed at the point at which the amnions meet the placenta.



Heterotopic pregnancy is a condition in which a normal uterine pregnancy co-exists with an extrauterine (ectopic) pregnancy. Although heterotopic pregnancy can occur spontaneously, it is more likely after artificial reproductive technology (ART) and ovulation induction.

The below image shows heterotopic pregnancy with 2 sacs, one inside the uterus (white arrow) and the other in the cervix (black arrows).



Solution to Question 11:

S phase is the most radioresistant phase of the cell cycle.

The replication of the DNA genome occurs only at a specified time period during the cell cycle. This period is referred to as the synthetic or S phase. The S phase is separated from the mitotic, or M phase, by nonsynthetic periods referred to as gap 1 (G₁) and gap 2 (G₂) phases, occurring before and after the S phase, respectively. The cell prepares for DNA synthesis in G₁, and for mitosis in G₂.

The most radiosensitive phase of the cell cycle is the G₂-M phase. Cells in the G₂ or M phase do not have time to repair DNA damage that occurs before they start to divide. This affects mitosis, leading to catastrophe.

Cells become gradually less sensitive to radiation in the G₁ phase to the early S phase and are least sensitive in the late S phase.

Radiology NEET 2021

Question 1:

A 15 year old patient with history of spinal tuberculosis was brought to the hospital with low grade fever, dull backache and persistent flexion of hip joint. Hip joint motion was limited and painful. Identify the muscle affected in the CT image.



- a) 1
- b) 2
- c) 3
- d) 4

Question 2:

A 12 week old child presented with fever, cyanosis and fatigue. His chest X ray is shown below. What is the most likely diagnosis?



- a) Total anomalous pulmonary venous circulation
- b) Tetralogy of Fallot
- c) Transposition of great arteries
- d) Pericardial effusion

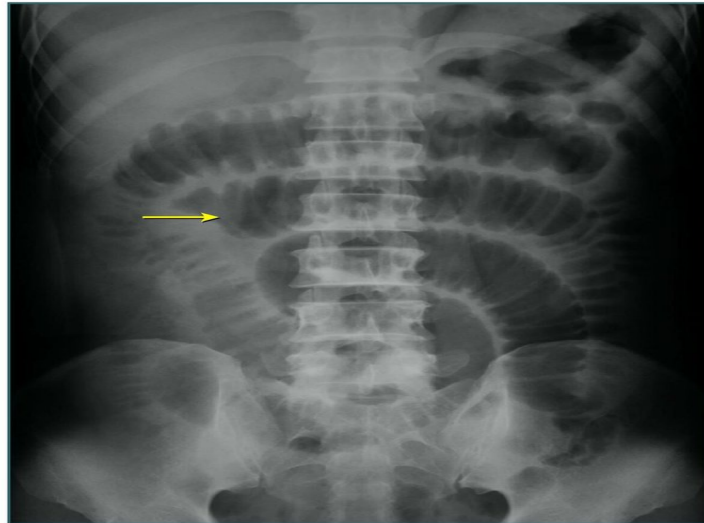
Question 3:

A patient following a skid resulting in a motor traffic accident was brought to the emergency room 2 hours later. On examination, he was stable with GCS 15/15. The pupil was reactive to light. Tenderness and bruising over the left lower chest wall with petechiae was seen. Severe tenderness was elicited in the left hypochondriac region and BP- 90/50 mm of Hg. What is the best investigation used in the ER?

- a) FAST
- b) X-ray
- c) Diagnostic peritoneal lavage
- d) CT scan

Question 4:

A patient presents with abdominal pain and distension. An X ray was taken which showed dilated bowel loops as given in the image below. Identify the segment involved.



- a) Jejunum
- b) Ileum
- c) Duodenum
- d) Transverse colon

Question 5:

A child with acute lymphoblastic leukemia undergoes prophylactic irradiation prior to autologous haematopoietic stem cell transplantation. Which of the following will be the least affected?

- a) Spermatogonia
- b) Intestinal epithelial cells
- c) Neurons
- d) Bone marrow/ erythroid precursor cells

Answer Key

Question No.	Correct Option
1	b
2	a
3	a

4	a
5	c

Detailed Explanations

Solution to Question 1:

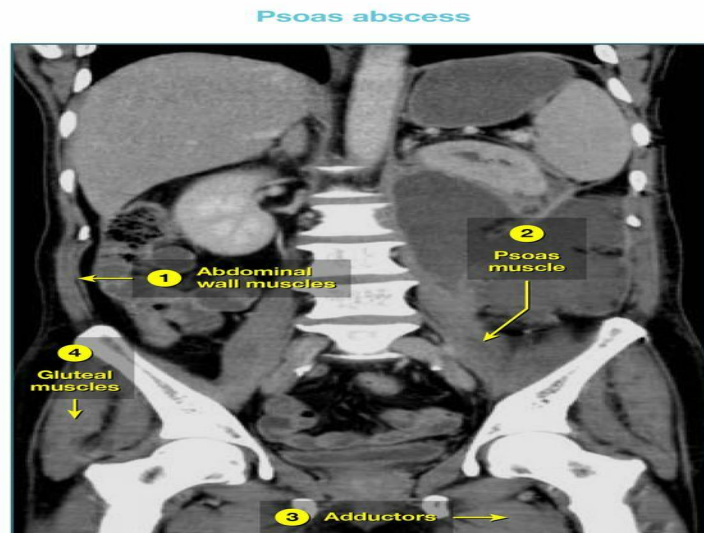
The clinical vignette of a patient with low-grade fever, dull backache, and persistent flexion of the hip joint with a history of spinal tuberculosis is suggestive of psoas abscess involving the psoas muscle, marked 2 in the given image. CT scan shows a mass in the psoas muscle lateral to the vertebral column which is diagnostic of the same.

Psoas abscess occurs due to a collection of pus in the iliopsoas compartment. The compartment is retroperitoneal and hence, the abscess is also called a retroperitoneal abscess. The compartment contains iliacus and iliopsoas muscles. The psoas compartment muscles lie in close proximity to major abdominal organs and is richly supplied by iliolumbar, external iliac and common femoral arteries. Owing to these reasons, there is hematogenous spread of infections like tuberculosis leading to abscess.

Clinical presentation is with back pain, fever, and lassitude. The hip may be in fixed flexion or there may be pain on passive extension.

CT is performed for diagnosis. Treatment is with percutaneous CT-guided drainage and antibiotic therapy.

Other muscles marked in the image are depicted below.



Solution to Question 2:

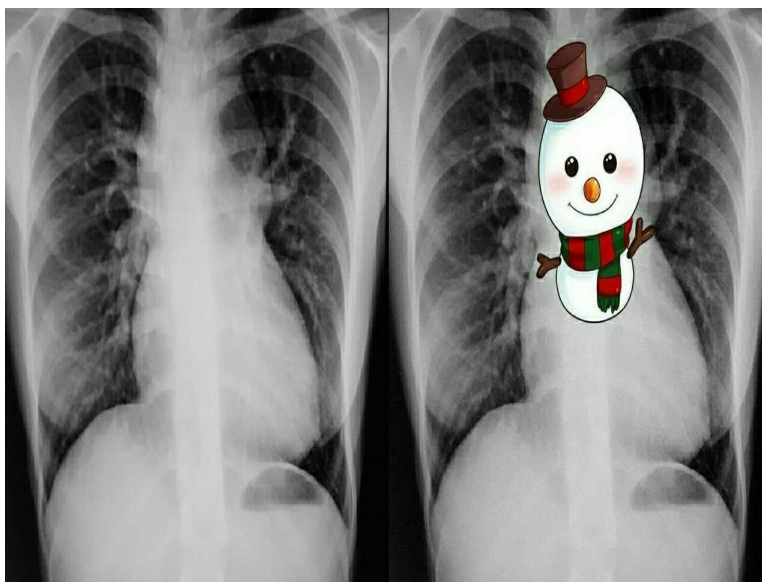
The given X-ray shows figure of 8 or snowman appearance characteristic of total anomalous pulmonary venous circulation (TAPVC).

TAPVC refers to a condition in which pulmonary veins instead of draining into the left atrium drain into the right atrium directly or indirectly. This results in anomalous drainage of pulmonary venous blood flow into the systemic venous circulation. It is a cyanotic heart disease with features of congestive heart failure. The patients are irritable and have failure to thrive.

The following types are seen:

- Supracardiac (most common): Common pulmonary vein drains into superior vena cava or the left innominate vein
- Cardiac: Pulmonary veins drain into the right atrium or coronary sinus.
- Infracardiac (always obstructive and most dangerous): Pulmonary veins drain into portal vein or inferior vena cava.

Snowman appearance is classically seen in supracardiac variant of TAPVC. The dilated vertical vein on the left, brachiocephalic vein on top, and the superior vena cava on the right form the head of the snowman; the body of the snowman is formed by the enlarged right atrium. It is shown in the image below.



Other options:

Option B: Tetralogy of Fallot comprises pulmonary stenosis, ventricular septal defect, overriding of the aorta over the ventricular septum, and right ventricular hypertrophy. On chest X-Ray, the cardiac silhouette appears like a boot ("coeur en sabot").

Option C: In the transposition of great arteries, the aorta arises from the right ventricle and the pulmonary trunk from the left ventricle. The X-ray shows cardiomegaly with cardiac contours which appear like an egg on a string.

Option D: Pericardial effusions occur when excessive fluid gets collected in the pericardial space. The X-ray shows globular enlargement of the cardiac shadow giving a water bottle configuration.

Solution to Question 3:

FAST (Focused Assessment with Sonography for Trauma) is the best investigation that can be performed in the emergency room (ER) for this patient with a blood pressure of 90/50 mm Hg who has sustained blunt abdominal trauma.

Focussed assessment with sonography for trauma is a non-invasive investigation that can be performed rapidly in the emergency room. It helps in identifying an abnormal collection of fluid or blood. FAST is indicated in:

- Haemodynamically unstable patients who had a blunt abdominal trauma
- Patients who sustained penetrating abdominal trauma with no other indications for immediate laparotomy (signs of haemodynamic instability, peritoneal irritation, peritoneal penetration and gunshot wound with transperitoneal trajectory).

FAST includes an examination of four regions:

- Pericardial sac
- Hepatorenal fossa
- Splenorenal fossa, and
- Pouch of Douglas

It is an operator-dependent procedure with limited visualization of the retroperitoneal structures.

Other options:

Option B - A chest X-Ray is used to exclude hemothorax or pneumothorax in a hemodynamically normal patient with penetrating trauma above the umbilicus or a suspected thoracoabdominal injury. Since the given patient is alert, awake, and with no mention of any pelvic pain or tenderness, a pelvic X-Ray is not indicated either.

Option C - Diagnostic peritoneal lavage is an invasive procedure and requires surgical expertise. For these reasons, it is done only in centres where FAST and/ CT are not available.

Option D - CT scan requires transporting the patient and is time-consuming. Hence, it cannot be performed in the ER in a haemodynamically unstable patient.

Solution to Question 4:

The given clinical scenario and X-ray suggest intestinal obstruction where the involved segment is the jejunum. The obstructed bowel loops are central in location and show valvulae connivente (mucosal folds that cross the entire diameter of the bowel) which is characteristic of jejunal involvement. The gas bubbles trapped between the valvulae connivente give a string of beads appearance.

Radiological features of bowel obstruction

- Small bowel - The involved dilated segments are straight and usually centrally located.
- Jejunum - It is characterized by the presence of valvulae connivente with gas bubbles trapped between them. This gives a string of beads/ ladder effect.

- Ileum - The distal ileal obstruction is usually described as featureless.
- Caecum - A rounded gas shadow in the right iliac fossa is suggestive of caecal obstruction.
- Large bowel - The presence of haustrations is a feature of large bowel obstruction. Haustral folds are irregular and do not cross the entire diameter of the bowel. These are absent in the caecum.

In intestinal obstruction, fluid levels appear as the obstruction progresses. When the fluid levels are prominent, it indicates that the obstruction is advanced. The number of fluid levels also increases if the lesion is more distal.

The most reliable criterion for the diagnosis of small bowel obstruction is the transition zone, seen on CT imaging. It identifies a definite point of obstruction. Dilated small bowel loops are seen proximal to the site of obstruction and collapsed loops are seen distal to the obstruction.

Valvulae connivente in jejunal obstruction



Large bowel obstruction - X-Ray appearance



Solution to Question 5:

Neurons are least susceptible to radiation. The radiosensitivity of biological tissue is directly proportional to its mitotic activity and inversely proportional to its degree of differentiation. This is known as the law of Bergonie and Tribondeau.

Hence, hematopoietic cells are the most radiosensitive, and the central nervous system is the least sensitive to radiation.

Acute Radiation Syndromes (ARS) is an acute illness that is caused by exposure of the entire body to a high dose of penetrating radiation in a very short period of time. It is also known as radiation toxicity or radiation sickness. It consists of three syndromes:

Stages of ARS:

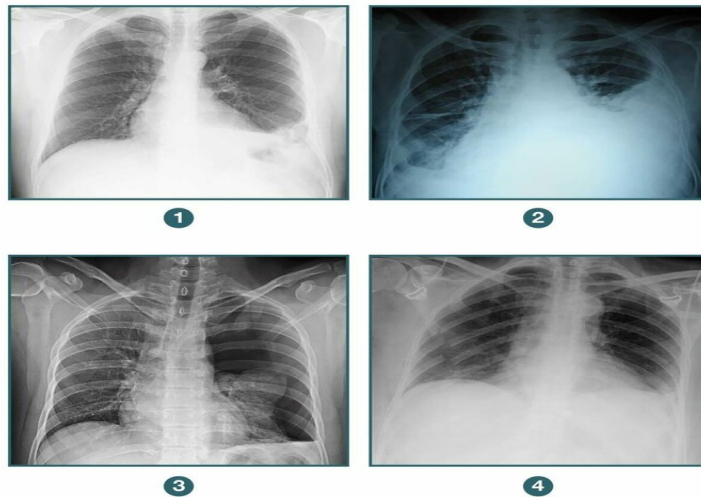
- Prodomal stage - This stage starts within minutes of exposure and may continue for a few days. Symptoms include nausea, vomiting, and anorexia.
- Latent stage - It begins within a few hours and continues up to a few weeks. The patient generally feels healthy.
- Manifest illness stage - This stage may continue up to months and symptoms vary according to the syndrome.
- Recovery/ death - The recovery process occurs over two years. Recovery cannot be expected in cardiovascular and central nervous system syndromes, which lead to death.

Syndrome	Radiation dose
Hematopoietic (Bone marrow)	>0.7 Gy
Gastrointestinal	>10 Gy
Cardiovascular/ Central nervous system	>50 Gy

Radiology INI-CET 2022

Question 1:

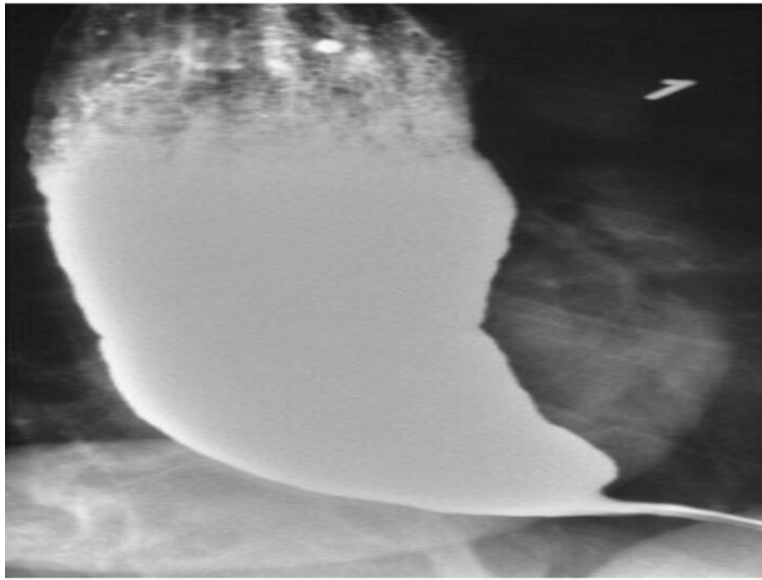
A patient presents to the casualty following blunt trauma to the chest. A chest X-ray was done. Among the following radiographs, in which case would you further evaluate the patient before putting a chest tube?



- a) 4
- b) 1
- c) 2
- d) 3

Question 2:

A 30-year-old patient presents with progressive difficulty in swallowing both solids and liquids for 8 months. A barium swallow was done and is shown in the image below. What other investigations should be done to confirm the diagnosis in this patient?



- a) UGIE + manometry
- b) UGIE + contrast enhanced CT
- c) UGIE + PET
- d) UGIE + 24 hr pH monitoring

Question 3:

A 28-year-old woman with a severe headache is scheduled to undergo a CT scan. During history taking, which of the following will you definitely ask for before the commencement of the scan?

- a) Use of cochlear implant
- b) LMP
- c) History of fall
- d) History of hypertension

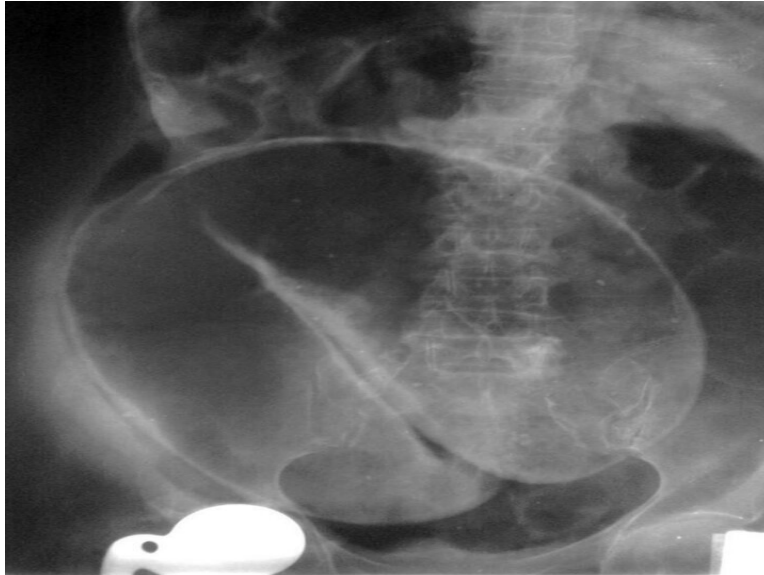
Question 4:

Which of the following findings are seen in a high-resolution CT scan of fungal pneumonia?

- a) 1, 2, 4
- b) 1, 3, 4
- c) 1, 2, 3
- d) 2, 3, 4

Question 5:

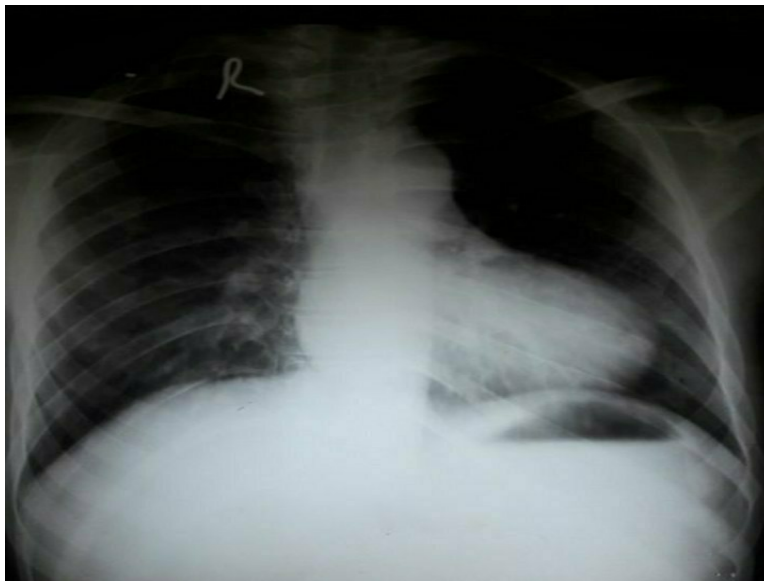
An elderly lady presents with abdominal pain and distension. She has a history of scoliosis. Given below is the erect X-ray abdomen radiograph of the lady. What is the most likely diagnosis?



- a) Large bowel obstruction due to sigmoid volvulus
- b) Large bowel obstruction due to fecolith
- c) Cecal volvulus
- d) Small bowel obstruction due to bowel perforation

Question 6:

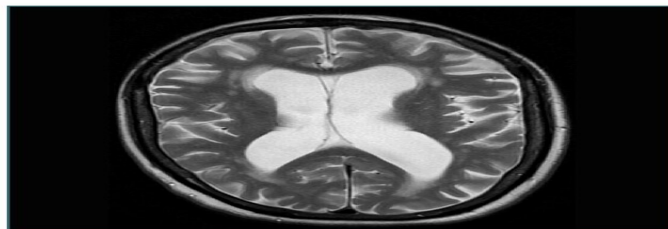
A child presents with cyanosis. His chest radiograph is shown below. What is the diagnosis?



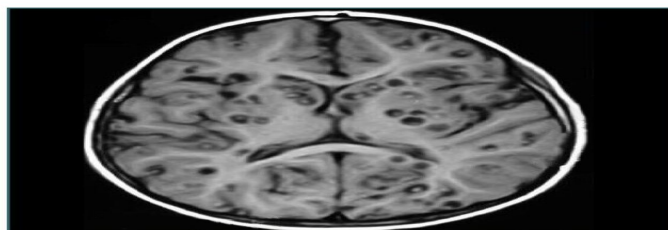
- a) Tetralogy of Fallot (TOF)
- b) Ebstein's anomaly
- c) Transposition of great arteries (TGA)
- d) Total anomalous pulmonary venous return (TAPVC)

Question 7:

An HIV patient presented with a headache and altered sensorium. Consider the following statements and identify the essential criteria for the diagnosis of neurocysticercosis.



1



2

- a) 1 and 2 only

- b) 1 and 4 only
- c) 1, 3, and 4
- d) 1, 2, and 4

Question 8:

Identify the condition based on the non-contrast CT scan of a patient given below.



- a) Hepatocellular carcinoma
- b) Fibronodular hyperplasia
- c) Liver abscess
- d) Hydatid cyst

Question 9:

Match the following:

- a) 1-b, 2-d, 3-a, 4-c
- b) 1-c, 2-a, 3-b, 4-d
- c) 1-d, 2-b, 3-a, 4-c
- d) 1-d, 2-a, 3-b, 4-c

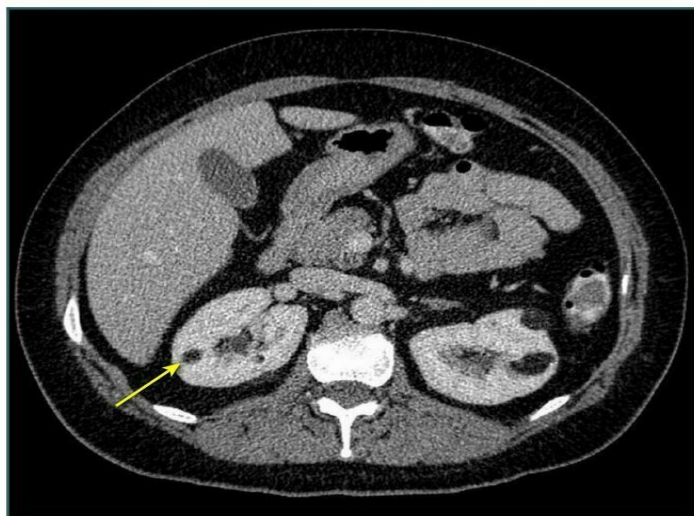
Question 10:

Whole-body CT scan protocol for a trauma patient includes all, except _____.

- a) CT Head
- b) CT Cervical spine
- c) CT Limbs
- d) CT Abdomen

Question 11:

A patient presents with hematuria and a mass in the right kidney on a CT scan, as shown below. What is the diagnosis?



- a) Renal cell carcinoma
- b) Renal oncocytoma
- c) Renal angiomyolipoma
- d) Renal cyst

Question 12:

Egg shell calcification is seen in all the following, except _____.

- a) Silicosis
- b) Sarcoidosis
- c) Post-radiation therapy lymph nodes
- d) Metastatic lymph nodes

Question 13:

A 55-year-old man presents with hematuria and a mass in the left kidney on a CECT scan, as shown below. What is the diagnosis?



- a) Renal angiomyolipoma
- b) Renal oncocytoma
- c) Renal cell carcinoma
- d) Renal cyst

Question 14:

A patient who was diagnosed with prostate cancer is being investigated. The bone scan is reported as a super scan with increased uptake in the bones and reduced activity in the kidney and soft tissue. What is the reason for this super scan appearance?

- a) Increased uptake due to diffuse metastasis
- b) Increased uptake by carcinoma prostate
- c) Increased uptake by the bone
- d) Decreased uptake by adrenal glands and kidney

Answer Key

Question No.	Correct Option
1	b
2	a
3	b
4	c
5	a
6	a
7	b
8	d
9	d
10	c
11	c
12	d
13	c
14	a

Detailed Explanations

Solution to Question 1:

Among the following radiographs, image 1 shows dilated bowel loops in the hemithorax, and is suggestive of diaphragmatic hernia and it needs to be evaluated further before putting a chest tube since it can cause injury to the intestinal contents in the thorax.

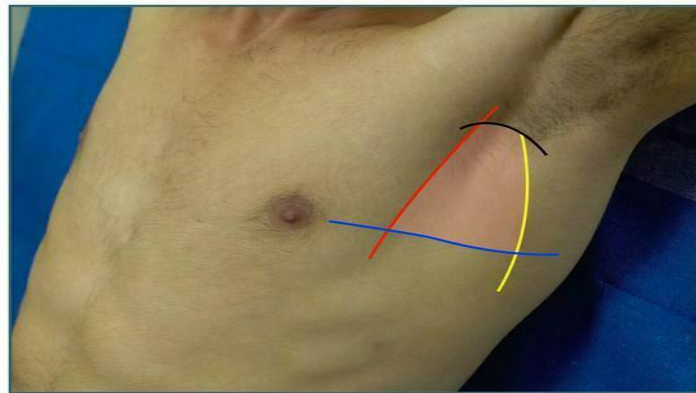
Blunt injury to the diaphragm can lead to rupture and therefore the abdominal contents can herniate into the chest. Nasogastric tube placement followed by chest radiography can be useful in case of herniation of the stomach into the chest. The most accurate investigation is video-assisted thoracoscopy or laparoscopy.

Operative repair should be done. Repair should be done via the abdomen in case of a diaphragmatic rupture.

Radiographs 2,3,4 are hemothorax, pneumothorax, and flail chest respectively. These conditions are life-threatening. Hence, the diagnosis and management have to be immediate. Rapid insertion of intercostal chest drains should be done. They are inserted into the fifth intercostal space. The safe landmark is identified as a triangle that lies anterior to the midaxillary line above the nipple, below, and lateral to the pectoralis major muscle.

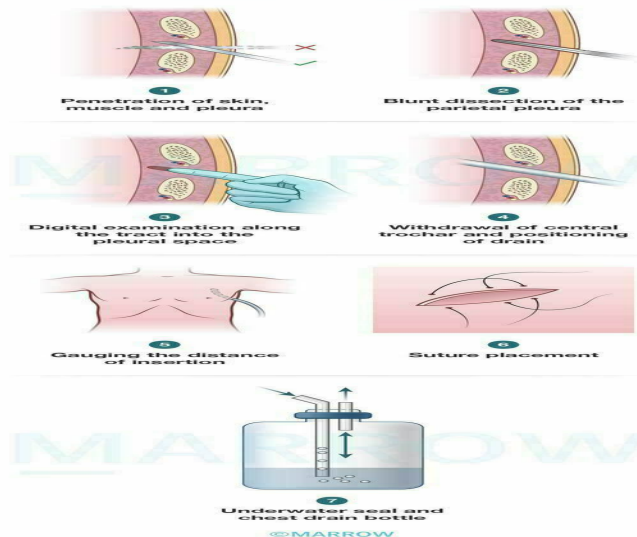
The below images show the triangle of safety and technique of chest drain insertion.

Triangle of safety



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Insertion of chest drain



Hemothorax can occur following blunt trauma. It occurs due to bleeding from the intercostal vessels or internal mammary artery secondary to rib fracture. Features include hemorrhagic shock, flat neck veins, unilateral absence of breath sounds, and dullness on percussion. Correction of hypovolemic shock, intercostal drain insertion, and intubation are useful in management.

Pneumothorax can be open or tension pneumothorax. In tension pneumothorax, the common etiology is penetrating or blunt trauma, and features such as tachypnea, dyspnea, distended neck veins, tracheal deviation, hyper resonance and decreased breath sounds are seen. Immediate decompression can be attained by large bore cannula insertion into the second intercostal space on the affected side in the midclavicular line after which a chest tube can be placed in the 5th intercostal space along the anterior axillary line. Chest radiograph shows mediastinal shift to the opposite side and diaphragmatic depression on the same side.

Flail chest occurs as a result of blunt trauma and is associated with rib fractures. Clinically, paradoxical movement of the chest can be present. Management includes oxygen administration, analgesia, chest drain, and surgery to stabilize the flail segment. Rib fractures can be seen on

chest radiograph.

Solution to Question 2:

The given clinical scenario and the barium swallow showing bird beak sign are suggestive of achalasia cardia. Upper gastrointestinal endoscopy (UGIE) and manometry should be done to confirm the diagnosis.

Achalasia is a motility disorder and is less common. It typically presents as dysphagia for solids and liquids. Other features such as regurgitation and chest pain are also present.

On fluoroscopy, a grossly dilated esophagus with tapering end towards the gastroesophageal junction is seen. This is called the bird beak sign.

Barium is retained in the esophagus for more than 5 minutes and is a significant finding.

Diagnosis is confirmed with endoscopy and manometry.

Solution to Question 3:

The most important information that should be ascertained during history taking in a 28-year-old woman of reproductive age group posted for a CT scan is the last menstrual period (LMP). CT scan is avoided during pregnancy due to the potential adverse effects of radiation on the fetus; hence, LMP is ascertained to rule out pregnancy.

CT is a high-resolution technique using x-rays to generate cross-sectional images of any area of the body. It is associated with high radiation doses.

Other options:

Option A: Use of cochlear implant is a contraindication for MRI and not CT. CT scan can be performed on a patient with a cochlear implant.

Option B: History of fall in a patient with a headache is asked for suspected head trauma. A CT scan in this patient can help in the diagnosis and assess extent of damage.

Option C: History of hypertension points to a possible intracranial bleed. A CT scan can be useful in ruling out the bleed.

Since it is imperative to rule out pregnancy before taking a CT scan, last menstrual period is the most appropriate answer to this question.

Solution to Question 4:

High-resolution computed tomography (HRCT) of fungal pneumonia shows interlobular septations, peripheral wedge-shaped consolidation, and pleural effusion. HRCT of fungal pneumonia shows a nodule (not a cavity) with surrounding ground-glass opacities.

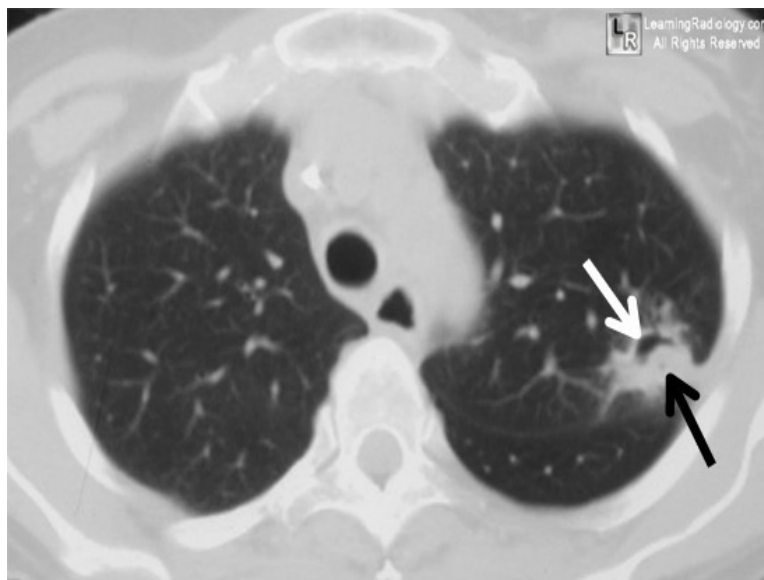
A nodule surrounded by ground-glass attenuation on HRCT imaging is seen in aspergillosis, candidiasis, *Pneumocystis jirovecii* pneumonia, coccidioidomycosis, and paracoccidioidomycosis. Cavitation on HRCT is seen in cases of community-acquired pneumonia due to *Mycobacterium tuberculosis*, *S. aureus*, and gram-negative bacilli.

Aspergillosis is a fungal disease caused by *Aspergillus* sp. Pulmonary aspergillosis has been categorized into saprophytic, allergic, and invasive forms.

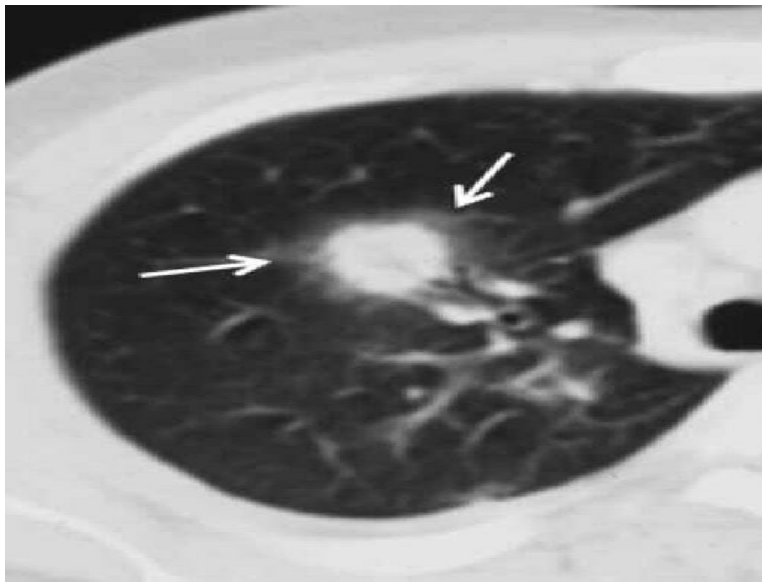
HRCT findings of various types of aspergillosis are as follows:

- Aspergilloma (saprophytic) shows a mass with soft-tissue attenuation, which is separated from the thoracic wall by airspace called the air crescent sign. Multiple nodules and bilateral involvement may be seen.
- Allergic bronchopulmonary aspergillosis (ABPA-allergic) shows hypodense mucous plugs.
- Angioinvasive aspergillosis shows multiple nodules surrounded by a halo of ground-glass attenuation ('halo sign') or pleural-based wedge-shaped areas of consolidation.

The following is an HRCT image of aspergilloma showing the air crescent sign:



The following CT image of aspergilloma shows the halo sign:



Candidiasis pneumonia caused by *Candida* sp. in immunocompromised patients shows multiple bilateral nodular opacities, areas of consolidation, and ground-glass opacities on HRCT. Pleural effusion, thickening of the bronchial walls, and cavitation are less commonly seen.

Paracoccidioidomycosis (PCM), is a disease caused by *Paracoccidioides brasiliensis*. On HRCT, ground-glass opacities, nodules, the 'halo' and 'reversed halo' signs, interlobular septal thickening, airspace consolidation, cavitation, and fibrosis.

Solution to Question 5:

The given clinical scenario with an old woman and the radiograph showing characteristic coffee bean appearance are highly suggestive of sigmoid volvulus causing a large bowel obstruction.

Sigmoid volvulus is a cause of large bowel obstruction and occurs when the sigmoid colon twists on the sigmoid mesocolon. Symptoms include constipation, abdominal bloating, nausea, and/or vomiting. The onset may be acute or chronic. It occurs more commonly in patients with a history of mental illness and chronic constipation.

X-ray abdomen shows a large, dilated loop of the colon, often with a few air-fluid levels. Specific signs include the coffee bean sign, Frimann-Dahl sign (three dense lines converge towards the site of obstruction), and the absence of rectal gas. CT may reveal a large gas-filled loop without haustral markings, forming a closed-loop obstruction.

Here are some pointers that help in differentiating the two volvulus:

Other options:

Option B: In large bowel obstruction due to fecolith, the presence of haustrations is a feature. Haustral folds are irregular and do not cross the entire diameter of the bowel. These are absent in the cecum. The given image shows haustrations in large bowel obstruction.

The following image shows an x-ray radiograph of large bowel obstruction:

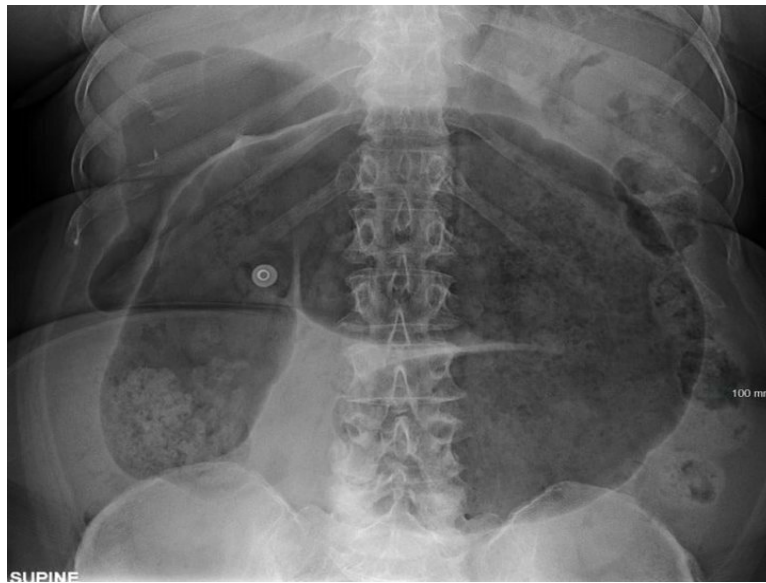
Sigmoid volvulus	Cecal volvulus
More common incidence	Less common incidence
Coffee bean sign is typical.	Coffee bean sign is occasionally seen
Seen in old age, psychiatric patients	Usually seen in those aged 30-60 years old
The top of the sigmoid volvulus usually lies very high in the abdomen with its apex on the left side.	In half the patients, the caecum twists and inverts so that the pole of the caecum and the appendix occupy the left upper quadrant. In other patients, the twist occurs in an axial plane without inversion and the caecum still occupies the right half of the abdomen.
Haustra aren't usually seen	Haustra usually seen
Multiple air-fluid level seen	One air-fluid level seen
	Attached air-filled appendix can be seen

Large bowel obstruction



Option C: Cecal volvulus shows characteristic radiological findings like cecal dilatation, single air-fluid level that may extend into the left upper quadrant, and absence of gas in the distal colon. The colonic haustrations are well maintained.

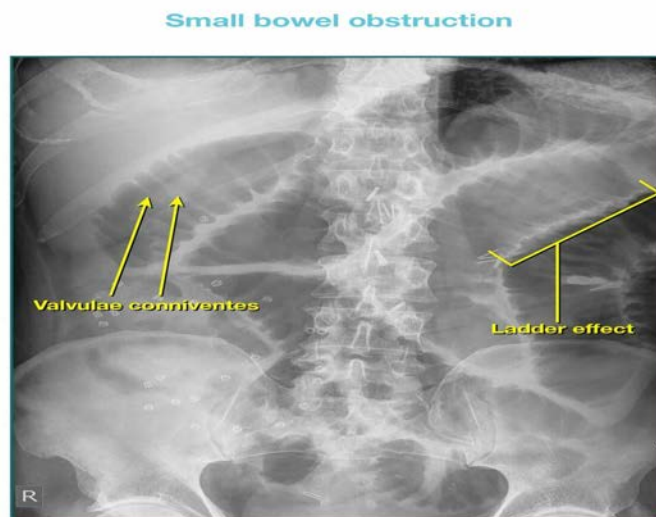
The following image shows an x-ray radiograph of cecal volvulus:



Option D: In small bowel obstructions, the obstructed bowel loops are central in location and lie transversely. The jejunal obstruction shows valvulae conniventes (mucosal folds that cross the entire diameter of the bowel) giving it a ladder effect. The distended cecum may show the presence of gas shadow in the right iliac fossa.

Common causes of small bowel obstruction are adhesions (~60%), neoplasms (~20%), hernias (~10%), Crohn's disease (~5%) and other miscellaneous conditions (~5%).

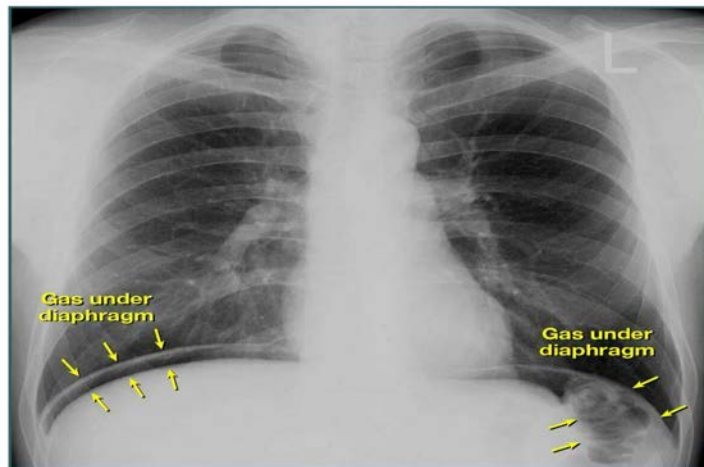
The following image shows an x-ray radiograph of small bowel obstruction:



Cases of bowel perforation have an erect abdominal radiograph showing free gas under the diaphragm. A supine radiograph may show dilated gas-filled loops of the bowel, suggestive of paralytic ileus.

The below radiograph image shows a hollow viscus perforation:

Hollow viscus perforation



Solution to Question 6:

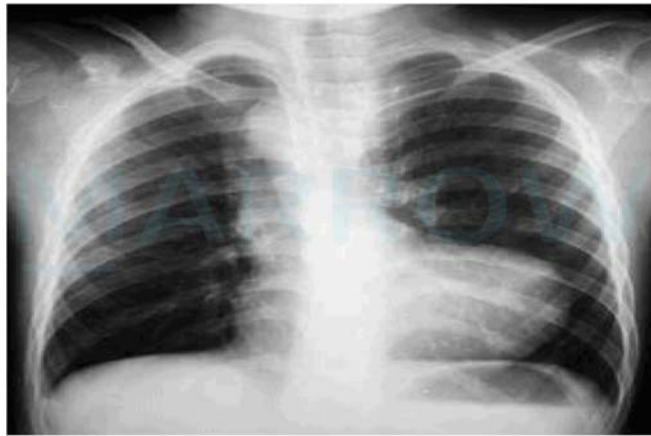
The given clinical scenario and chest X-ray showing a "coeur en sabot" or boot-shaped heart are suggestive of tetralogy of Fallot (TOF).

TOF is a cyanotic congenital heart disease with the following four components:

- Obstruction to right ventricular outflow (pulmonary stenosis)
- Malalignment type of ventricular septal defect (VSD)
- Dextroposition of the aorta so that it overrides the ventricular septum (overriding of the aorta), and
- Right ventricular hypertrophy (RVH)

In TOF, the hypertrophied right ventricle causes the rounded apical shadow to be uptilted so that it is situated higher above the diaphragm than normal. On x-ray, this appears as a boot-shaped heart or coeur en sabot, as shown below:

Boot-shaped heart

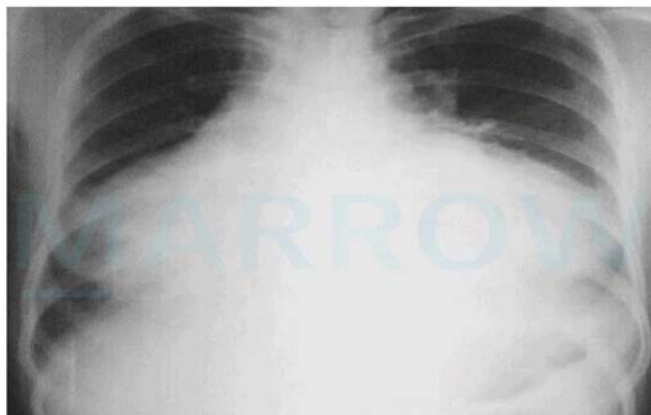


©Marrow

Option B: Ebstein anomaly is a rare cyanotic congenital heart disease with diminished pulmonary blood flow due to abnormality of the tricuspid valve.

In the Ebstein anomaly, there is severe right-sided cardiomegaly due to an elongated and enlarged right atrium, which leads to an elevated apex giving a box-shaped heart, as shown below:

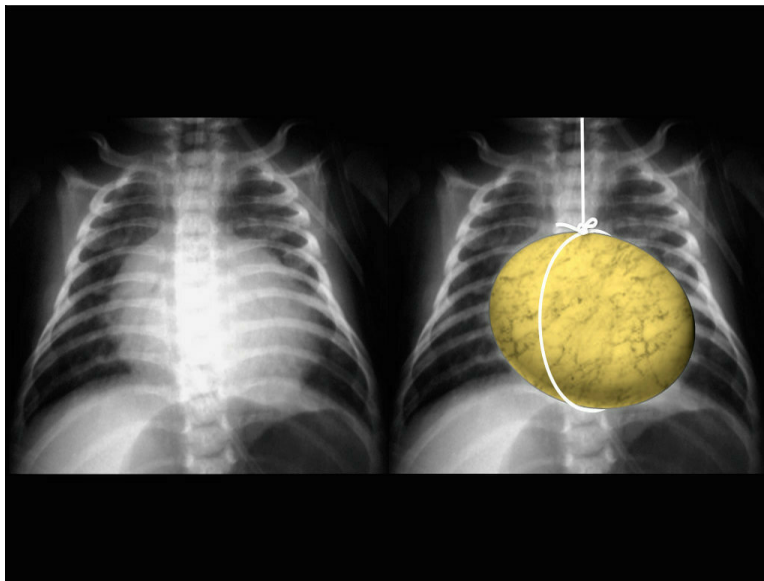
Box-shaped heart seen in Ebstein anomaly



©Marrow

Option C: Transposition of great arteries (TGA) can present from birth to infancy with cyanosis, decreased exercise tolerance, and poor physical development. TGA is the most specific heart disease in infants of diabetic mothers.

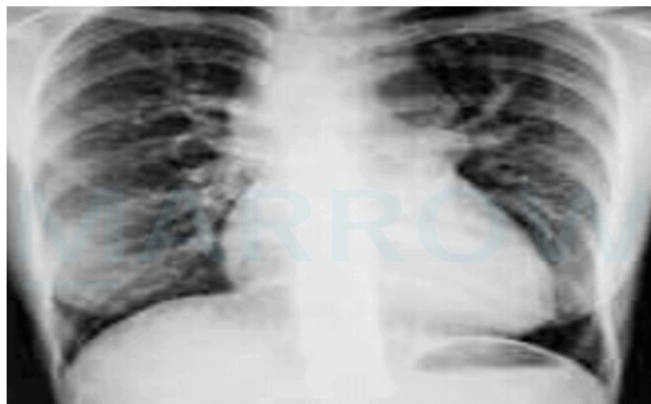
The following is a radiograph of TGA showing the egg-on-a-string appearance/egg-on-side appearance of the heart:



Option D: Total anomalous pulmonary venous return (TAPVC) occurs due to the anomalous connection of pulmonary veins instead of joining to the left atrium resulting in pulmonary venous blood reaching the right atrium.

In the supracardiac type of TAPVC, the configuration of the heart and the superior mediastinal borders resemble a snowman. It is also referred to as the figure of 8 sign, as shown below:

Snowman appearance seen in total anomalous pulmonary venous connection



©MARROW

Solution to Question 7:

The MRI brain labeled as B shows multiple cysticerci within the brain suggesting neurocysticercosis. Among the given statements, the essential criteria for the diagnosis of neurocysticercosis include 1 and 4 (option B), i.e.,

- Brain biopsy showing the parasite.

- Neuroradiologic demonstration of cystic lesions with scolex (MRI labeled B in this case).

Neurocysticercosis is a condition caused by the larvae of *Taenia solium*.

Criteria for diagnosis of neurocysticercosis:

1. Absolute criteria

- Demonstration of cysticerci by histologic or microscopic examination of a biopsy taken from the brain or spinal cord lesion.
- Visualization of the parasite in the eye by funduscopy
- Neuroradiologic demonstration of cystic lesions containing a characteristic scolex

2. Neuroimaging criteria

- Major neuroimaging criteria: Cystic lesions without a discernible scolex, multilobulated cystic lesions in the subarachnoid space, typical small enhancing lesions, typical parenchymal brain calcifications
- Confirmative neuroimaging criteria: Resolution of cystic lesions spontaneously or after cysticidal drug therapy and migration of ventricular cysts documented on sequential neuroimaging studies
- Minor neuroimaging criteria: Obstructive hydrocephalus or abnormal enhancement of basal leptomeninges

3. Clinical/exposure criteria

- Major clinical/exposure criteria: Demonstration of antibodies to cysticerci or cysticerci antigen in cerebrospinal fluid by enzyme-linked immunosorbent assay; Household contact with an individual infected with *T. solium*; Evidence of cysticercosis outside the central nervous system (e.g., cigar-shaped soft-tissue calcifications)
- Minor clinical/exposure criteria: Clinical manifestations suggestive of neurocysticercosis; Residence in a cysticercosis-endemic area; Frequent travel to a cysticercosis-endemic area

Diagnosis is confirmed by:

- One absolute criterion
- Two major neuroimaging criteria plus any clinical/exposure criteria
- One major and one confirmative neuroimaging criterion plus any clinical/exposure criteria
- One major neuroimaging criteria plus two clinical/exposure criteria (including at least one major clinical/exposure criterion), with the exclusion of other pathologies producing similar neuroimaging findings

A probable diagnosis is supported by the fulfillment of

- One major neuroimaging criterion plus any two clinical/exposure criteria
- One minor neuroimaging criterion plus at least one major clinical/exposure criterion

Other options: The MRI labeled as A shows ventriculomegaly in brain atrophy.

Solution to Question 8:

The given CT image shows a cyst in the liver with the serpent sign and areas of calcification, which are suggestive of hydatid disease.

Hydatid disease or echinococcosis is a zoonotic infection caused by the Echinococcus species of flatworms. Its life cycle consists of dogs as definitive hosts and humans as accidental intermediate hosts.

Primary hydatid cysts most commonly occur in the liver. Most patients are asymptomatic but when symptomatic, the patient presents with hepatomegaly, pain, and obstructive jaundice.

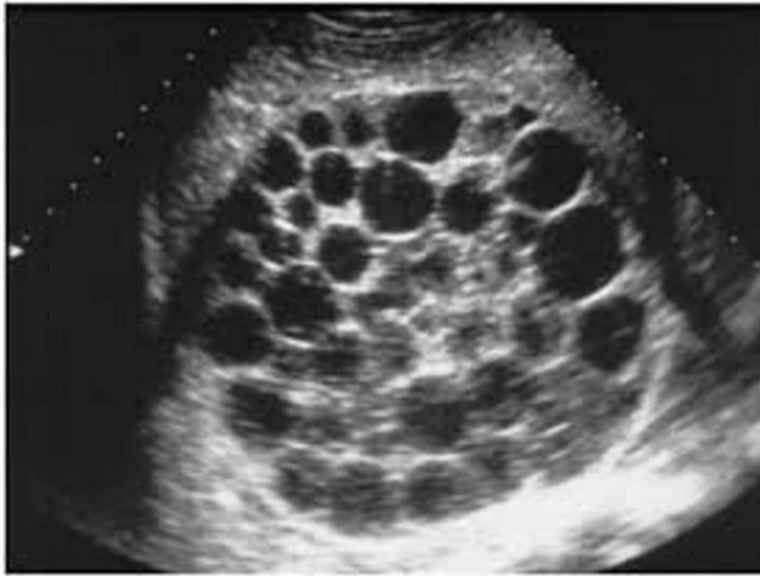
The first/initial imaging test is USG, and CT scan can be used to diagnose the condition.

CT scan shows the presence of a space-occupying lesion with a smooth outline and septa. CT scan has an accuracy of 98% in the detection of hydatid cysts and high sensitivity to the detection of daughter cysts and extra-hepatic cysts.

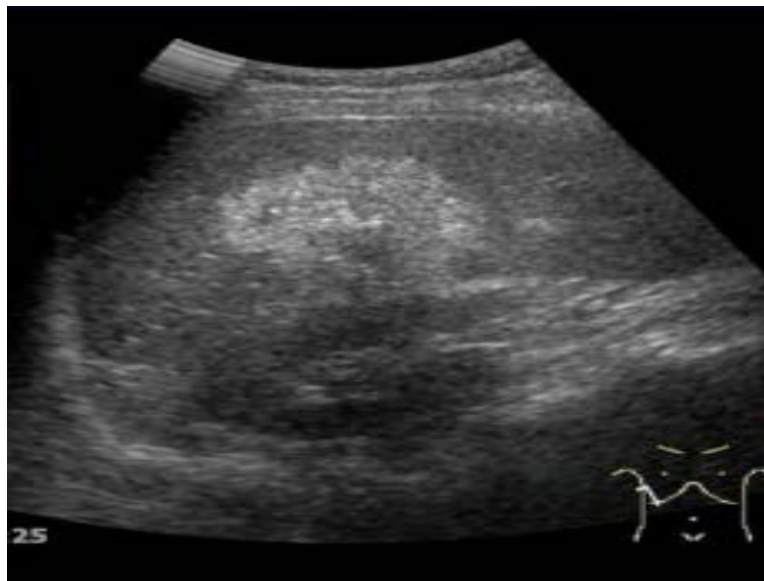
On CT imaging, serpent sign is seen as wavy membranes within the cyst:



The USG of an active hydatid cyst shows multiple septations and a rosette or honeycomb appearance:



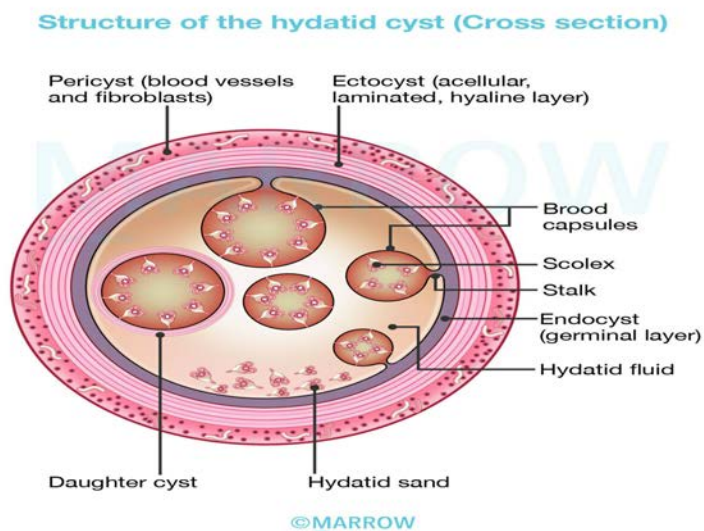
The ball of wool sign is seen on USG when the contents settle in the dependent portion of the cyst, as seen below:



On USG, the water-lily sign is seen in hydatid infections when there is detachment of the endocyst membrane, which results in floating membranes within the pericyst that mimic the appearance of a water lily.



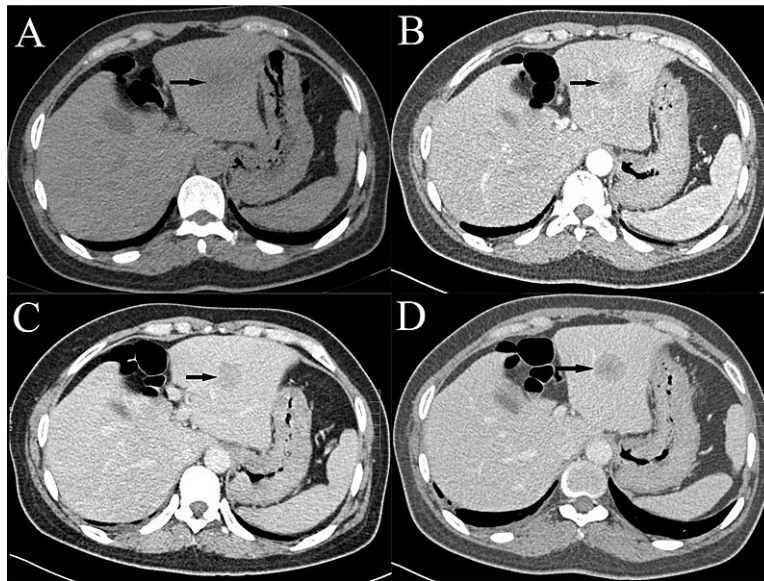
The image below shows the structures present within a cyst.



Other options:

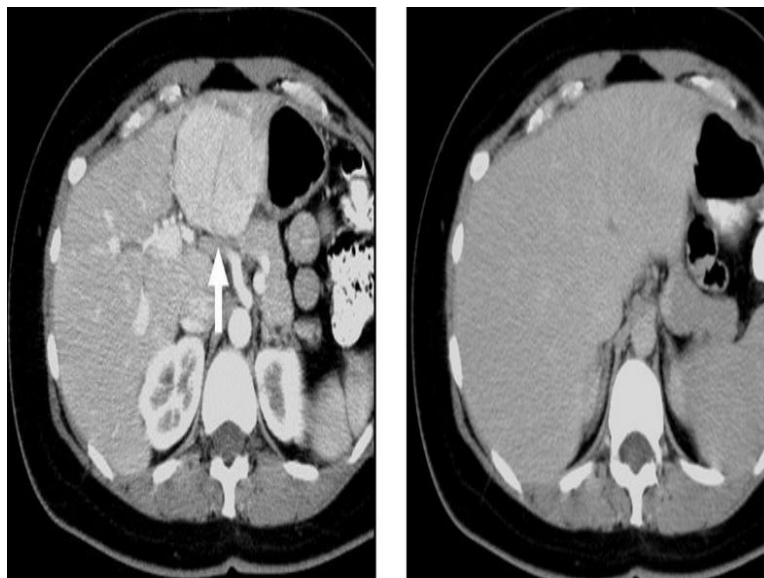
Option A: Hepatocellular carcinoma (HCC) is diagnosed with contrast-enhanced CT (CECT) or MRI. On a non-contrast CT, it appears as a low-attenuated lesion with occasional calcification. On CECT, HCC lesions show enhancement in the arterial phase and washout in the portal phase.

The following image shows the triphasic CECT of HCC:



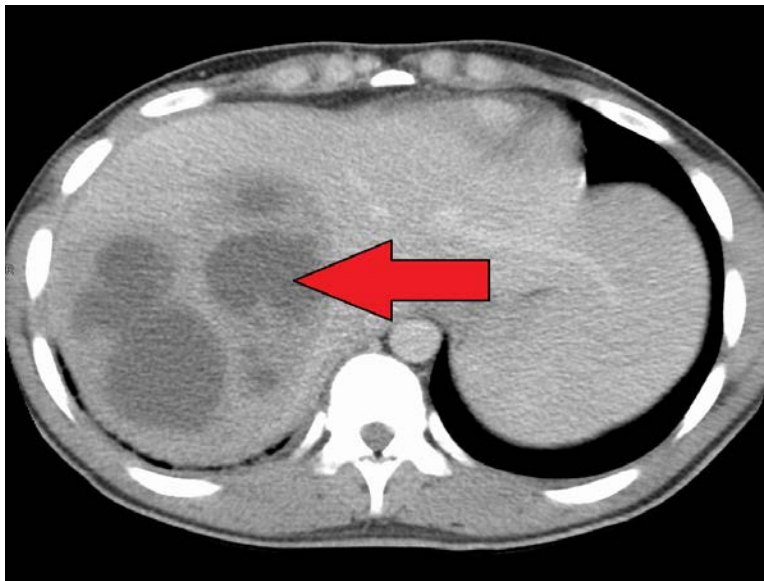
Option B: Focal nodular hyperplasia is a benign solid lesion in the liver. Histologically, it is composed of normal liver elements, but the biliary elements are not connected to the normal biliary tree. On CECT, they exhibit homogeneous enhancement in the arterial phase of IV contrast.

The image below shows late arterial and portal venous phase CECT of focal nodular hyperplasia:



Option C: Liver abscess may be caused by various infections. It begins as a local inflammatory reaction and later develops liquefactive necrosis in the center, with an inflammatory margin.

On a non-contrast CT, it appears as an ill-defined lesion with low attenuation, as seen in the image below:



The given image shows a liver abscess with rim enhancement on CECT.



Solution to Question 9:

The correct match is 1-d, 2-a, 3-b, 4-c (option D).

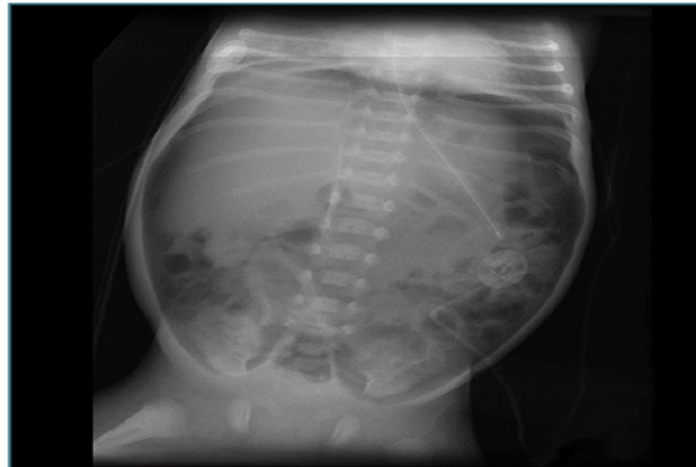
Football sign is seen in cases of massive pneumoperitoneum in abdominal x-ray film taken in the supine position.

The following image shows football sign in a patient with pneumoperitoneum:

1. Football sign	d. Pneumoperitoneum
2. String sign	a. Hypertrophic pyloric stenosis

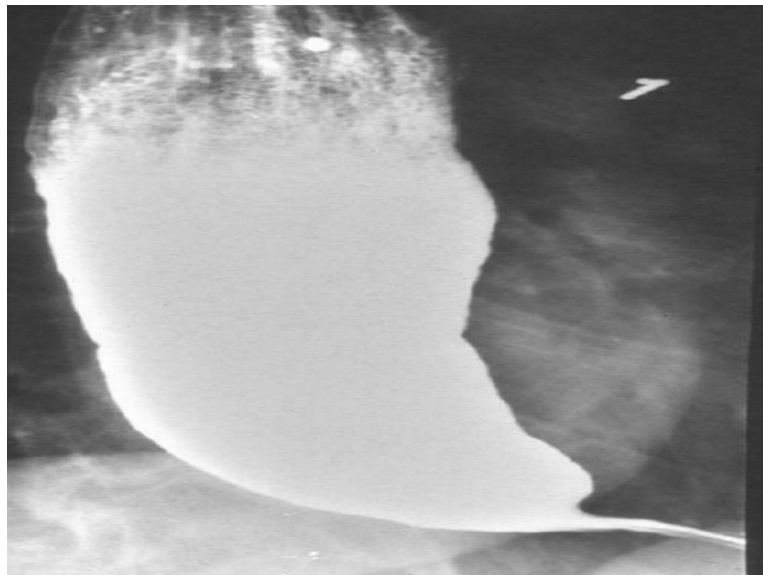
1. Football sign	d. Pneumoperitoneum
3. Rat tail sign	b. Achalasia cardia
4. Inverted umbrella sign	c. Ileocecal TB

Football sign- Pneumoperitoneum

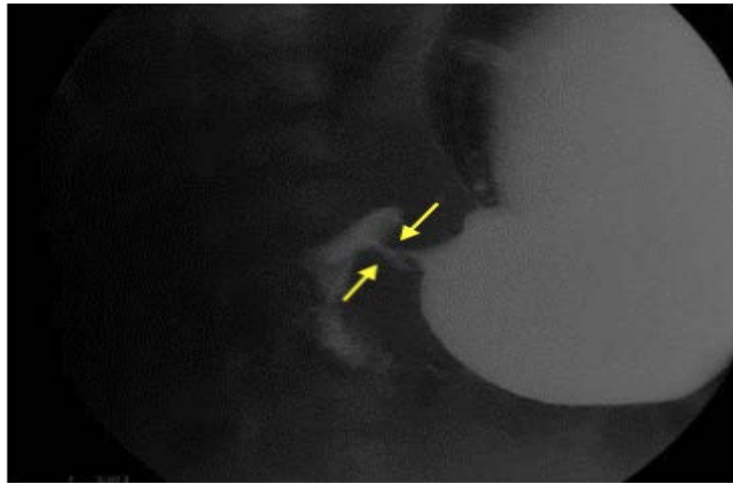


Rat tail sign refers to dilatation of the esophagus with narrowing at the gastro-esophageal junction in barium swallow/ fluoroscopy studies. It is also known as bird beak sign and is typically seen in achalasia cardia. In the case of a rat tail sign with irregular contour, esophageal cancer should be suspected.

The following image shows rat tail sign or bird beak sign in a patient with achalasia cardia:



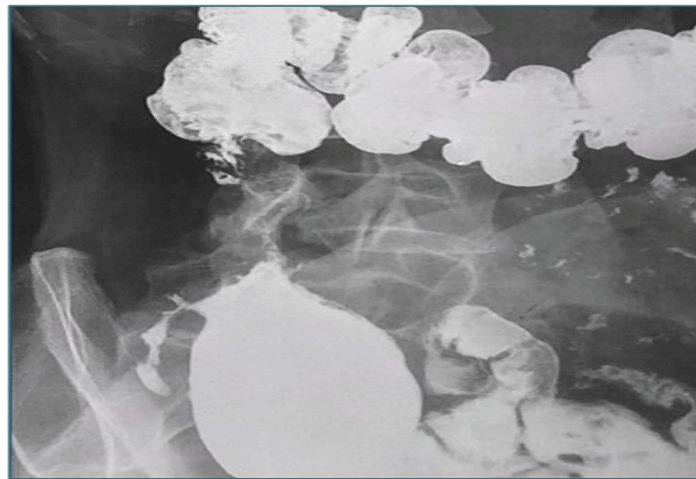
String sign refers to the string-like appearance of severely narrowed bowel lumen in fluoroscopy/barium studies. It is seen in hypertrophic pyloric stenosis, gastrointestinal tuberculosis, colon cancer, and carcinoid tumor. String sign of Kantor is classically described in Crohn's disease.



String Sign in Pyloric Stenosis.

Inverted umbrella sign, also known as Fleischner sign, is seen in barium studies of ileocecal tuberculosis. It refers to a widely gaping, thickened, patulous ileocecal valve and a narrowed, ulcerated terminal ileum.

Fleischner sign/Inverted umbrella sign



Solution to Question 10:

The whole-body CT of a trauma patient does not include CT limbs.

Whole-body CT or trauma CT or panscan is increasingly being used in the context of trauma patients with a dangerous mechanism of injury:

- Non-contrast CT of head
- Non-contrast CT of neck
- Contrast-enhanced CT of chest (arterial phase)

- Contrast-enhanced CT of abdomen (portal venous phase)

The purpose of the scan is to rapidly evaluate life-threatening injuries and accurately diagnose known and unknown injuries.

Solution to Question 11:

The CT image showing renal lesions that has hypo-attenuating components consistent with macroscopic fat is highly suggestive of renal angiomyolipoma.

Angiomyolipomas are benign tumors that occur sporadically or in association with tuberous sclerosis. The mean age of presentation is 43 years and there is a female predilection. It can present as spontaneous retroperitoneal hemorrhage, especially with larger tumors. The patient can present with flank pain, palpable mass, hematuria, urinary tract infections, hypertension, or renal failure.

Imaging in renal angiomyolipoma: Demonstration of macroscopic fat is the main part of making a diagnosis. USG may show hyperechoic lesions in the cortex with posterior acoustic shadowing. In patients with tuberous sclerosis, the entire kidney is affected and produces an echogenic appearance with loss of normal corticomedullary junction.

CT findings:

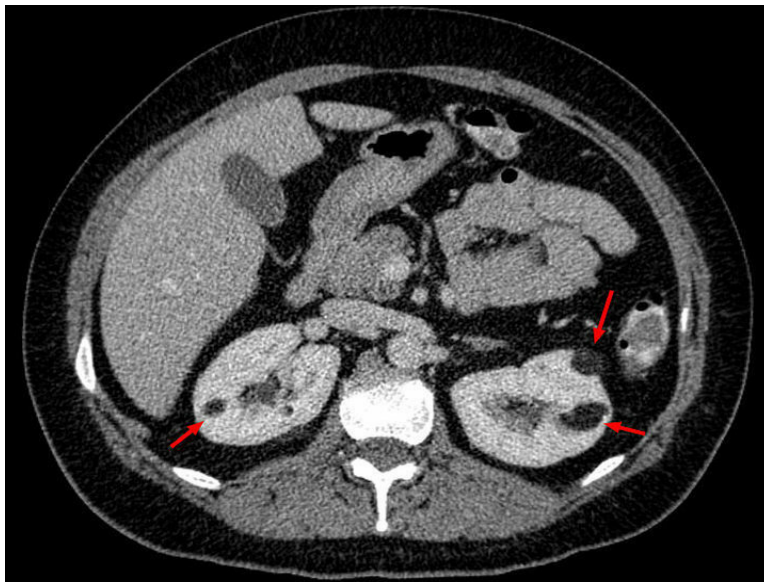
- Cortical lesions that demonstrate macroscopic fat (less than -20 HU)
- Absence of ossification or calcification
- The presence of macroscopic fat is strongly suggestive of this condition, but not pathognomonic as RCC may rarely demonstrate macroscopic fat.
- Approximately 5% may be fat-poor.

MRI: It is an excellent tool to demonstrate intramural fat components. Two sequences are used:

- Fat-saturated techniques
- Phase-in and phase-out imaging that shows India ink artifacts at the junction between fat and non-fat components.

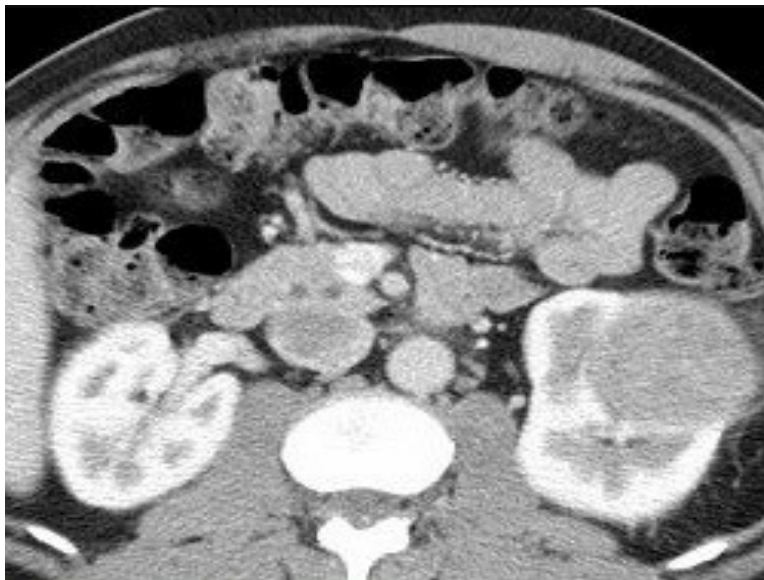
Digital subtraction angiography: They are hypervascular lesions showing a 'sunburst appearance' in the arterial phase and an 'onion peel appearance' in the venous phase.

They usually do not require any intervention. mTOR inhibitors like everolimus can be used to decrease the size of the lesion.

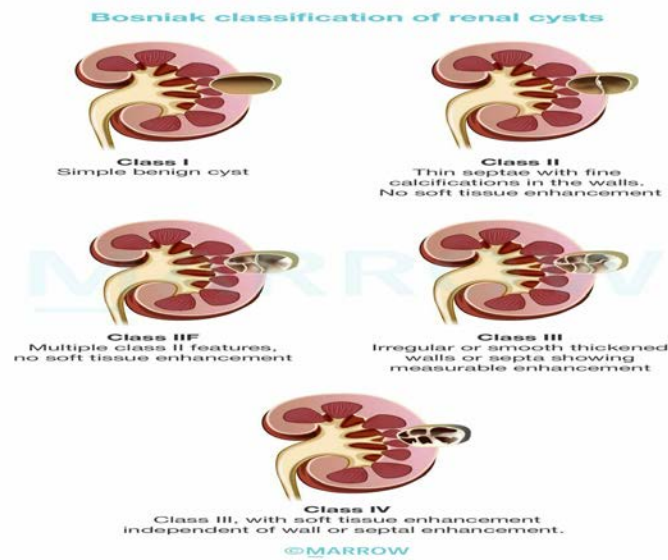
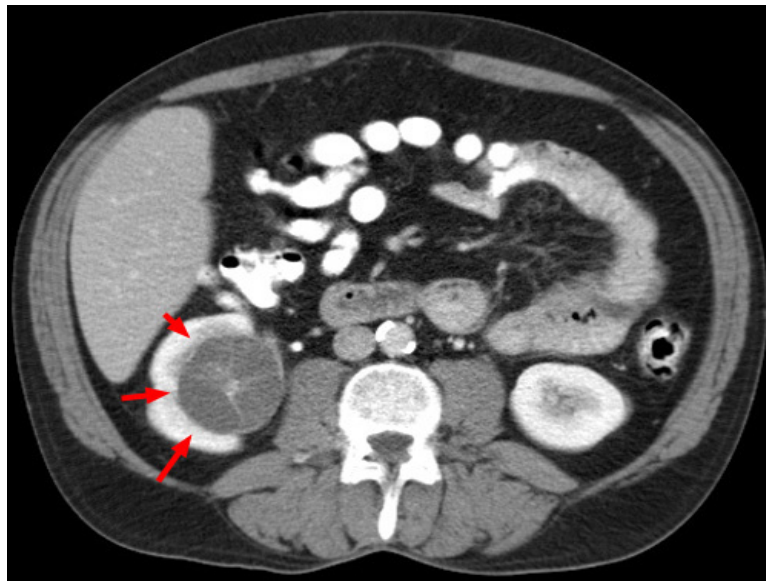


Other options:

Renal cell carcinoma (RCC): It is the most common malignant tumor of the kidney. It usually occurs in the age group 50-70 years and macroscopic hematuria is seen in about 60% of the cases. It is a differential for angiomyolipoma but it is important to note that macroscopic fat is almost always associated with ossification or calcification.

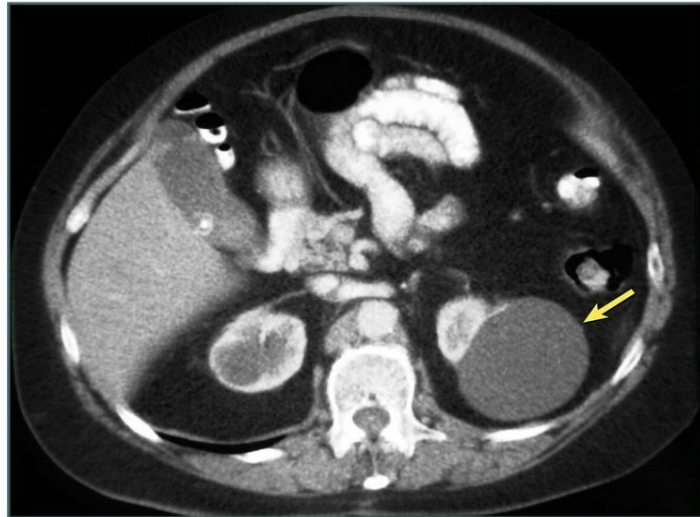


Renal oncocytoma: They are relatively benign in nature and usually present in the 6th to 7th decade. An important point regarding this tumor is its similarity with RCC and hence it is very difficult to differentiate between them preoperatively. A sharp central stellate scar may be seen in one-third of the cases. But RCC may also show this stellate scar appearance in some cases.



Renal cyst: It is a broader term that includes the cystic lesions of the kidney. They can be simple cysts that are benign. Complex or complicated cystic lesions need evaluation and can be malignant. Bosniak classification of renal cysts is given above.

Image showing a simple renal cyst:



Solution to Question 12:

Eggshell calcification is not seen in the case of metastatic lymph nodes.

A ring of calcification at the periphery of a lymph node is known as eggshell calcification.

Solution to Question 13:

The history of an elderly male presenting with painless hematuria, renal mass, and an axial renal cortical phase CECT image showing renal lesions with irregular enhancement is suggestive of renal cell carcinoma (RCC).

Renal cell carcinoma (RCC) is the most common malignant tumor of the kidney. It usually occurs in the age group 50–70 years and macroscopic hematuria is seen in about 60% of the cases. It usually presents with painless gross hematuria, flank pain, or painless flank mass.

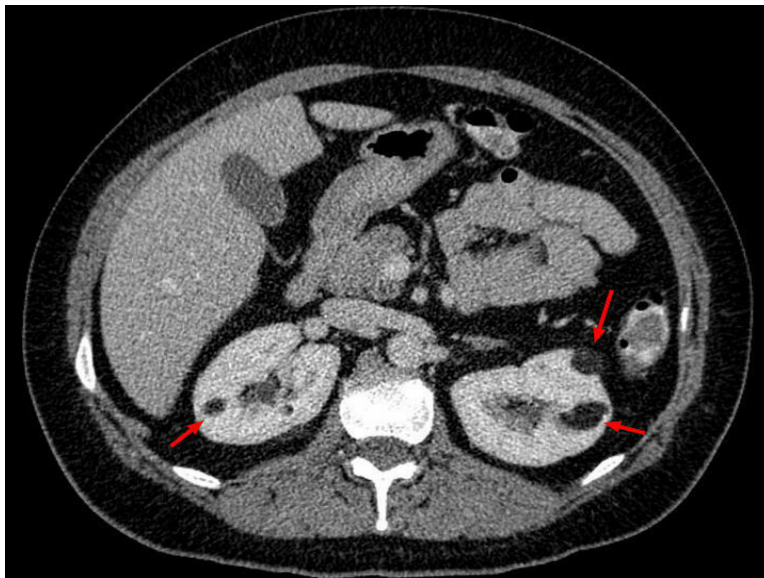
Contrast-enhanced CT is the investigation of choice for RCC. On CT, small lesions enhance homogeneously, whereas larger lesions have irregular enhancement due to areas of necrosis. The clear cell subtype may show much stronger enhancement is a differential for angiomyolipoma. Still, it is essential to note that macroscopic fat is almost always associated with ossification or calcification.

T1-weighted MRI is often heterogeneous due to necrosis, hemorrhage, and solid components. T2-weighted MRI is hyperdense for clear cell RCC and hypodense for papillary RCC. Thus, a T2-weighted MRI can be used for histological diagnosis. Contrast-enhanced MRI is the investigation of choice for detecting vascular invasion of RCC.

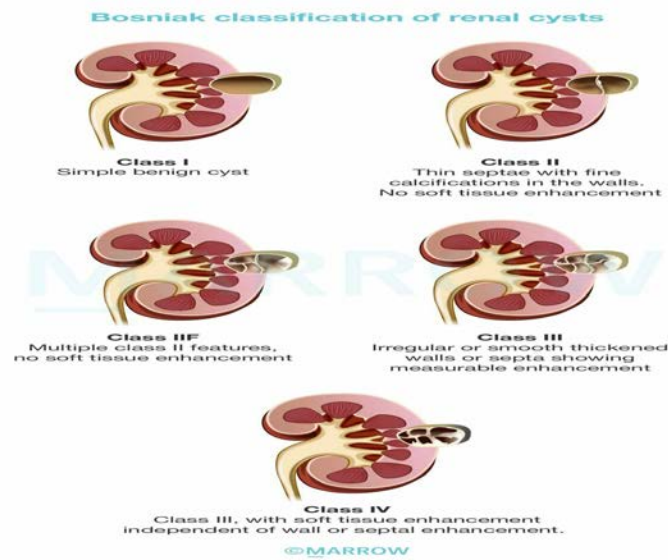
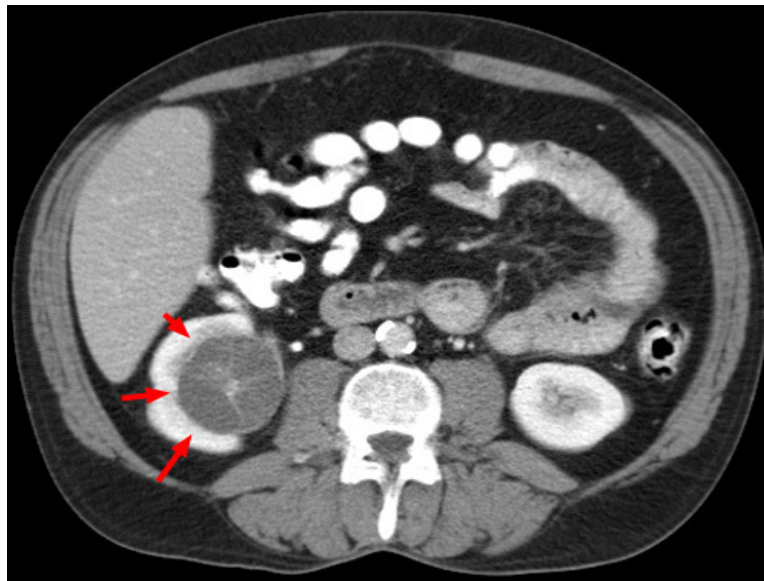


Other options:

Renal angiomyolipoma: Angiomyolipomas are benign tumors that occur sporadically or in association with tuberous sclerosis. The mean age of presentation is 43 years, and there is a female predilection. CT findings show cortical lesions that demonstrate macroscopic fat (less than -20 HU) and absence of ossification or calcification.

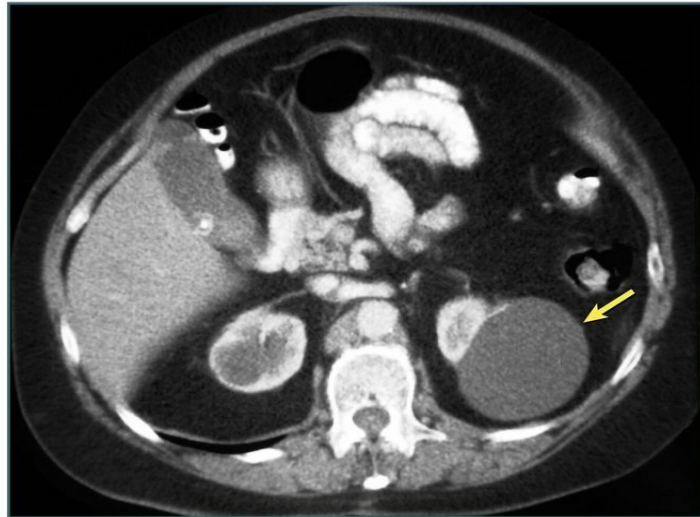


Renal oncocytoma: They are relatively benign in nature and usually present in the 6th to 7th decade. An important point regarding this tumor is its similarity with RCC, and hence, it is very difficult to differentiate between them preoperatively. A sharp central stellate scar may be seen in one-third of the cases. But RCC may also show this stellate scar appearance in some cases.



Renal cyst: It is a broader term that includes the cystic lesions of the kidney. They can be simple cysts that are benign. Complex or complicated cystic lesions need evaluation and can be malignant. Bosniak classification of renal cysts is given above.

Image showing a simple renal cyst:



Solution to Question 14:

The super scan appearance is due to increased uptake due to diffuse metastasis.

Diffuse osteoblastic metastatic diseases such as carcinoma breast and prostate carcinoma lead to a generalized increase in the skeletal uptake relative to soft tissue and absent or faint renal activity on a bone scan. This is known as a super scan. It may also be seen in generalized bone diseases such as hyperparathyroidism.

Carcinoma of the prostate is the most common malignant tumor in men over the age of 65 years. Most cancers are clinically silent and are detected on routine rectal examination or PSA measurements. Grossly, malignant tissue is gritty and firm in consistency (in contrast to the spongy appearance of benign tissue) and thus more readily palpable on digital rectal examination. They usually arise in the peripheral zone of the gland.

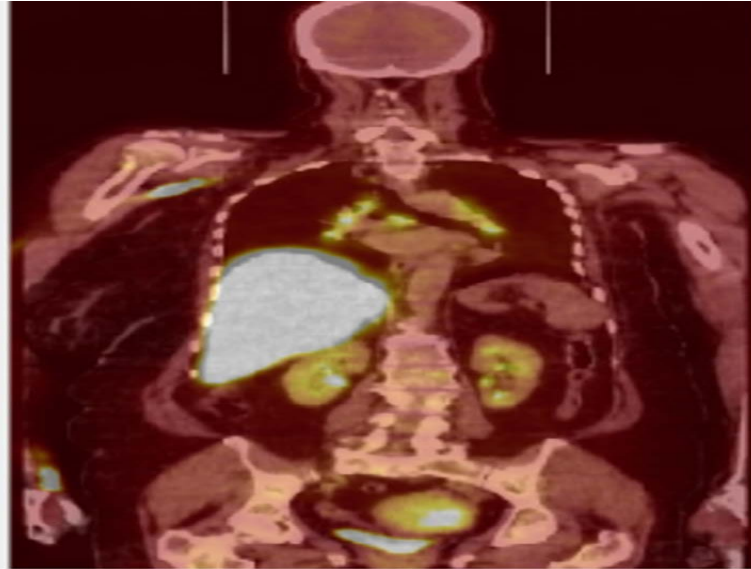
Local extension most commonly involves the periprostatic tissue, seminal vesicles, and the base of the urinary bladder. Metastasis spreads via the lymphatics to the obturator nodes and the para-aortic nodes. Hematogenous spread occurs chiefly through the bones to the axial skeleton and rarely to the viscera. The bony metastasis is blastic in origin and involves the lumbar spine, proximal femur, pelvis, thoracic spine, and ribs in decreasing order of frequency.

Prognosis and staging: The Gleason grading system correlates with the pathological stage, spread, and prognosis. Measurement of serum PSA levels may be useful to assist with the diagnosis and management.

Radiology NEET 2022

Question 1:

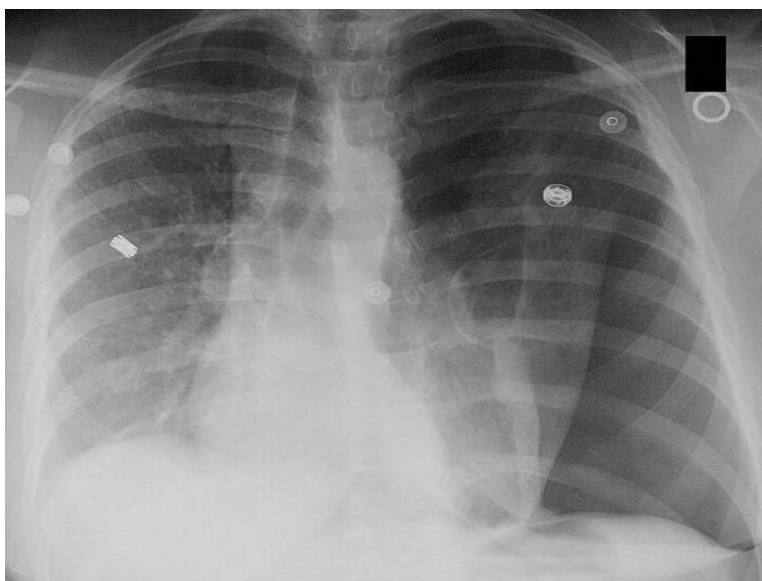
Identify the imaging modality given below.



- a) Cerebral blood flow scan
- b) PET CT
- c) Sestamibi scan
- d) Full body MDCT

Question 2:

A patient is brought to the casualty following a road traffic accident. On examination, the patient is conscious, BP is 90/60 mm Hg, respiratory rate is 40 breaths per minute, and pulse rate is 120 bpm. The X-ray is shown below. What is the next best step in management?



- a) Chest tube insertion
- b) Pericardiocentesis
- c) Thoracotomy
- d) Pleurodesis

Answer Key

Question No.	Correct Option
1	b
2	a

Detailed Explanations

Solution to Question 1:

The above imaging modality is a PET CT scan.

PET (positron-emission tomography) CT scan is a hybrid technique that combines PET and CT scan that is used for targeted imaging of tumor physiology and biology. It is based on the principle that tumor cells generally have high metabolic activity compared with normal tissues.

A small dose of the radiopharmaceutical agent ^{18}F -fluorodeoxyglucose (FDG). This enters cells and is phosphorylated to FDG 6-phosphate which cannot be metabolized further. Since it is a highly polar molecule, it cannot easily diffuse back out of the cell. After intravenous injection of

FDG, it continuously accumulates in the metabolically active cell.

The main advantage of PET-CT is the identification of distant metastatic disease not seen on CT alone. The disadvantages of PET scans are radiation burden, poorer spatial resolution when used alone, and high cost.

Other options:

Option A: Cerebral blood flow scan refers to the perfusion MRI that uses gadolinium as contrast and provides information about the cerebral blood flow and volume.

Option B: The sestamibi scan is used for parathyroid localization and is now considered the most accurate and reliable method for imaging the parathyroid glands. Sestamibi accumulates in mitochondria. Parathyroid adenomas often have a high concentration of oxyphilic cells with high mitochondrial content. These retain sestamibi tracer and adenomas are therefore associated with a slow washout when compared with the thyroid gland.

Option D: The full body MDCT scan refers to the multidetector CT scan which is an advanced CT imaging technique that uses multiple detector elements that provide improvement in speed, quality, and scope.

Solution to Question 2:

The given X-ray features showing a well-demarcated area devoid of lung markings on the left side with tracheal deviation and the mediastinal shift to the right side, along with the clinical scenario is suggestive of tension pneumothorax. The next best step in the management is chest tube insertion.

Tension pneumothorax is a clinical diagnosis, and treatment should never be delayed while waiting for radiological confirmation. Imaging modalities used are ultrasound—extended focused assessment with sonar for trauma (eFAST), chest x-ray, and spiral CT.

A tension pneumothorax develops when there is a tear either in the parietal or visceral pleura. This tear acts as a one-way valve i.e., air enters from the lung or chest wall through it during inspiration but has no way to escape during expiration. This ultimately leads to the collapse of the affected lung.

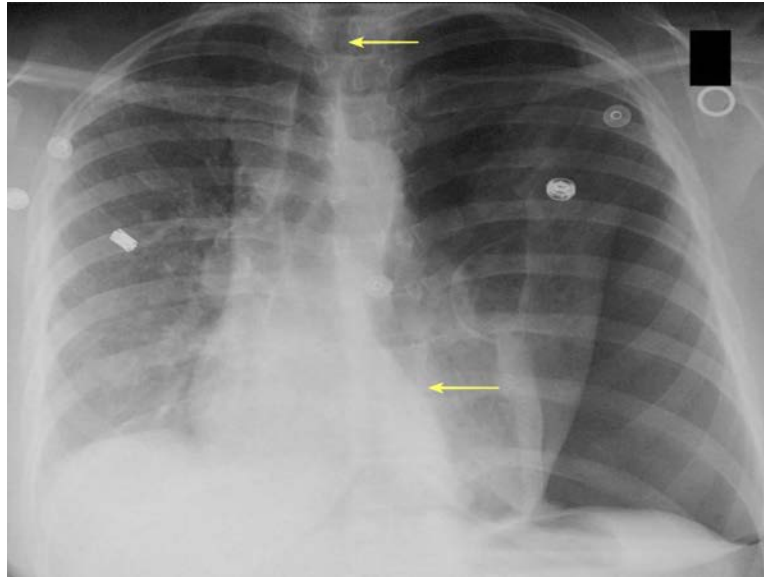
It usually occurs following penetrating or blunt chest trauma, iatrogenic lung injury, or mechanical positive pressure ventilation.

It is characterized by:

- Chest pain
- Tachypnea and acute dyspnea
- Sudden onset hemodynamic compromise (tachycardia, hypotension, low O₂ saturation, loss of consciousness)
- Neck vein distention
- Tracheal deviation away from the side of the injury
- Decreased chest movements on the affected side

- Hyper-resonance on percussion on the affected side
- Absence or decreased breath sounds on the affected side

The X-ray below shows a left-sided tension pneumothorax. There is a large, well-demarcated area devoid of lung markings on the left side. The yellow arrows are pointing to the tracheal deviation and the mediastinal shift to the right side.

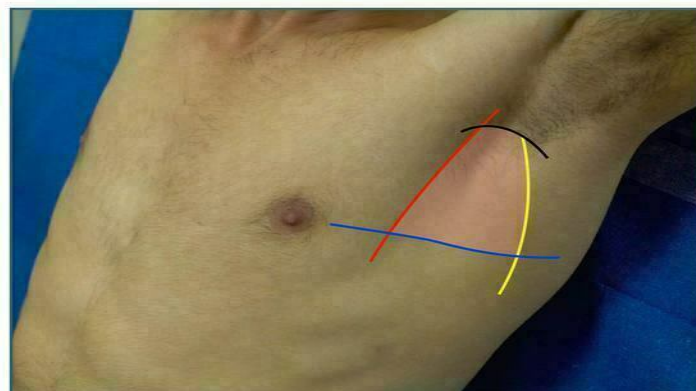


Management of tension pneumothorax:

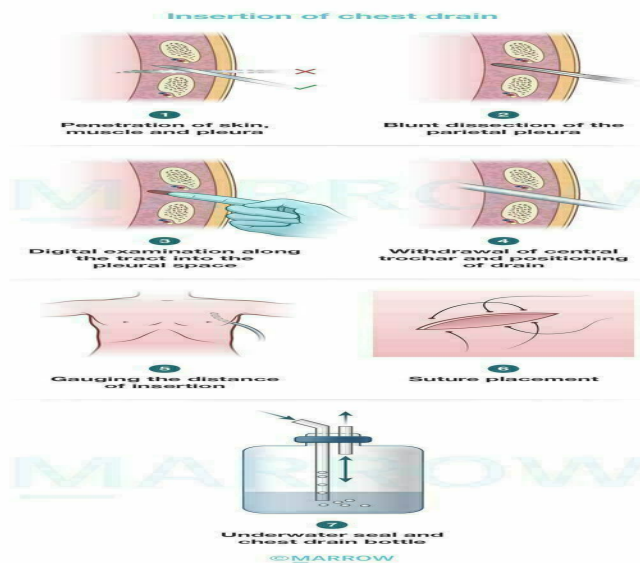
- Immediate management- needle decompression or needle thoracostomy i.e. insertion of a wide-bore needle in the 5th ICS anterior to the midaxillary line
- Definitive management- tube thoracostomy or chest tube insertion in the 5th ICS in the triangle of safety, identified as a triangle that lies anterior to the midaxillary line above the nipple, below, and lateral to the pectoralis major muscle.

The images below show the triangle of safety and technique of chest drain insertion.

Triangle of safety



©Marrow



Other options:

- Pericardiocentesis is done in cases of cardiac tamponade which is characterized by the presence of hypotension, distended jugular veins, and muffled heart sounds (Beck's triad). The breath sounds are absent in tension pneumothorax while they are present in cardiac tamponade.
- Surgery for pneumothorax can be performed by video-assisted thoracoscopic surgery (VATS) or as an open procedure (thoracotomy) in cases of spontaneous pneumothorax failing to settle despite chest drainage. Pleural adhesion is achieved by chemical pleurodesis where talc is used and is insufflated into the chest cavity.

Radiology NEET 2023

Question 1:

A 2-month-old infant is brought to the OPD with a parietal swelling present since birth. The X-ray image is given below. What could be the probable diagnosis?



- a) Subgaleal hematoma
- b) Cephalhematoma
- c) Caput succedaneum
- d) Encephalocele

Question 2:

A patient presented with abdominal pain and sterile pyuria, and the x-ray showed the following features. What is the likely diagnosis?



- a) Putty kidney
- b) Nephrocalcinosis
- c) Staghorn calculus
- d) Psoas calcification

Question 3:

A woman with endometrial carcinoma is undergoing radiotherapy. Which of the following is true?

- a) Intensity is inversely proportional to the square of the distance from the source
- b) Small blood vessels are most radioresistant
- c) Rapidly proliferating cells are most radioresistant
- d) Small intestinal mucosa is most radio-resistant

Question 4:

A patient presents with a history of fever and cough for the past 3-5 days. His examination findings reveal crepitations. Chest X-ray is given below. What is the probable diagnosis?



- a) Apical segment of Right lower lobe consolidation
- b) Right middle lobe consolidation
- c) Posterior segment of right lower lobe consolidation
- d) Loculated pleural effusion

Answer Key

Question No.	Correct Option
1	b
2	a
3	a
4	b

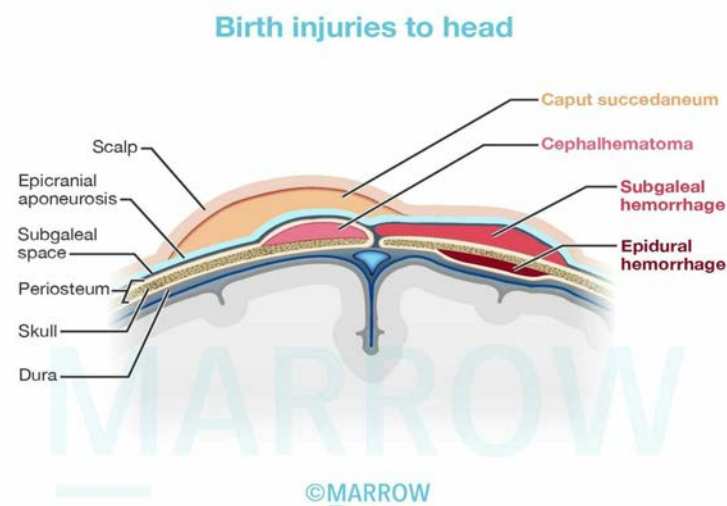
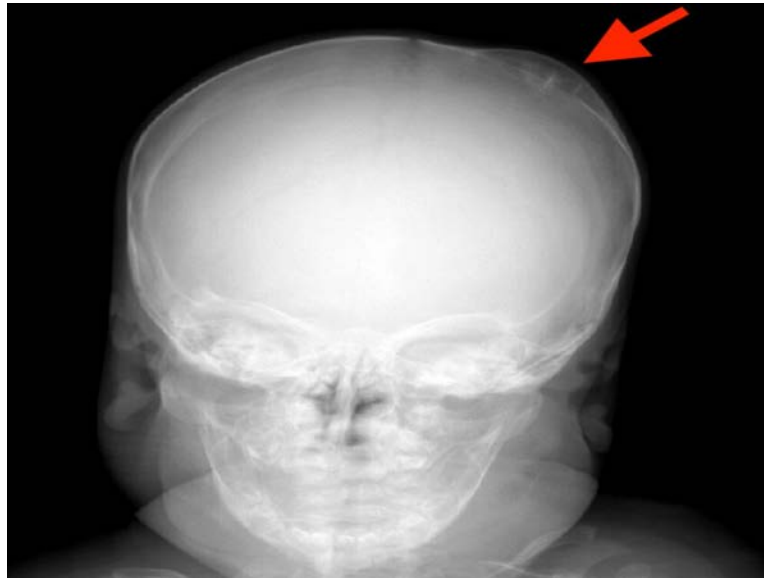
Detailed Explanations

Solution to Question 1:

In the given clinical picture, the swelling over the infant's head is likely to be a cephalhematoma. Cephalohematoma is a collection of blood in between the pericranium and the flat bone (subperiosteal) of the skull, usually unilateral and over a parietal bone. It is never present at birth but gradually develops after 12–24 h. The swelling is limited by the suture lines of the skull as the pericranium is fixed to the margins of the bone.

The arrow mark in the image represents cephalohematoma.

Note: Cephalhematoma is more likely to be identified soon after birth, and its presence as a defined area of swelling that does not cross suture lines is more consistent with this diagnosis.



Subgaleal hematoma occurs between the galea aponeurotica and the periosteum. It is commonly caused by birth trauma, particularly forceps or vacuum-assisted delivery. Subgaleal hematoma can cross suture lines and typically presents as a soft, fluctuant swelling on the scalp.

Caput succedaneum is an edema of the scalp that can cross suture lines. It is usually present at birth or develops shortly after due to pressure on the presenting part of the head during labor. It generally resolves within a few days.

Encephalocele is a neural tube defect that results in a sac-like protrusion of the brain and its meninges through an opening in the skull. It is a congenital condition and is typically diagnosed at birth or via prenatal ultrasound.

Solution to Question 2:

A patient with abdominal pain and sterile pyuria, with the given x-ray showing diffuse alveolar calcification in the right renal region point towards a diagnosis of a putty kidney.

The putty kidney is a characteristic finding of genitourinary tuberculosis (TB), specifically renal TB. The given image shows a homogenous and ground-glass-like calcification of the kidney appearing like putty. It is caused by the dissemination of the organism through the bloodstream and is thus always a secondary TB. There is either reinfection or reactivation of old TB.

Genitourinary TB is a cause of sterile pyuria. Diagnosis is confirmed with a tuberculin test, or at least 3 consecutive early-morning specimens of urine are examined for acid-fast bacilli. They are stained with a Ziehl-Neelsen stain and subsequently cultured.

CT is the most sensitive modality and is the preferred means of visualizing renal calcifications. CT urography is done to assess the various stages of renal TB and its manifestations, which are as follows:

- Early:
- Papillary necrosis
- Progressive:
- Multifocal strictures
- Generalized or focal hydronephrosis
- Mural thickening and enhancement
- Poorly enhancing renal parenchyma
- End-stage:
- Progressive hydronephrosis
- Amorphous dystrophic calcification that involves the entire kidney - putty kidney

Treatment involves short-course therapy with antitubercular drugs like pyrazinamide, isoniazid, and rifampicin. Partial nephrectomy, nephrectomy, or reconstructive surgery is often done.

Option B: Nephrocalcinosis - This condition is characterized by the deposition of calcium in the kidney parenchyma or tubules, often seen in hypercalcemic conditions like hyperparathyroidism. However, it would not explain the clinical presentation of sterile pyuria.



Option C: Staghorn calculi are large kidney stones that fill the renal pelvis and extend into the renal calyces, often caused by urinary tract infections with organisms such as *Proteus*, *Pseudomonas*, and *Staphylococcus*. While these calculi can cause abdominal pain, they usually do not result in sterile pyuria, as they are associated with infection.

The image below shows a staghorn calculus of the left kidney.



Option D: Psoas calcification - This refers to calcification within the psoas muscle and is often seen after acute renal failure and rhabdomyolysis. It is not associated with sterile pyuria and is not the cause of the patient's symptoms.

Solution to Question 3:

The true statement is that the intensity of radiation is inversely proportional to the square of the distance from the source in brachytherapy.

Radiotherapy is a common treatment modality for various cancers, including endometrial carcinoma. It works by targeting rapidly dividing cancer cells with ionizing radiation, which can damage the DNA of these cells and lead to cell death. Radiotherapy can be delivered externally (external beam radiotherapy) or internally (brachytherapy). In brachytherapy, radioactive sources are placed within or adjacent to the tumor, leading to a high radiation dose delivered directly to the tumor while minimizing damage to surrounding healthy tissue.

Other options:

Option B: Larger blood vessels are more radioresistant than microvasculature.

Option C: Permanent non-dividing cells are more radioresistant than rapidly proliferating cells.

Option D: Small intestinal epithelial cells (intestinal crypts), which divide rapidly with no differentiation are the most radiosensitive.

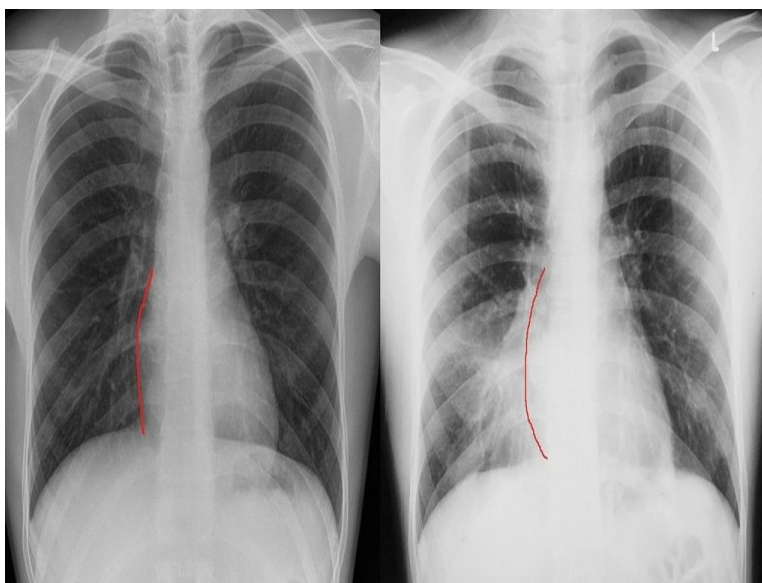
Solution to Question 4:

The given clinical scenario of fever, cough, and crepitations suggests features of pneumonia. The chest X-ray showing opacities with obscuration of the right cardiac border is suggestive of right middle lobe consolidation.

This loss of clear demarcation between the normal lung and its surrounding soft structures (heart, diaphragm) is called the silhouette sign. It occurs due to loss of contrast between lungs and soft tissue when the lung parenchyma is no longer filled with air. This sign may be seen in consolidation, atelectasis, or mass lesions.

In the following images:

- Left image - red line indicates the normal demarcation between the right heart border and middle lobe of the right lung
- Right image - loss of demarcation i.e, positive silhouette sign due to right middle lobe consolidation



Positive silhouette sign and corresponding anatomical site of pathology:

- Right heart border - right middle lobe or medial right lower lobe
- Left heart border - lingula segments of the left upper lobe
- Left hemidiaphragm or descending aorta - left lower lobe
- Aortic knuckle - left upper lobe (posterior segment)
- Right hemidiaphragm - right lower lobe
- Right paratracheal stripe - right upper lobe

Other options:

Apical segment of Right lower lobe consolidation: In this case, the chest X-ray would show an opacity or consolidation in the apical segment of the right lower lobe. The consolidation would be located in the lower part of the right lung, towards the apex. The right heart border, however, would remain visible, as it is not adjacent to the apical segment of the right lower lobe.

Posterior segment of right lower lobe consolidation: A chest X-ray revealing consolidation in the posterior segment of the right lower lobe would show an opacity in the lower part of the right lung, more posteriorly. This would not affect the visibility of the right heart border, as it is not adjacent to the posterior segment of the right lower lobe. In a posteroanterior (PA) chest X-ray, this consolidation may be difficult to appreciate, but it would be more evident in a lateral view.

Loculated pleural effusion: In a chest X-ray showing a loculated pleural effusion, there would be a fluid collection within the pleural space. This fluid collection would appear as a well-defined opacity, typically with a convex border facing the lung parenchyma. It may appear as a lenticular shape, conforming to the space between the lung and chest wall or diaphragm. The loculated pleural effusion would not create a silhouette sign with the right heart border, as it is located in the pleural space rather than the lung parenchyma.

Radiology INI-CET 2023

Question 1:

A patient presented with dysuria and pain in the back. His lumbar spine X-ray shows the below finding. What is the likely diagnosis?



- a) Wilson's disease
- b) Addison's disease
- c) Cystic fibrosis
- d) Hyperparathyroidism

Question 2:

Identify the finding in the given ultrasound image.



- a) Mercedes-Benz sign
- b) Gall bladder polyp
- c) Gall bladder cancer
- d) Phrygian cap

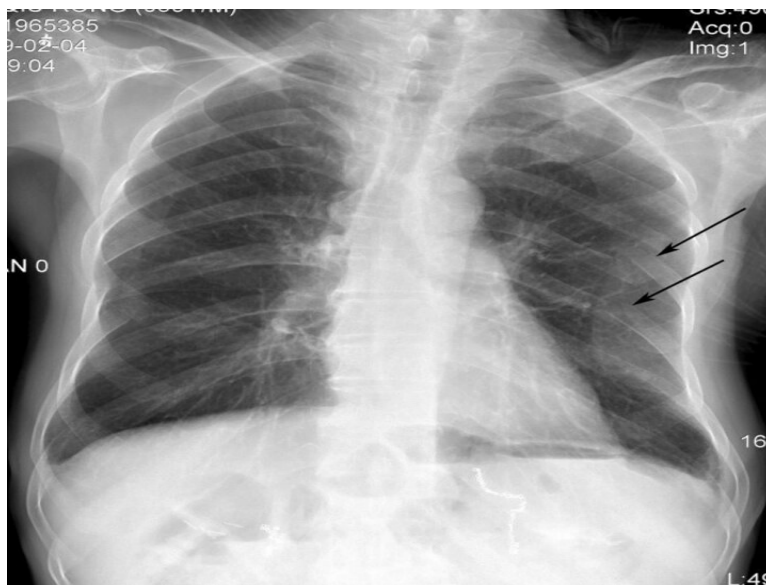
Question 3:

Banana sign on USG is associated with which of the following conditions?

- a) Spina bifida
- b) Omphalocele
- c) Downs syndrome
- d) Klinefelter syndrome

Question 4:

Identify the fracture site in the X-ray shown below.



- a) 5th, 6th rib
- b) 5th, 6th & 7th rib
- c) 7th and 8th rib
- d) 6th and 7th rib

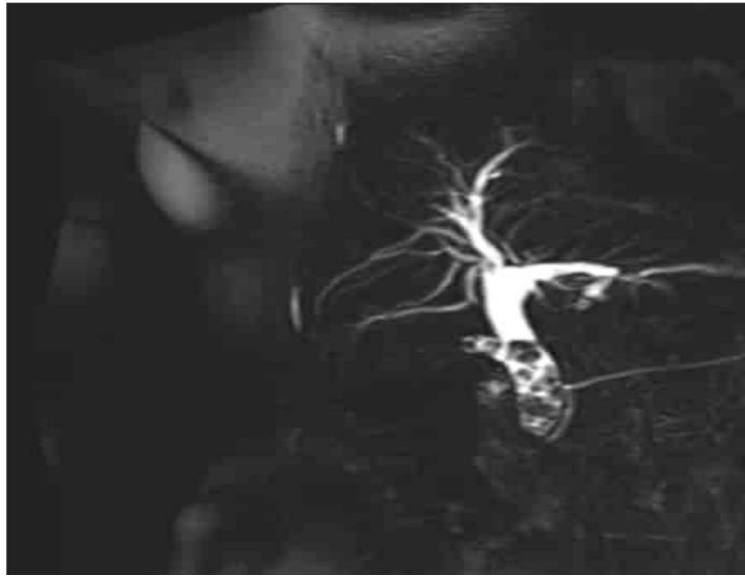
Question 5:

Which of the following is not an advantage of USG over Mammography?

- a) Can be used for screening mammographically undetectable clinically non palpable breast mass
- b) Useful in young females with dense breast
- c) Differentiates between solid and Cystic lesions in the Breast
- d) Guided biopsy can be done

Question 6:

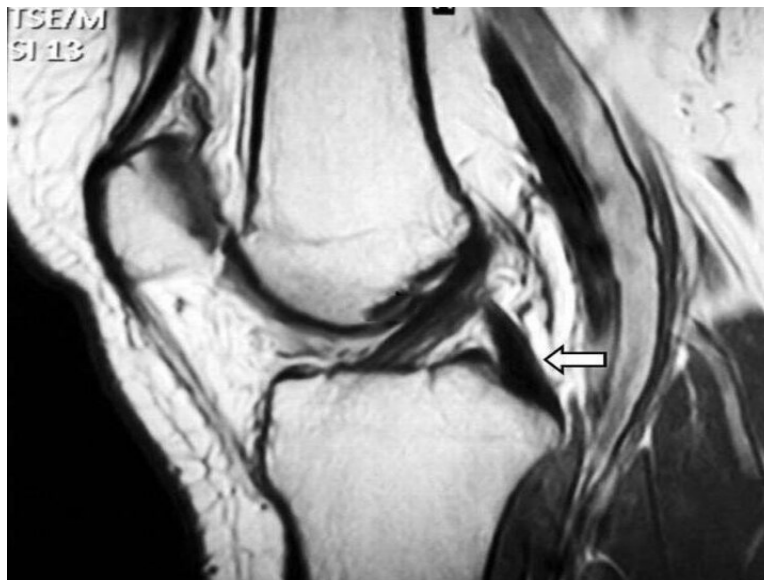
An MRCP image is shown below. Which among the following is the most likely diagnosis?



- a) Choledocholithiasis
- b) Cholecystitis
- c) Pancreatic duct stricture
- d) Malignancy

Question 7:

The image given below shows the MRI of the knee. Identify the marked structure.

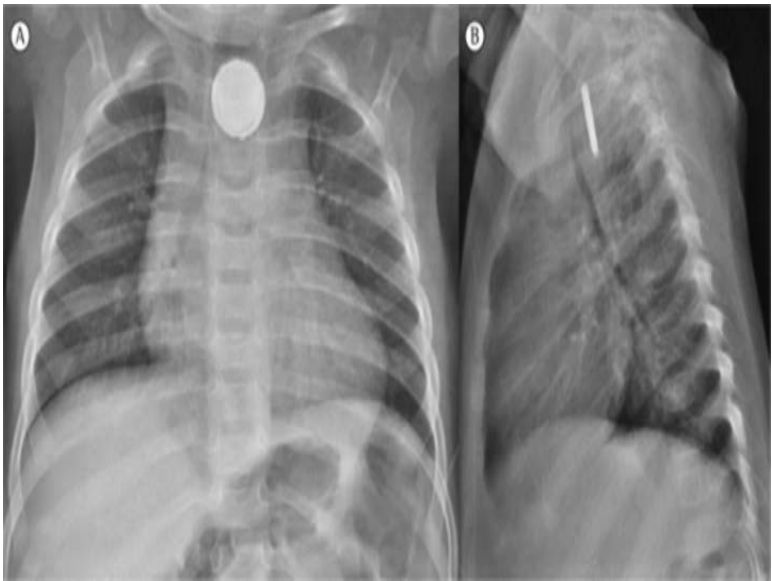


- a) ACL
- b) PCL

- c) MCL
- d) Patellar tendon

Question 8:

A child was brought to the emergency with drooling and choking following ingestion of a foreign body. Based on the given X-ray, what object was most likely swallowed by the child?



- a) Cap of a cool drink bottle
- b) Coin
- c) Button battery
- d) Metal ball

Answer Key

Question No.	Correct Option
1	d
2	d
3	a
4	d
5	a
6	a

7	b
8	c

Detailed Explanations

Solution to Question 1:

The given clinical scenario and X-ray showing diffuse calcification of both kidneys are suggestive of nephrocalcinosis which is seen in hyperparathyroidism.

Nephrocalcinosis is the deposition of calcium in the kidney parenchyma or tubules which is seen in conditions causing hypercalcemia like hyperparathyroidism.

Clinical features of hyperparathyroidism:

- Bones:
- Pathological fractures
- Subperiosteal bone resorption - 2nd and 3rd digits, most common in the radial aspect
- Salt and pepper skull
- Osteitis fibrosa cystica or VonReckling Hausen's disease of bone
- Stones
- Multiple and recurrent renal stones - most common
- Nephrocalcinosis
- Abdominal groans - colicky due to pancreatitis
- Psychiatric overtones - raised PTH causing CNS depression and psychiatric symptoms
- Band keratopathy
- Features of hypercalcemia -
- Vomiting
- Confusion
- ECG changes: increased PR interval, shortened QT interval

Investigations done in primary hyperparathyroidism:

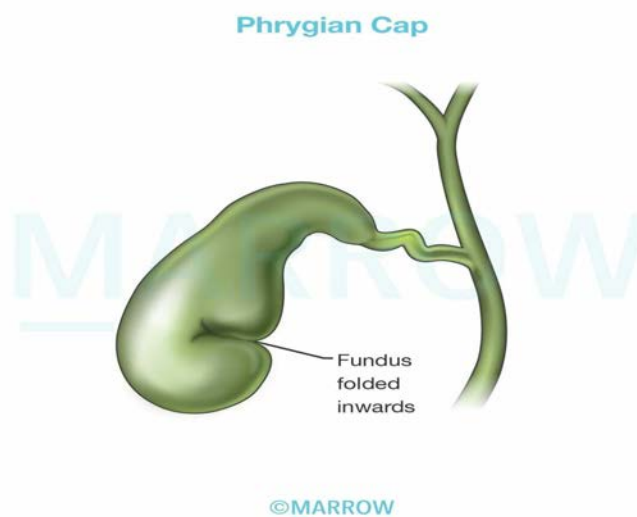
- Ultrasonography of the neck is a safe and widely used technique for the localization of abnormal parathyroid glands.
- Nuclear medicine scanning with radiolabeled Sestamibi is also a widely used technique. Sestamibi accumulates in parathyroid adenomas. This radionuclide is concentrated in the thyroid and parathyroid tissue but usually washes out of normal thyroid tissue in under an hour. It persists in abnormal parathyroid tissue.
- Dual-energy radiographic absorptiometry is a useful tool to demonstrate skeletal involvement in primary hyperparathyroidism.

Solution to Question 2:

The ultrasound examination of the gallbladder shows a Phrygian cap. It is a normal anatomical variant.

Phrygian cap refers to an anatomical variant of the gallbladder, where, the fundus of the gallbladder folds on itself, or is separated from the body by a partial or complete septum. A true septum is uncommon and is usually an apparent septum due to the gallbladder wall folding. It is an incidental finding that may be seen on CT, MRI, or ultrasound. It causes no symptoms.

The Phrygian cap is named so, due to the resemblance of the ultrasonographic appearance to a hat or cap that is worn by the inhabitants of ancient Phrygia circa 1200 BC.



Solution to Question 3:

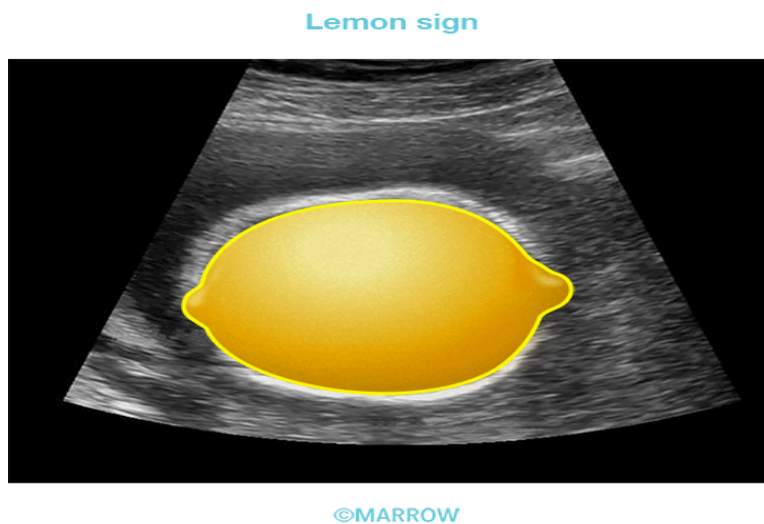
The banana sign on USG is seen in spina bifida.

Banana sign is an abnormal anterior curvature of the cerebellum along with effacement of the cisterna magna that produces this appearance. It is seen in spina bifida and is also associated with Arnold-Chiari malformation.

Spina bifida is a congenital anomaly occurring due to incomplete development of the neural tube. The most commonly cited signs present in spina bifida are the banana sign and the lemon sign.



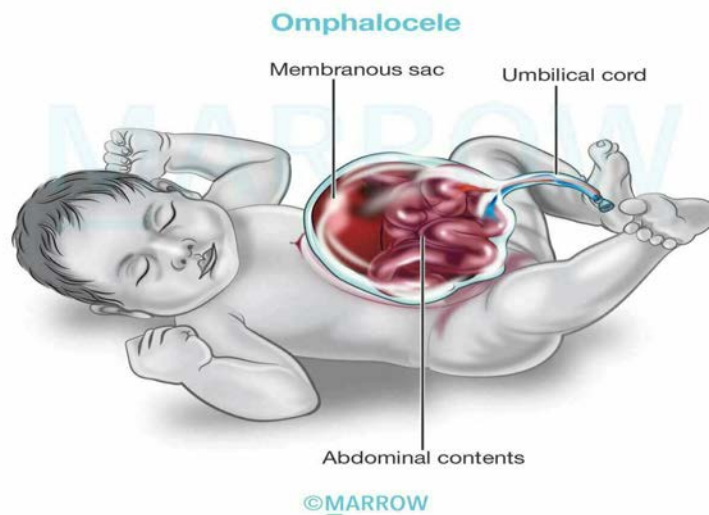
The lemon sign occurs due to the overlapping of the frontal bones due to the posterior shift of intracranial contents as shown below.



Other options:

Option B: Omphalocele is a congenital midline anterior abdominal wall defect through which the solid abdominal viscera or bowel herniates. The herniated abdominal contents are covered by a

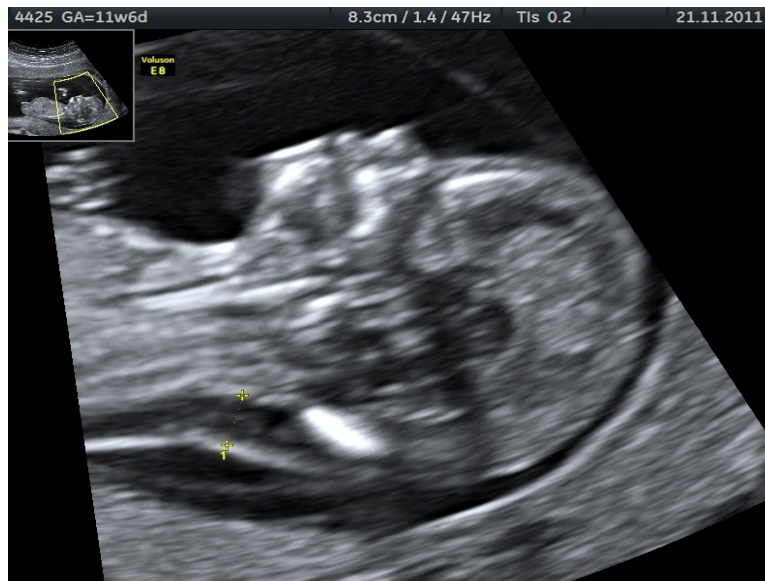
membranous sac as shown below.



The image below shows an ultrasound image of the fetus suggestive of an omphalocele (asterisk).



Option C: Nuchal translucency (NT) is used to antenatally screen Down's syndrome (Trisomy 21) during the first trimester. NT is the maximum thickness of the translucent area between the skin and soft tissue over the fetal spine at the back of the neck. When the NT measurement is increased, it is a marker for fetal aneuploidy and structural anomalies. The ultrasound image below shows increased NT thickness and absent nasal bone of a fetus with Trisomy 21 at 11 weeks gestation.

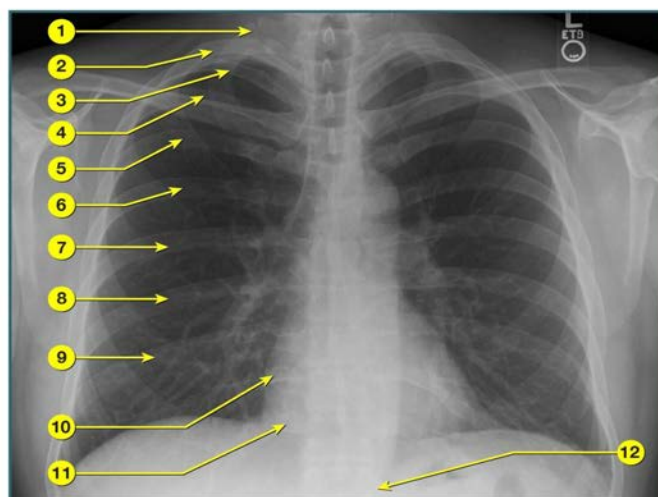


Solution to Question 4:

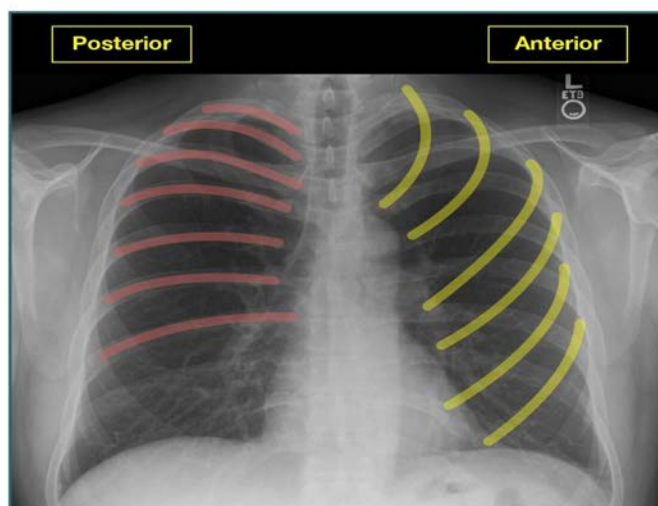
The given X-ray shows the fracture of the 6th and 7th rib

In the chest X-ray, ribs appear as light grey strips. The anterior part of the rib is more oblique and the medial part cannot be demonstrable as it ends with costal cartilage. The posterior part of the rib is more horizontal and the attachment of the rib with the vertebra can be easily made out.

Normal Chest X-ray Rib Identification



The anterior(yellow) and posterior(red) ribs are highlighted in the image given below.



Solution to Question 5:

USG cannot be used for screening mammographically undetectable clinically nonpalpable breast mass. Although ultrasound has its advantages in certain scenarios, mammography remains the primary screening tool for breast tumor detection.

Mammography is complementary to physical examination and can detect clinically non-palpable breast mass. The primary purpose of mammography is to detect small breast tumors. While mammography is typically the initial screening tool for breast cancer detection, there are specific scenarios where ultrasound may be preferred or used in conjunction with mammography.

Advantages of USG over mammography include:

- USG is preferred in pregnant women, lactating women, and young women since breast tissue is dense (Option B) in these populations.
- USG can differentiate between solid and cystic lesions (Option C) of the breast and is able to detect abnormal axillary lymph nodes.
- Ultrasound can assist in guided breast biopsy (Option D) procedures.

Solution to Question 6:

The given MRCP image shows multiple round filling defects within the common bile duct indicative of choledocholithiasis.

Choledocholithiasis or common bile duct stones are classified based on their point of origin. Choledocholithiasis is mostly clinically silent but may present with biliary colic and even obstructive jaundice. Management of choledocholithiasis includes ERCP +/- sphincterotomy followed by laparoscopic cholecystectomy.

MRCP uses the fluid which is present in the biliary and pancreatic ducts as a contrast agent and does not require external contrast material administered into the ductal system. It is a non-invasive investigation for evaluating extrahepatic, intrahepatic bile duct, and pancreatic duct. No interventions are performed under MRCP.

MRCP uses T2 weighted MRI and it recognizes slow-flowing or stationary fluid within ducts, which appear brighter relative to adjacent solid tissues. MRCP can be used to evaluate various conditions of the pancreaticobiliary ductal system, examples include:

- Congenital anomalies of the cystic and hepatic ducts
- Post-surgical biliary anatomy
- Pancreas divisum
- Anomalous pancreaticobiliary junction
- Choledocholithiasis
- Biliary structures
- Chronic pancreatitis
- Pancreatic cystic lesions
- Trauma to the biliary system

MRCP in benign strictures would typically show smooth elongated tapering of the bile duct, while malignancy-associated strictures typically show abrupt shouldering.

Solution to Question 7:

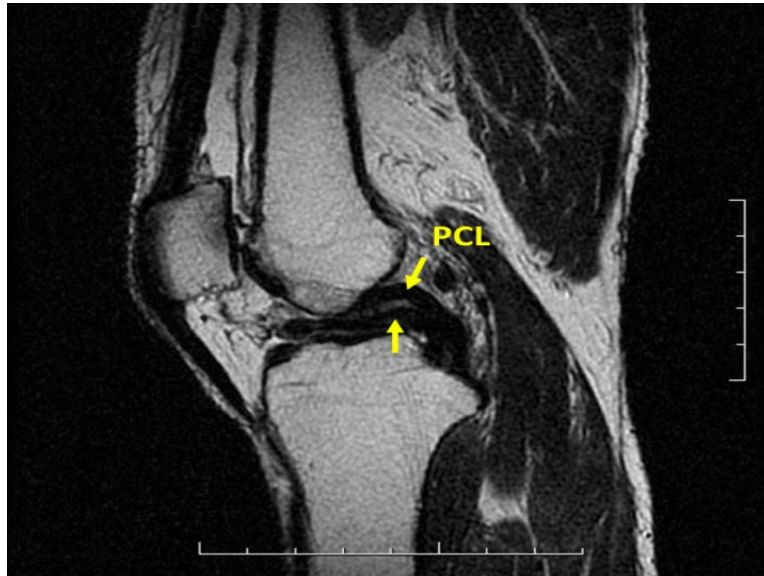
In the above given image of an MRI of the knee joint, the arrow shows the posterior cruciate ligament.

The posterior cruciate ligament (PCL) is the only primary restraint to posterior translation throughout the entire knee range of motion. It prevents the internal rotation of the tibia on the femur when there is varus stress on the knee.

It acts as a secondary restraint to valgus/varus, external rotation.

Anterior cruciate ligament	Posterior cruciate ligament
Attached to the anterior part of the intercondylar area of the tibia	Posterior part of an intercondylar area of the tibia
Direction – Upwards, backwards & laterally & attached to posterior aspect of medial surface of lateral condyle.	Upwards, forwards & medially & attached to the anterior aspect of the lateral surface of the medial condyle.
Prevents posterior dislocation of femur on tibia and forwards dislocation of tibia on femur.	Prevents anterior dislocation of femur on tibia and posterior dislocation of tibia on femur

Anterior cruciate ligament	Posterior cruciate ligament
Limits hyperextension of knee joint (taut during extension)	Limits hyperflexion of knee joint. (taut during flexion)
Weaker	Stronger



Solution to Question 8:

The presence of a halo sign or double ring sign is indicative of the foreign body being a button battery. Furthermore, the presence of irregularities along the periphery of the battery indicates the possibility of corrosion.

Button batteries and sharp objects in the oesophagus and foreign bodies associated with respiratory symptoms all mandate urgent endoscopic removal.

Button batteries, in particular, are very dangerous foreign bodies, which must be expediently removed because they can cause:

- Mucosal injury in 1-hour contact time
- Involve all layers of the oesophagus in 4 hours

Injury is thought to be secondary to electrochemical burns from electrical discharge. Chemical burns from electrolyte alkali leakage and pressure necrosis also may contribute to mucosal injury.

Asymptomatic blunt objects and coins lodged in the oesophagus can be observed for up to 24 hr in anticipation of passage into the stomach.

Another AP and lateral X-ray demonstrating the presence of a button battery in the oesophagus is as shown below.



Radiology INI-CET 2024

Question 1:

Which of the following statements is correct about CT scan?

- a) CT score of water is 0
- b) CT consists of unfiltered x-ray beams
- c) It is the investigation of choice for gallstones
- d) 20mSV for NCCT head

Question 2:

'Mount Fuji sign' is seen in _____

- a) Tension pneumomediastinum
- b) Tension pneumothorax
- c) Tension pneumoperitonium
- d) Tension pneumocephalus

Question 3:

A patient presented with a stab wound over the right chest. The supine X-ray image of the patient is given below. What is the diagnosis?



- a) Right pneumothorax
- b) Pulmonary contusion
- c) Right hemothorax
- d) Right diaphragmatic rupture

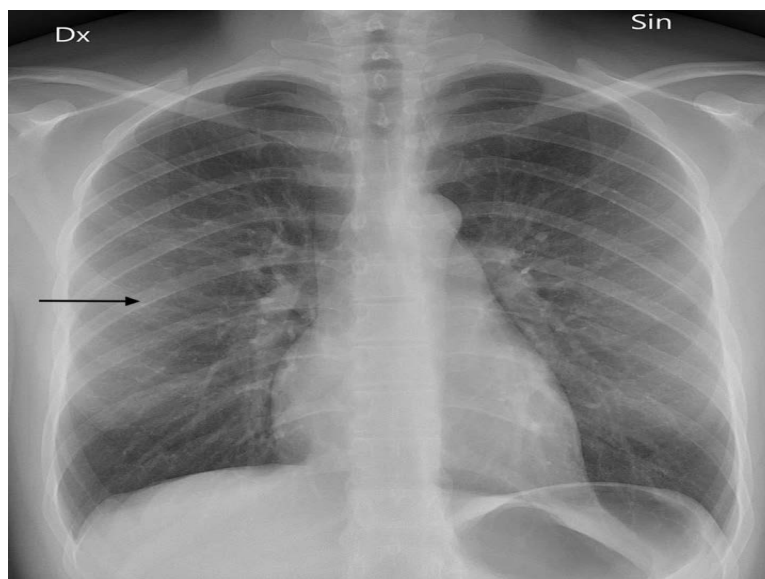
Question 4:

Which of the following contrast agents is safe to use in renal failure patients?

- a) Gadolinium contrast used in MRI
- b) Iodinated ionic contrast agent
- c) Iodinated non-ionic contrast agent
- d) Ultrasound contrast agent

Question 5:

Which of the following structures is the arrow pointing to?



- a) Anterior end of rib
- b) Costal cartilage
- c) Posterior end of rib
- d) Right pulmonary artery

Question 6:

Which of the following radiological interventions are used in the management of cerebral aneurysm?

- a) Coiling
- b) Sclerotherapy
- c) Stenting
- d) Clipping

Answer Key

Question No.	Correct Option
1	a
2	d
3	c
4	d
5	c
6	a

Detailed Explanations

Solution to Question 1:

The correct statement is that the CT score of water is 0.

On CT, each image consists of a matrix of data points (pixels) that are assigned a number that represents the X-ray attenuation(μ) at that point in the body. CT number is a scaled version of μ that is normalised to the X-ray attenuation of water (μ_W).

$$CT = 1000 \times (\mu/\mu_W - 1)$$

The resulting numbers are called 'Hounsfield units'(HU). Due to this normalisation, water is assigned 0 HU, and air is assigned a CT number of -1000 HU.

CT numbers:

- Water = 0 HU
- Solid soft tissues are centred around 0 HU
- Lungs = -1000 HU

- Air = -1000 HU
- Dense cortical bone = +1000 HU
- Acute Blood Clot = 60-90 HU
- Fat = -100 HU

Other options:

Option B: Computed tomography (CT) imaging involves the use of a narrow beam of X-rays that are emitted from a rotating X-ray tube. CT scanners typically use a combination of X-ray beams at different energies and filters to reduce scatter and improve image quality. Therefore, the X-ray beams used in CT imaging are often filtered to enhance image contrast and reduce radiation dose to the patient.

Option C: The investigation of choice for gallstones is an Abdominal ultrasound and not a CT Abdomen. An ultrasound can assess for complications of gallstones, such as inflammation of the gallbladder (cholecystitis) or obstruction of the bile ducts. In cases where ultrasound findings are inconclusive or further characterisation is needed, other imaging modalities such as CT scan or magnetic resonance cholangiopancreatography (MRCP) may be used.

Option D: The mSv (millisievert) value for a non-contrast computed tomography (NCCT) scan of the head typically ranges from around 1 to 2 mSv. This radiation dose can vary depending on factors such as the specific CT scanner used, the scanning protocol, and the individual patient characteristics. However, in general, NCCT head scans are associated with relatively low radiation doses compared to other medical imaging procedures.

Solution to Question 2:

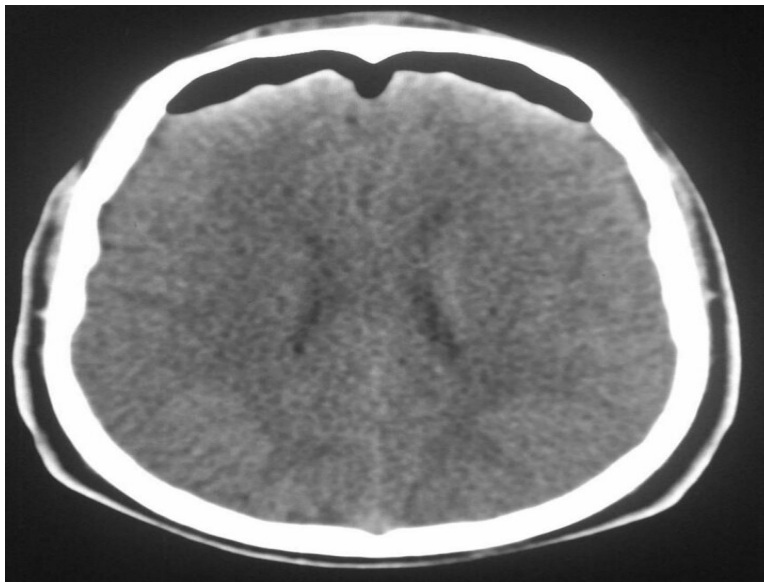
'Mount Fuji sign' is seen in tension pneumocephalus.

Mount Fuji sign refers to the presence of air (pneumocephalus) between the tips of the frontal lobes giving the appearance of Mount Fuji.

Tension pneumocephalus occurs with subdural air causing a mass effect over the underlying brain parenchyma, often from a ball valve mechanism causing one-way entry of air into the subdural space.

The appearance of this sign often warrants immediate surgery to prevent permanent damage.

The CT image below shows the Mount Fuji sign.



Solution to Question 3:

The history of stab injury to the right chest and the X-ray showing opacity of the right hemithorax, with a slight tracheal deviation towards the opposite side, indicates a diagnosis of right hemothorax.

In a plain chest radiograph, it's difficult to differentiate an acute hemothorax from other collections of pleural fluid as they are indistinguishable. As blood clots, it often leads to localisation, occasionally forming a fibrin body. Pleural thickening and calcification are acknowledged as potential consequences.

The following are the findings commonly seen in a case of hemothorax:

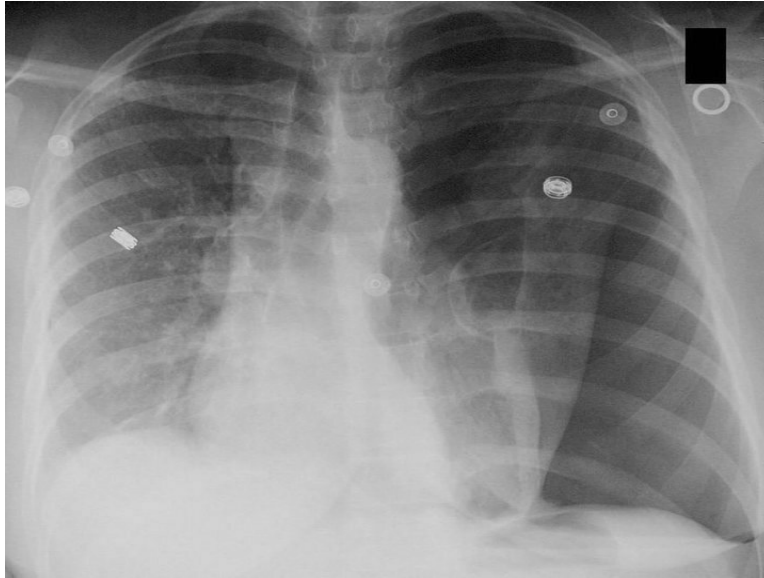
- **Blunting of Costophrenic Angle:** Hemothorax results in the accumulation of blood in the pleural cavity, causing the lung to collapse partially. This can lead to the blunting or obliteration of the costophrenic angle on the affected side.
- **Increased Opacity:** The presence of blood in the pleural space leads to increased opacity or density in the affected hemithorax. This can manifest as a homogenous shadowing that obscures the lung markings.
- **Mediastinal Shift:** Depending on the volume of blood in the pleural cavity, there may be a mediastinal shift towards the opposite side. However, in mild to moderate hemothorax, this may not be readily apparent on supine X-rays.
- **Air-Fluid Level:** In cases where there is concurrent pneumothorax (air in the pleural cavity), an air-fluid level may be visible, with the blood layering dependently and the air layer superiorly.
- **Unilateral Findings:** The abnormalities will typically be limited to one side of the chest, corresponding to the side of the hemothorax.

In cases of suspected hemothorax, a lateral decubitus or chest X-ray in erect position may be more sensitive for detection. Additionally, other imaging modalities such as ultrasound or computed tomography (CT) scans can provide more detailed information about the presence and extent of a

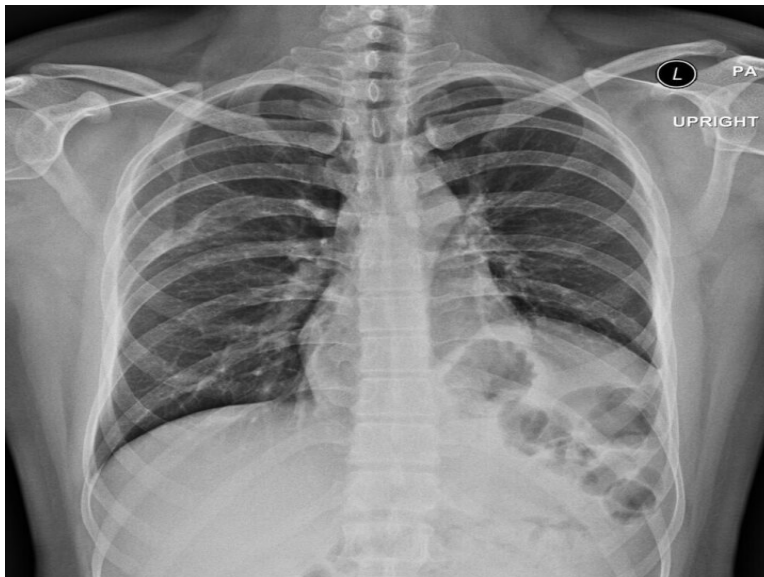
hemothorax.

Other options:

The following image is characteristic of a pneumothorax (left-sided) with radiolucency over the left lung, lung collapse, and mediastinal shift on x-ray.



The given radiograph is a case of diaphragmatic hernia that shows elevation of the left hemidiaphragmatic line along with the bowel loops and stomach seen in the pleural cavity.



The following is an image of a right-sided pulmonary contusion post-blunt chest trauma. Note the increased opacity just below the horizontal fissure.



Solution to Question 4:

Ultrasound contrast agents are gas-filled microbubbles that are administered intravenously to the systemic circulation. Hence, considered as low risk in patients with renal impairment.

Ultrasound contrast agents are used to enhance the visibility of blood vessels, organs, and tissues during ultrasound imaging. These agents typically consist of tiny gas-filled microbubbles encapsulated within a shell. When injected intravenously, they flow through the bloodstream and resonate in response to ultrasound waves, producing strong echoes that enhance the ultrasound image.

Perflutren/Optison is an ultrasound contrast agent. Levovist (consisting of galactose microparticles, air bubbles, and palmitic acid) is another example of an ultrasound contrast agent. Sulfur hexafluoride with a phospholipid shell is also used as an ultrasound contrast agent.

Other options:

Option A: Gadolinium contrast used in MRI is predominantly excreted through the kidney. In renal failure, it gets accumulated in the body and releases gadolinium ions. Since the ion is toxic, the body tries to contain it by fibrosis resulting in multisystem painful fibrosis called nephrogenic systemic fibrosis.

Option B: Iodinated ionic contrast agents increase the risk of Contrast-induced nephropathy(CIN). Hence alternative imaging methods, without iodinated contrast media, should be chosen in patients with considerably impaired renal function (CKD 4 or 5).

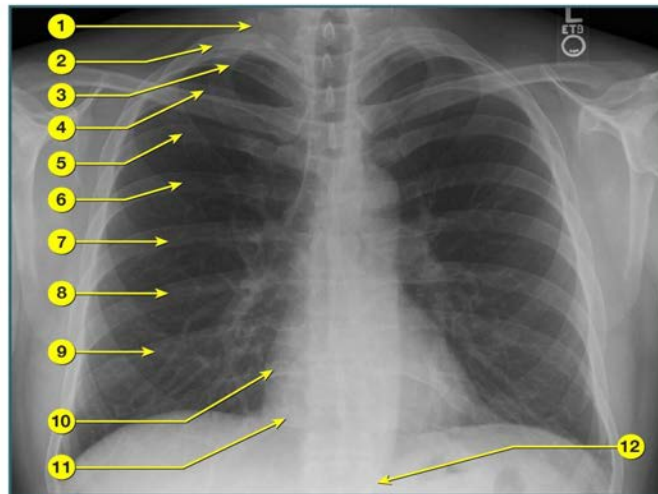
Option C: Non-ionic contrast agents have a lower risk of causing adverse reactions compared to ionic contrast agents. They are generally better tolerated by patients, with fewer incidences of allergic reactions and nephrotoxicity. However, USG contrast agents are among the safest in renal failure patients.

Solution to Question 5:

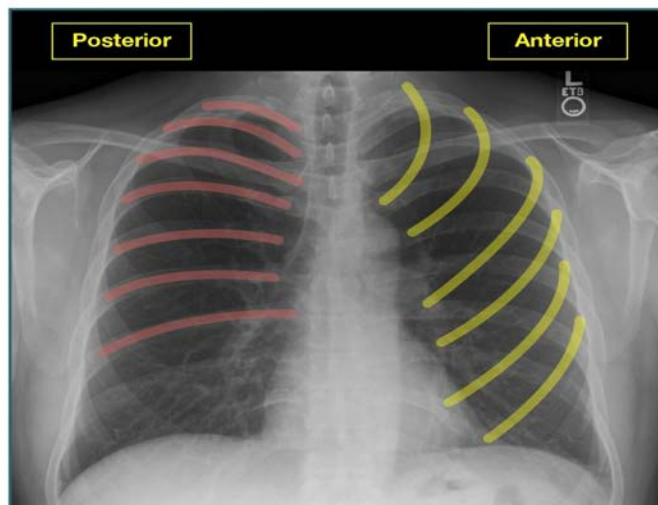
The structure pointed by the arrow mark in the given chest X-ray PA view is the posterior end of the rib.

In the chest X-ray PA view, ribs appear as light grey strips. The anterior part of the rib is more oblique and the medial part cannot be demonstrable as it ends with costal cartilage. The posterior part of the rib is more horizontal, hyperdense and the attachment of the rib with the vertebra can be easily made out.

Normal Chest X-ray Rib Identification



The anterior(yellow) and posterior(red) ribs are highlighted in the image given below.



Solution to Question 6:

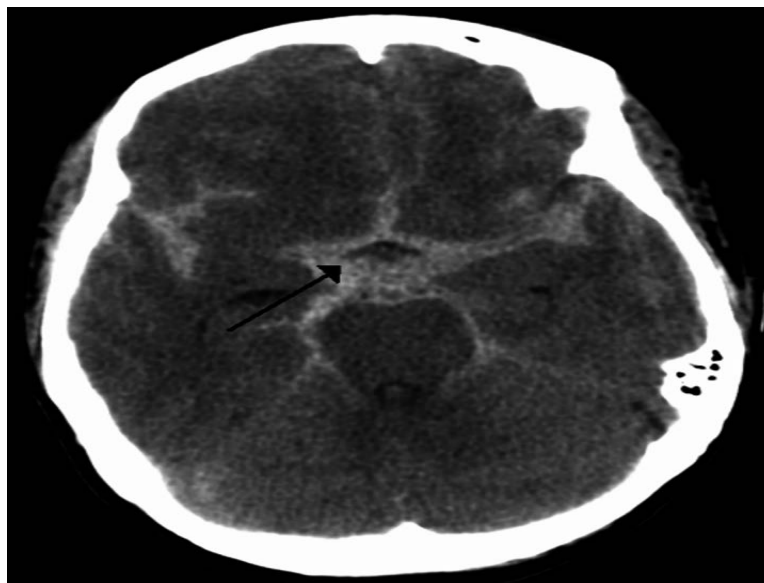
Coiling is the radiological intervention used in the management of cerebral aneurysm.

Aneurysms occur at the bifurcations of the large to medium-sized intracranial arteries. Rupture is into the subarachnoid space in the basal cisterns. Approximately 85% of the aneurysms occur in the anterior circulation, mainly on the circle of Willis.

Most unruptured intracranial aneurysms are completely asymptomatic. When the aneurysm ruptures, 45% of cases present with sudden onset severe headache, described as the worst headache of the patient's life. Other features include neck rigidity and transient loss of consciousness.

Diagnosis -

CT shows hyperattenuating material filling the subarachnoid space. CT angiography has a high sensitivity for aneurysms and AVMs, but digital subtraction angiography remains the gold standard.



Treatment -

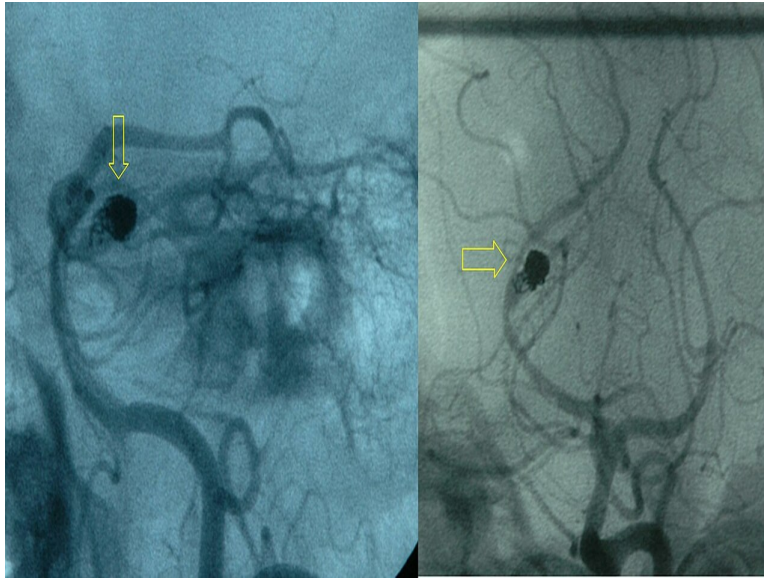
Early aneurysm repair prevents re-rupture. At many centers, definitive repair is carried out within 24 h of the bleed in all patients who are stable enough to tolerate the procedure. An aneurysm can be “clipped” by a neurosurgeon or “coiled” by an endovascular approach. Surgical repair involves placing a metal clip across the aneurysm neck, thereby immediately eliminating the risk of rebleeding. This approach requires craniotomy (option D) and brain retraction, which is associated with neurologic morbidity.

Endovascular techniques involve placing platinum coils within the aneurysm via a catheter that is passed from the femoral artery. The aneurysm is packed tightly to enhance thrombosis and over time is walled off from the circulation.

Intracranial pressure should be measured and the cerebral perfusion pressure targeted at 60–70 mmHg.

Delayed cerebral ischemia remains the leading cause of morbidity and mortality following aneurysmal SAH. Treatment with the calcium channel antagonist nimodipine (60 mg PO every 4 h) improves outcome, perhaps by preventing ischemic injury rather than reducing the risk of vasospasm.

The image below is an arteriogram showing a coiled aneurysm (indicated by yellow arrows) of the posterior cerebral artery with a residual aneurysmal sac.



Other options :

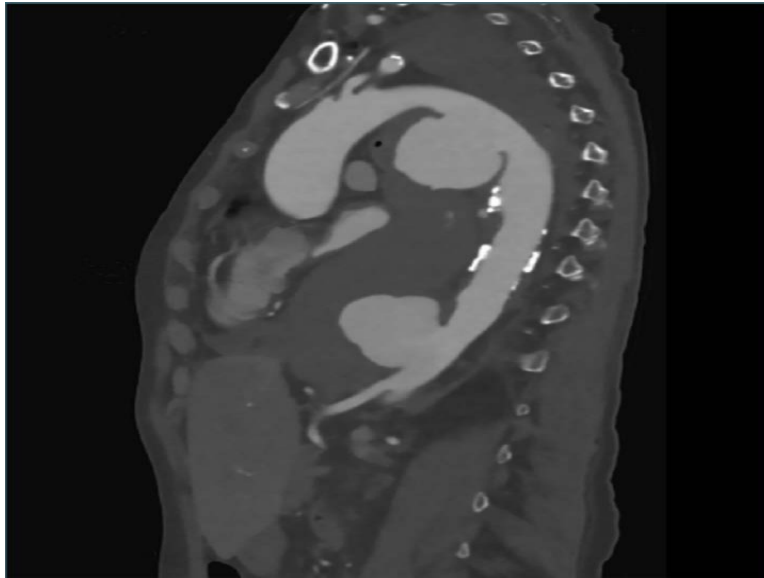
Option B - Sclerotherapy is used in the management of venous insufficiency.

Option D - Stenting is employed mainly for stenotic vascular lesions.

Radiology NEET 2024

Question 1:

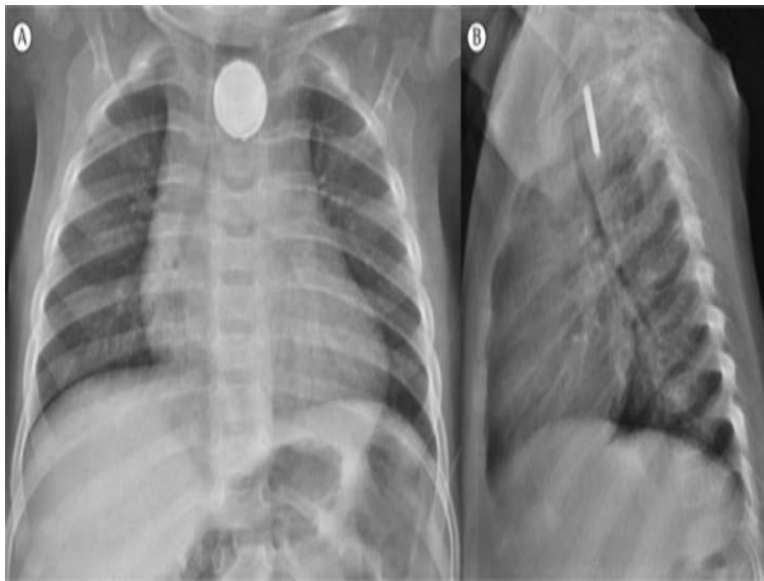
A patient presented with chest pain and hoarseness for one month, and on a CT angiogram, the following image is seen, what is the possible diagnosis?



- a) Stenosis of aortic arch
- b) Aortic dissection
- c) Saccular aneurysm of distal aorta
- d) Coarctation of aorta

Question 2:

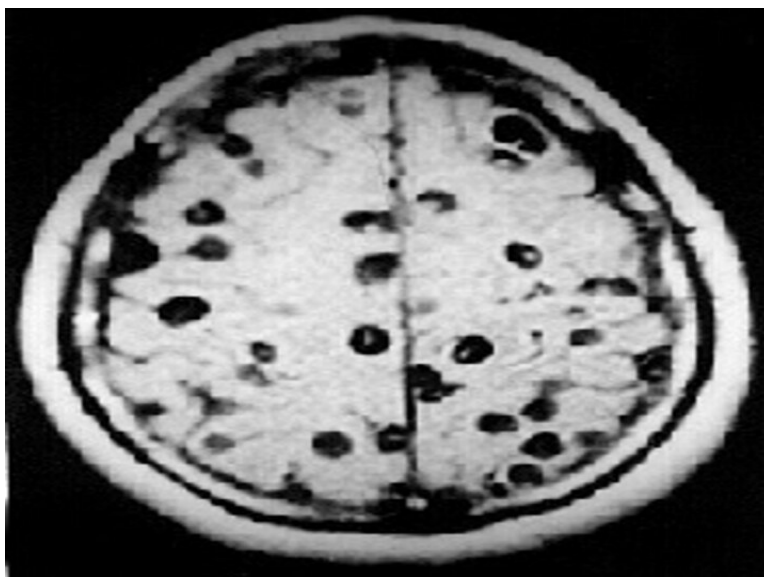
This is the radiograph of a child with a foreign body in the oesophagus. What is the sign seen in this?



- a) Thumb sign
- b) Steeple sign
- c) Double ring sign
- d) String sign

Question 3:

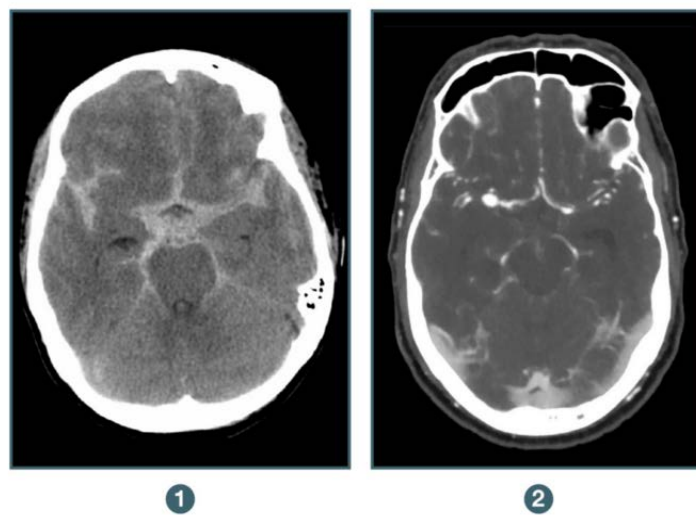
A patient presents with a one-month history of lethargy, headache, fever, and vomiting. Shortly after being admitted to the hospital, the patient tragically passes away. Radiological imaging reveals multiple cystic lesions in the cerebrum with invasion into the ventricles and subarachnoid space. What is the likely cause?



- a) Neurocysticercosis
- b) Astrocytoma
- c) AV malformation
- d) Brain metastasis

Question 4:

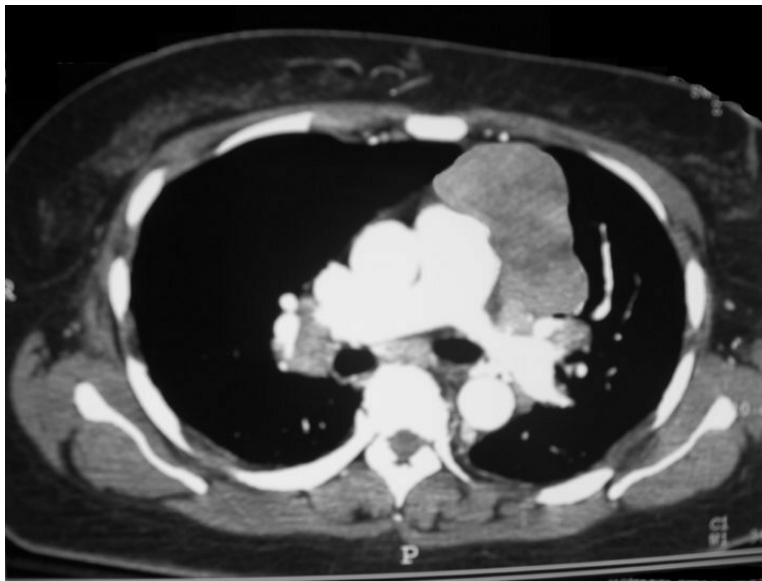
A patient complains of severe headache. NCCT and CT angiography images are as below. What is the likely etiology?



- a) AV malformation
- b) Subarachnoid hemorrhage
- c) Intraparenchymal hemorrhage
- d) Subdural hemorrhage

Question 5:

An elderly woman with a known history of Myasthenia Gravis presents with symptoms including fatigue, facial congestion, dilated neck veins, and breathlessness, which are indicative of superior vena cava (SVC) obstruction. What is the likely diagnosis?



- a) Lipoma
- b) Ectopic thyroid
- c) Thymoma
- d) Germ cell tumor

Question 6:

A 4-week-old infant presented to the ER with projectile vomiting. USG abdomen findings of the infant are given below. Probable diagnosis is _____

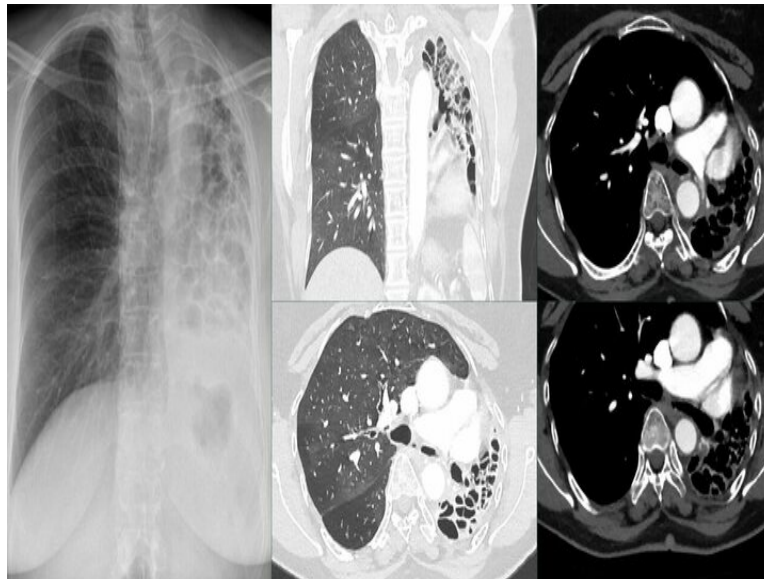


- a) Sigmoid volvulus

- b) Duodenal atresia
- c) Congenital hypertrophic pyloric stenosis
- d) Intussusception

Question 7:

An infant presents with respiratory distress. He has a previous history of recurrent respiratory infections. What is the diagnosis?



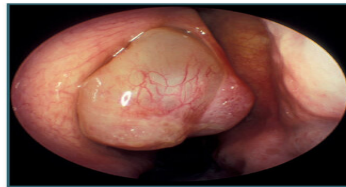
- a) Lung abscess
- b) Congenital malformations of the airway
- c) Congenital diaphragmatic hernia
- d) Pyopneumothorax

Question 8:

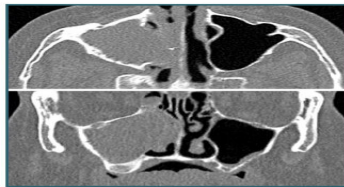
A young patient presents with nasal polyps and rhinosinusitis. X-ray and clinical images are as below. What is the diagnosis?



1



2



3

- a) Young's syndrome
- b) Kartagener's syndrome
- c) Churg Strauss syndrome
- d) Cystic fibrosis

Question 9:

Given below is a USG taken at approximately 8 weeks of gestation. Which anomaly can be observed here?

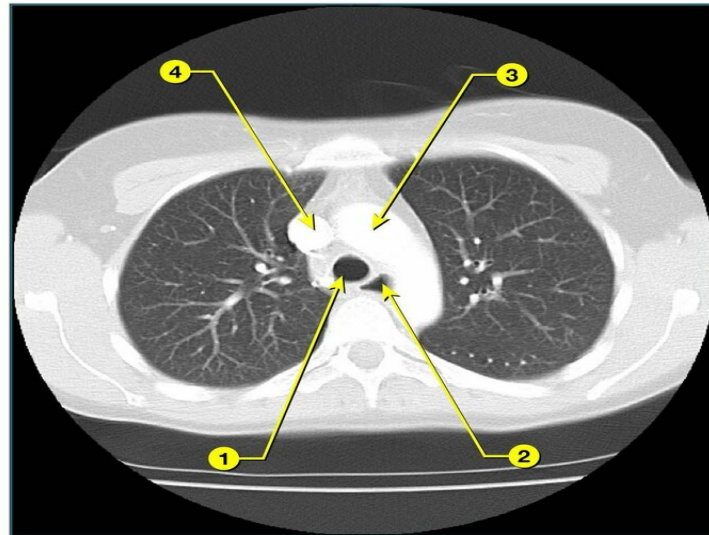


- a) Anencephaly

- b) Hydranencephaly
- c) Meningocele
- d) Encephalocele

Question 10:

Match the following using the radiological image given below:



- a) 3-A, 2-B, 1-C, 4-D
- b) 2-A, 3-B, 4-C, 1-D
- c) 1-B, 2-C, 3-A, 4-D
- d) 2-A, 1-B, 3-C, 4-D

Question 11:

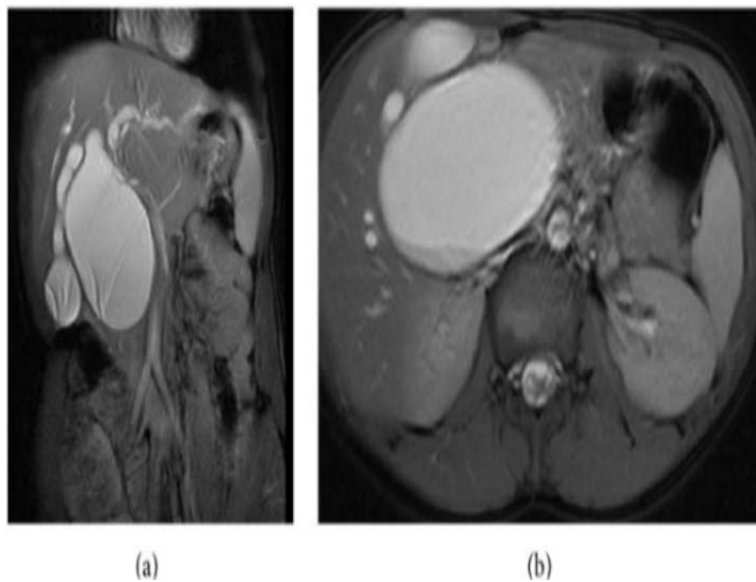
Identify the diagnosis from the given CT scan.



- a) PUJ obstruction
- b) Renal cyst
- c) Renal tumour
- d) Multicystic kidney

Question 12:

What does this image below indicate?

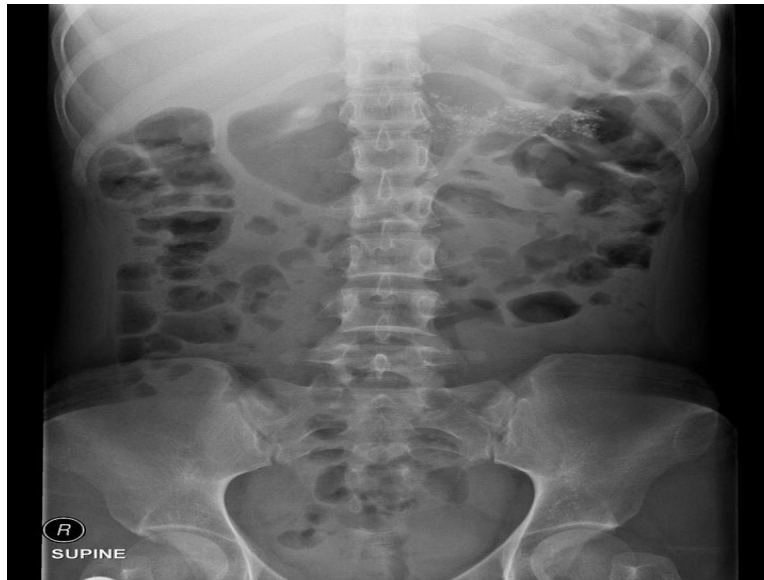


- a) Choledochal cyst
- b) Gall bladder carcinoma

- c) Cholangiocarcinoma
- d) Choledocholithiasis

Question 13:

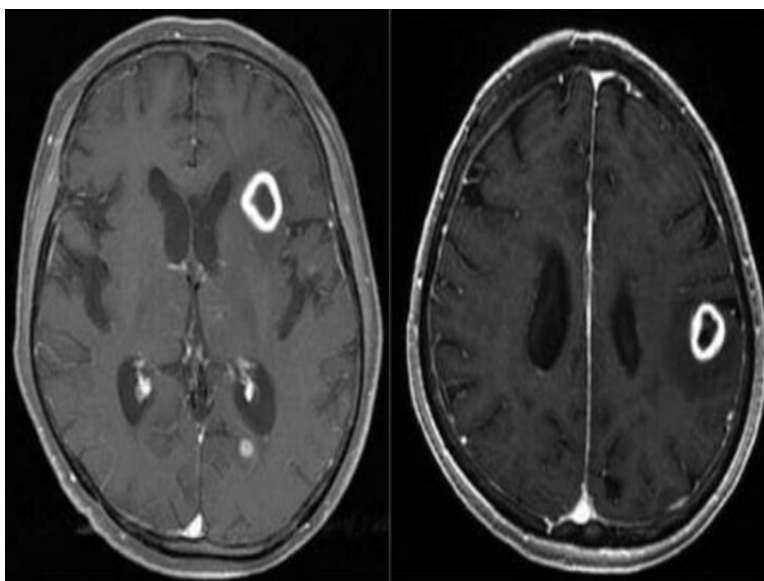
Identify the condition shown in the image:



- a) Chronic calcific pancreatitis
- b) Mesenteric calcification
- c) Horseshoe kidney
- d) Jejunal fecolith

Question 14:

A 6-year-old child presented with seizures which subsided with injection Midazolam. Blood investigations and lumbar puncture studies were normal. MRI is done and the image is given below. What is the next best investigation of choice for further assessment?



- a) MR spectroscopy
- b) Ocular B scan
- c) CBNAAT of CSF
- d) PET scan

Answer Key

Question No.	Correct Option
1	c
2	c
3	a
4	b
5	c
6	c
7	b
8	b
9	a
10	c
11	a
12	a
13	a
14	a

Detailed Explanations

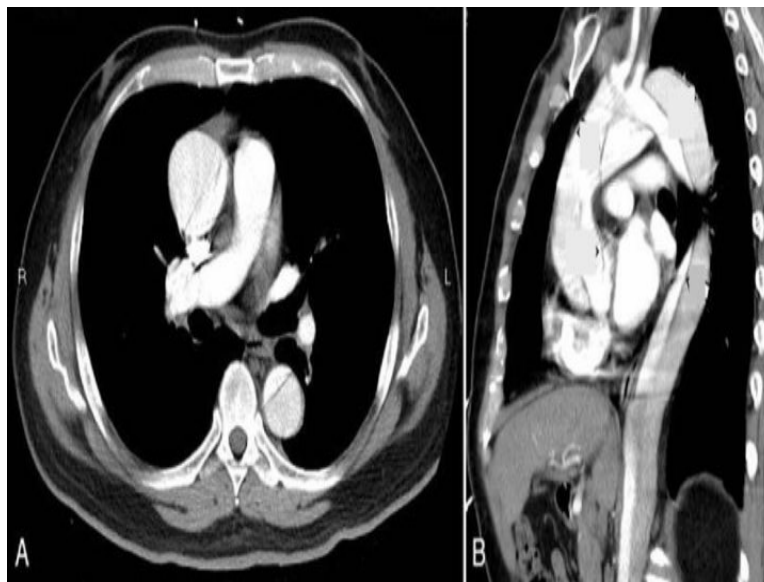
Solution to Question 1:

The given CT shows saccular aneurysm of the distal aorta. The clinical presentation of chest pain and hoarseness supports the diagnosis.

Other options:

Option A: Stenosis of the aortic arch would show a narrowing rather than an aneurysmal dilation, and it is more likely to present with symptoms related to decreased blood flow to the upper body.

Option B: Aortic dissection would usually present with a tearing chest pain that radiates to the back and would show a double lumen or false lumen on imaging. The below image shows Stanford type A aortic dissection:



Option D: Coarctation of the aorta typically presents with hypertension in the upper limbs and weak or absent pulses in the lower limbs.

Solution to Question 2:

The presence of a halo sign or double ring sign is indicative of the foreign body being a button battery.

Button batteries and sharp objects in the oesophagus and foreign bodies associated with respiratory symptoms all mandate urgent endoscopic removal.

Button batteries, in particular, are very dangerous foreign bodies, which must be expediently removed because they can cause:

- Mucosal injury in 1-hour contact time
- Involve all layers of the oesophagus in 4 hours

Injury is thought to be secondary to electrochemical burns from electrical discharge. Chemical burns from electrolyte alkali leakage and pressure necrosis also may contribute to mucosal injury.

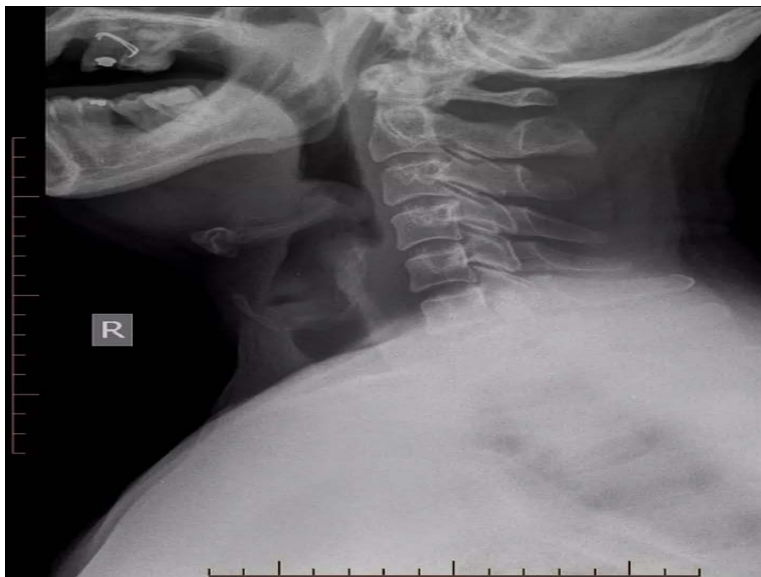
Asymptomatic blunt objects and coins lodged in the oesophagus can be observed for up to 24 hr in anticipation of passage into the stomach.

Another AP and lateral X-ray demonstrating the presence of a button battery in the oesophagus is as shown below.

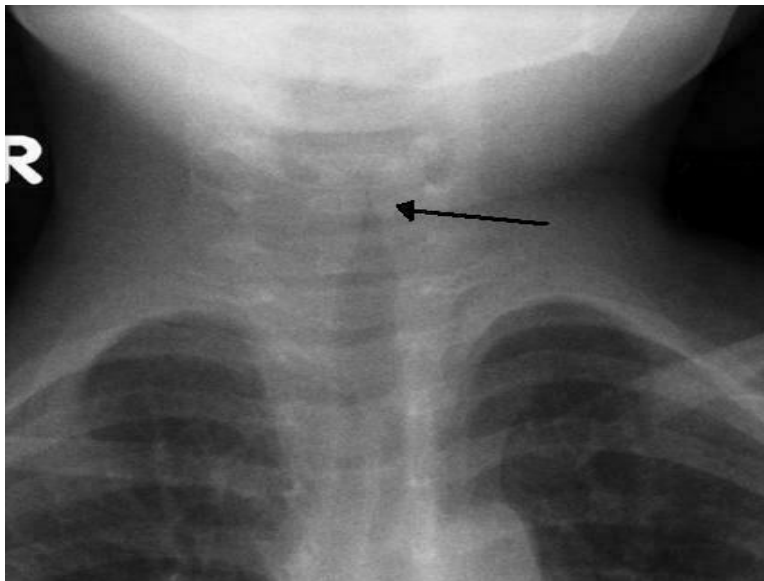


Other options

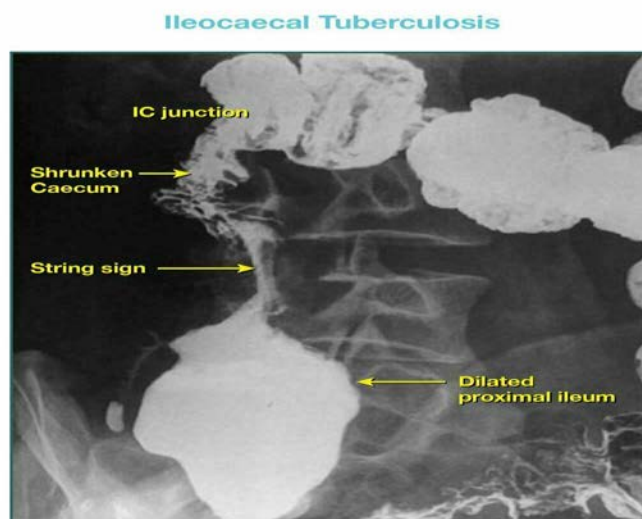
Option A: Thumb sign is seen in acute epiglottitis.



Option B: A steeple sign is seen in croup or laryngotracheobronchitis.



Option D: String sign on the barium enema is in ileocecal tuberculosis.



The image below shows the string sign of Kantor associated with Crohn's disease.



Solution to Question 3:

The given clinical scenario along with CT showing cystic lesions in the brain, particularly within the cerebrum, ventricles, and subarachnoid space is clearly suggestive of neurocysticercosis.

Neurocysticercosis is a form of cysticercosis caused by *Taenia solium*.

It usually presents with seizures and symptoms of increased intracranial pressure, including headache, nausea, vomiting, vision changes, dizziness, ataxia, or confusion. Papilledema may develop secondary to hydrocephalus.

Diagnosis is done by demonstrating eggs of *Taenia* in faecal examination.

A biopsy of the brain or spinal cord lesion or fundoscopic visualization of subretinal cysticercus can also be done.

Imaging shows a scolex within a cystic lesion, cystic lesions with or without enhancement (e.g., ring enhancement), nodular calcifications, focal enhancing lesions, or multilobulated cystic lesions in the subarachnoid space.

Treatment of intestinal *T. solium* infection is by a single dose of praziquantel. Neurocysticercosis is treated with anti-epileptic drugs, anticysticidal drugs like albendazole or praziquantel, and steroids given before anticysticidal therapy.

Other options:

Option B: Astrocytoma is a type of glial cell tumour in the brain, ranging from low to high grade. They do not typically present as multiple cystic lesions and are more likely to appear as a single mass with variable enhancement on imaging.

Option C: Arteriovenous malformation (AVM) is a congenital vascular anomaly characterized by a tangle of abnormal blood vessels connecting arteries and veins. AVMs are more likely to cause hemorrhage or seizures and typically appear as highly vascular lesions on imaging, not as cystic lesions.

Option D: While metastases can present as multiple lesions on imaging, they usually appear solid rather than cystic masses and often enhance with contrast.

Solution to Question 4:

The clinical manifestations of severe headache with CT revealing blood in the basal cistern point to a diagnosis of subarachnoid haemorrhage (SAH).

Subarachnoid haemorrhage (SAH) is bleeding into the subarachnoid space, which is the area between the arachnoid membrane and the pia mater surrounding the brain. The diagnosis is suspected when a hyperattenuating material is seen filling the subarachnoid space. Most commonly this is apparent around the circle of Willis, on account of the majority of berry aneurysms occurring in this region (65%), or in the Sylvian fissure (30%).

The modified Fisher scale is a method for grading SAH as seen on non-contrast CT (NCCT). The Hunt and Hess scale describes the severity of SAH and is used as a predictor of survival. Xanthochromia on CSF is indicative of SAH in case of a negative CT scan.

Other options:

Option A: Arteriovenous malformation (AVM) is a congenital vascular anomaly characterized by a tangle of abnormal blood vessels connecting arteries and veins. AVMs are more likely to cause haemorrhage or seizures and typically appear as highly vascular lesions on imaging, not as cystic lesions.

Option C: Intraparenchymal hemorrhage results from damage to lenticulostriate arteries. Hyperdensity within the brain parenchyma is seen on CT.

Option D: Acute subdural haemorrhage results from the accumulation of blood between the arachnoid membrane and the dura. The tearing of a bridging vein from the cerebral cortex to the dural sinuses is the most common cause of acute SDH. On head CT scan, acute SDH appears as a hyperdense, crescent-shaped (concavo-convex).

Solution to Question 5:

The diagnosis of myasthenia gravis along with the CT scan image showing an anterior mediastinal mass, is suggestive of thymoma.

Thymoma is the most common mediastinal tumor and is derived from the thymus gland.

Thymomas vary in behaviour from benign to aggressively invasive. Superior vena cava obstruction is a possible complication of mediastinal mass.

Masoaka classification system is used to stage thymomas.

Myasthenia gravis is often related to thymomas. Symptoms of myasthenia gravis can improve after excision of the thymus gland even in the absence of a thymoma.

Complete thymectomy is the treatment for thymoma. It can be performed by two approaches:

- A median sternotomy approach

- A transcervical approach with or without an additional VATS procedure (Video-Assisted Thoracoscopic Surgery) for smaller thymomas

Other options:

Option A: Lipomas are benign and slow-growing, less likely to cause significant compression of the SVC and produce the acute symptoms described.

Option B: Ectopic thyroid tissue can be found in the mediastinum, but it is typically asymptomatic and does not usually cause SVC obstruction.

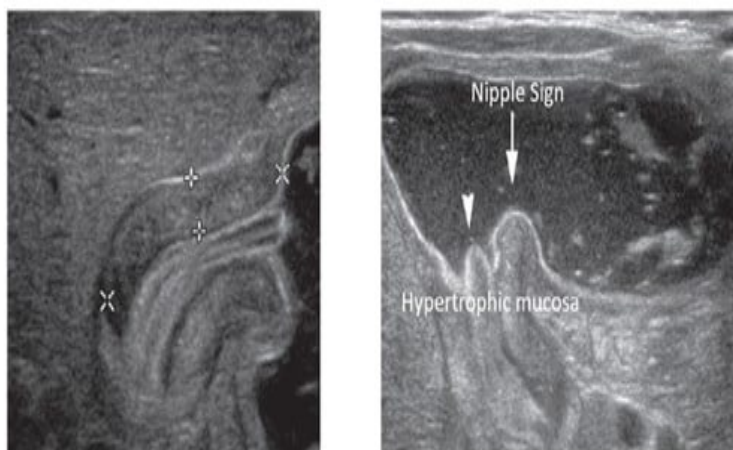
Option D: Germ cell tumours can occur in the mediastinum and cause SVC obstruction, but they are less commonly associated with Myasthenia Gravis compared to thymomas.

Solution to Question 6:

The probable diagnosis is Congenital Hypertrophic Pyloric stenosis since the cervix sign and nipple sign are signs of Congenital Hypertrophic Pyloric stenosis on USG.

- Double-layered hypertrophic mucosa protrudes into the stomach, creating the so-called Nipple sign.
- The sonographic appearance of the hypertrophied pylorus resembles that of the uterine cervix and is sometimes referred to as the cervix sign.

Congenital Hypertrophic Pyloric Stenosis - USG



Hypertrophic pyloric stenosis most presents with non-bilious projectile vomiting. It is common between 2 and 8 weeks of age. Hypertrophy of the circular muscle of the pylorus results in constriction and obstruction of the gastric outlet. Male infants are at a higher risk.

Clinically, an olive-shaped mass is palpable in the right upper quadrant or epigastrium. When the infant is given a test feed, visible peristalsis is usually observed moving from left to right across the abdomen.

Infants with hypertrophic pyloric stenosis develop a characteristic metabolic abnormality-hypochloreaemic hypokalaemic metabolic alkalosis. Persistent vomiting results in loss of

hydrochloric acid leading to the development of metabolic alkalosis (due to loss of H⁺) and hypochloremia (due to loss of Cl⁻).

Ultrasound examination is the investigation of choice for this condition. The diagnostic criteria for pyloric stenosis are a muscle thickness of ≥ 4 mm and a pyloric length of ≥ 16 mm. The cervix sign and nipple sign are signs seen on USG.

As for the management, the child should be rehydrated and the metabolic abnormality has to be corrected before proceeding to surgery. The Ramstedt pyloromyotomy remains the standard surgical procedure for its treatment.

Solution to Question 7:

The given clinical scenario along with the image is suggestive of congenital malformations of the airway.

Congenital pulmonary airway malformation (CPAM) is a condition where more normal lung tissue is mixed with hamartomatous or dysplastic lung tissue, usually confined to one lobe. The lesion most likely originates from a maldevelopment of terminal bronchiolar structures caused by an embryologic injury that occurred prior to the 35th day of gestation.

Pneumothorax, recurrent respiratory infections, and respiratory distress can all manifest in patients during the newborn or early infancy period. One could mistake the lesion for a diaphragmatic hernia. Until mid-childhood, when episodes of recurrent or persistent pulmonary infection or chest pain occur, patients with smaller lesions typically are asymptomatic.

Treatment includes aspiration of macrocystic lesions, excision of the affected lobe for microcystic lesions, and infrequently, open foetal surgery for severely affected infants. For patients with symptoms in the postnatal period, surgery is recommended.

Other options:

Option A: Lung abscess represents necrosis and cavitation of the lung following microbial infection.

Chest X-ray features of lung abscess include a lung cavity with an air-fluid level and a thick and irregular cavity wall with surrounding pulmonary infiltrate. The image showing the lung abscess is given below:



Option C: Congenital diaphragmatic hernia (CDH) is a birth defect of the diaphragm. Malformation of the diaphragm allows the abdominal organs to push into the chest cavity, hindering proper lung formation and resulting in pulmonary hypoplasia.

The most common type of CDH is a Bochdalek hernia, which in most cases involves the left side. CDH is a life-threatening pathology in infants and a major cause of morbidity and mortality due to two complications: pulmonary hypoplasia and pulmonary hypertension. The image of Bochdalek hernia is given below:



Option D: Pyopneumothorax is a pleural collection of pus and gas, also referred to as empyemic hydropneumothorax or infected hydropneumothorax. It could be compared to a thoracic empyema variation that contains gas. Chest pain and fever are typically the patient's initial complaints.

X-ray shows gas-fluid level. It could have loculations. There appears to be a loss of silhouette around the diaphragm dome. The thick pleural lining is strongly indicative of the

pyopneumothorax.

Solution to Question 8:

The given clinical scenario along with an X-ray showing dextrocardia, CT showing maxillary polyp, and endoscopy showing nasal polyp is suggestive of Kartagener's syndrome.

Primary ciliary dyskinesia (also called immotile cilia syndrome) is an autosomal recessive disorder affecting ciliary motility. When a patient has bronchiectasis, chronic sinusitis, and situs inversus triad, he is known to have Kartagener syndrome. It occurs due to the absence of protein dynein in the A tubules of the respiratory mucosa.

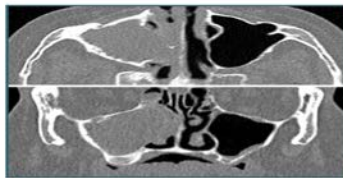
Clinical findings in Kartagener syndrome



X ray showing dextrocardia



FESS showing middle meatus polyp



CT showing maxillary sinus polyp

The disease is usually diagnosed based on the clinical picture and should be evaluated in later stages when a child presents with recurrent lower and upper respiratory tract infections. Other features of primary ciliary dyskinesia include congenital heart diseases and neonatal hydrocephalus. Male patients with this condition tend to be infertile, due to asthenospermia (immotile spermatozoa due to impaired sperm flagella function).

Transmission electron microscopy is the gold standard investigation to analyze cilia. There is no specific treatment for this condition and is managed according to the symptoms.

Other options:

Option A: Churg Strauss syndrome clinically presents with asthma, fever and eosinophilia.

Option C: Young's syndrome clinically presents with infertility (azoospermia) and bronchiectasis.

Option D: Cystic fibrosis is an autosomal recessive condition due to a mutation of the CFTR gene on chromosome 7. Clinical features include recurrent sinopulmonary infection, intestinal obstruction, pancreatic insufficiency, and male infertility. Diagnosis is made by elevated sweat chloride levels, abnormal nasal potential difference, and genetic testing for CFTR mutation.

Solution to Question 9:

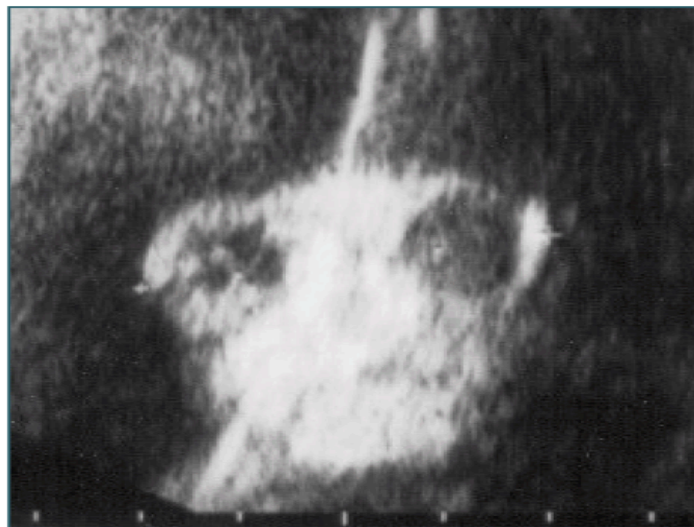
The given ultrasound image shows an open defect in the calvaria and skin. This is consistent with the diagnosis of anencephaly.

Anencephaly results from incomplete closure of the neural tube. It is one of the earliest detectable congenital defects that can be detected as early as 10 to 11 weeks of gestation. Screening can also be done by serum estimation of maternal serum alpha-fetoprotein (MSAFP). The following are the radiological signs seen in anencephaly.

- Frog eye sign - Absence of cranial calvaria, prominent orbits thereby resembling the face of a frog.
- Shower cap sign - The absence of the cranium could lead to the protrusion of a disorganized mass of brain tissue. This resembles a shower cap giving rise to a characteristic triangular facial appearance.

Neural-tube defects can be prevented with pre-conceptional supplementation of folic acid. A dose of 4 mg daily is recommended in subsequent pregnancies.

The frog eye and showercap signs are shown below:



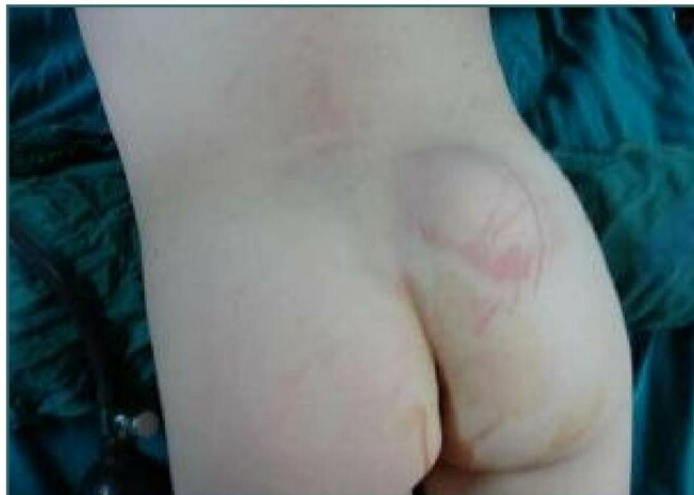


Other options:

Option B: Hydranencephaly: This condition involves the absence of cerebral hemispheres, replaced by fluid-filled sacs.

Option C: Meningocele: It involves herniation of the meninges through a defect in the skull or spinal column, forming a sac filled with cerebrospinal fluid.

The image below shows meningocele:



Option D: Encephalocele is caused due to defects in the skull through which the meninges and/or neural tissue herniate. The squamous part of the occipital bone is the most frequently affected bone.

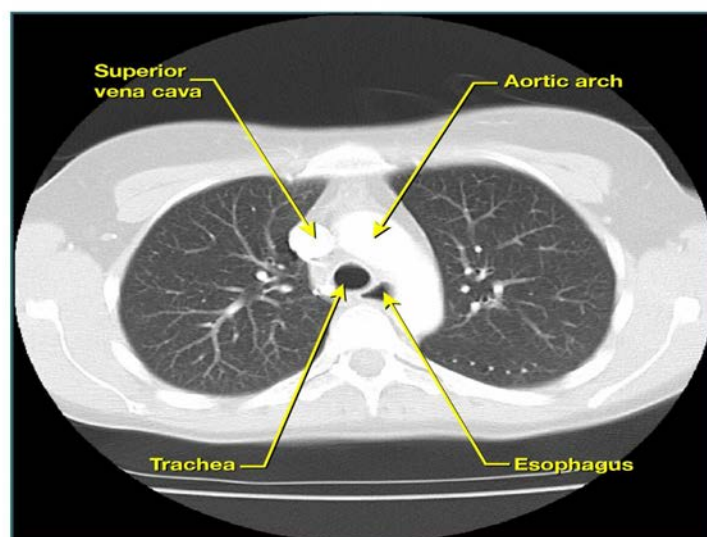
The image shows a newborn with encephalocele.



Solution to Question 10:

Here are the correctly matched structures:

1	B. Trachea
2	C. Esophagus
3	A. Arch of Aorta
4	D. Superior vena cava



Solution to Question 11:

The CT scan showing ballooning of the left renal pelvis with moderate hydronephrosis of the affected kidney indicating a likely diagnosis of ureteropelvic junction obstruction (UPJ) of the left side.

PUJ obstruction refers to a blockage or narrowing at the junction where the renal pelvis transitions into the ureter, leading to impaired drainage of urine from the kidney.

Indications for surgery in patients with UPJ obstruction include:

- On the diuretic renogram, UPJO with less than 40% in the split function of the afflicted kidney.
- Renal parenchymal atrophy in the presence of severe bilateral UPJO.
- Despite preventive medicines, recurring urinary tract infections.
- Obstructive UPJO that is symptomatic or linked with an abdominal mass

Hydronephrosis may be detected on antenatal ultrasound; it can, however, be transient or mild and resolves on its own. Re-evaluation should be done after birth and in cases presenting with symptoms of recurrent UTI, palpable flank mass or pain, urinary obstruction, and in cases of failure to thrive.

If the hydronephrosis is persistent, its severity is graded by the extent of the dilatation of the pelvicalyceal system and parenchymal thickness. It needs to be ascertained whether the other kidney is involved, the lower ureter is dilated, the bladder wall is thickened, and whether emptying of the bladder is incomplete. Based on these findings, a diagnosis can be made.

In cases of hydronephrosis, long-standing urine stasis and pressure may damage the kidneys.

The causes of UPJ obstruction are

- Intrinsic (due to an adynamic or stenotic segment of the proximal ureter) or
- Extrinsic (e.g. compression by accessory renal arteries). This needs to be corrected primarily.

The renal anatomy and function can be assessed using radioscintigraphy studies.

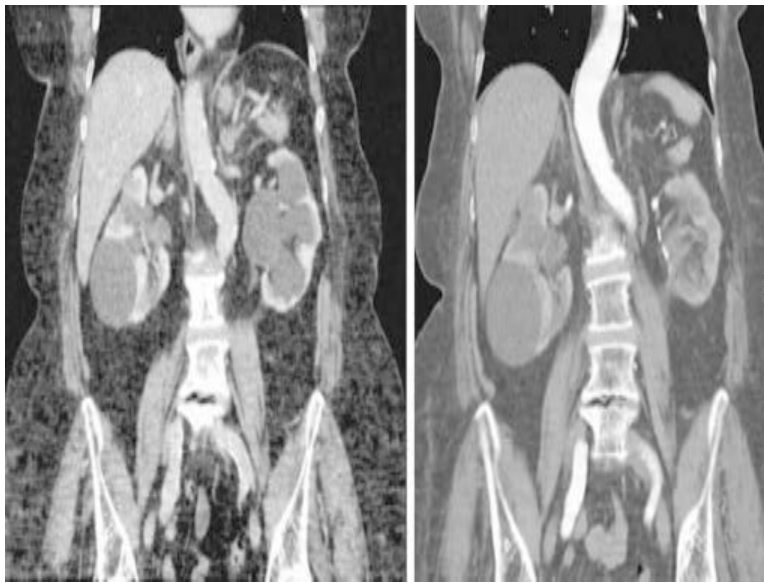
The investigation of choice for UPJ obstruction is ^{99m}Tc-MercaptoAcetyl-triGlycine (MAG3) scan. Micturating cystourethrogram is the investigation of choice in vesicoureteric reflux.

^{99m}Tc-dimercaptosuccinic acid (DMSA) scan is used for assessing renal scarring.

Treatment of intrinsic UPJ obstruction is done surgically if renal function is compromised or if there is bilateral involvement of the kidneys. In other cases, serial follow-up with ultrasonography is done to assess the spontaneous resolution of hydronephrosis.

Surgical options include pyeloplasty which entails the surgical division of the obstruction and the creation of a funnel-shaped anastomosis. Additionally, a temporary stent and a drain may be inserted. It is necessary to do further serial ultrasounds and MAG-3 imaging in order to ensure proper monitoring and evaluation. This procedure is now commonly being performed via laparoscopy and also with robotic assistance. Congenital PUJ obstruction is treated surgically by Anderson Hyen's pyeloplasty.

Below are CT scans of a left PUJ obstruction before(left) and after(right) surgery.



Other options:

Option B: A renal cyst would appear as a fluid-filled, well-demarcated structure within the kidney but does not involve the PUJ or cause obstruction.

Option C: A renal tumor would present as a solid, abnormal mass within the kidney, not typically involving the PUJ unless there is direct extension.

Option D: Multicystic kidney refers to the presence of multiple cysts within the kidney, often congenital, but is not specific to the PUJ and would not result in obstruction at the PUJ.

Solution to Question 12:

The given MRCP image showing saccular dilatation of the common bile duct suggests a diagnosis of type I choledochal cyst.

Todani classification:

- Type I- saccular dilatation of the whole (Ia) or a segment (Ib) of the common bile duct (CBD)
- Type II- diverticulum from the CBD
- Type III- choledochoceles that opens into the duodenum
- Type IV- multiple cysts (both intrahepatic and extrahepatic)
- Type V- multiple cysts that are only intrahepatic (Caroli's disease)

The diagnosis is made initially with a right upper quadrant ultrasound, following which a confirmatory MRCP can be carried out to help in pre-operative planning.

Treatment is usually surgical with a Roux-en-Y hepaticojejunostomy with radical resection of all the biliary tissue. There is a risk of malignant transformation in long-standing disease, so surgical removal is usually recommended.

Other options:

Option B: Gall bladder carcinoma would show abnormal growths or masses in the gall bladder rather than cystic dilations of the bile ducts.

Option C: Cholangiocarcinoma is a malignancy of the bile ducts and would present as irregular masses or bile duct strictures rather than cystic structures.

Option D: Choledocholithiasis refers to the presence of stones in the bile duct, which would appear as filling defects within the bile ducts rather than cystic dilations.

Solution to Question 13:

As the given image shows, there are multiple calcifications within the region of the pancreas, which is characteristic of chronic calcific pancreatitis.

This condition results from long-standing inflammation of the pancreas, leading to the deposition of calcium. Chronic pancreatitis is often associated with alcohol abuse, genetic factors, or recurrent acute pancreatitis. The calcifications are a hallmark feature and help differentiate it from other abdominal pathologies.

It presents with symptoms like chronic abdominal pain with a history of alcoholism and bulky stools which indicates a lack of pancreatic lipases seen in chronic disease leading to impaired fat digestion and absorption.

Chronic pancreatitis is a progressive inflammatory disease in which there is an irreversible destruction of pancreatic tissue. It is more common in males, with a male: female ratio of 4:1.

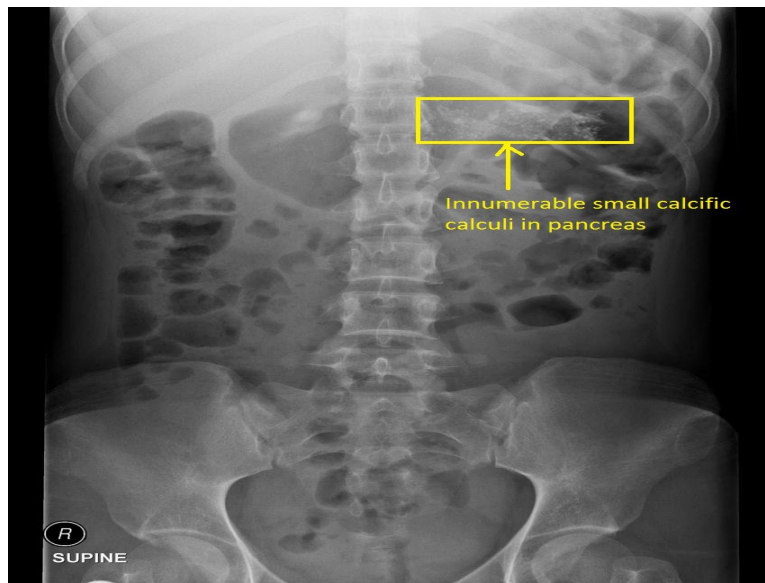
The etiology of chronic pancreatitis and pancreatic exocrine insufficiency can be classified based on the TIGAR-O classification system:

- Toxic-metabolic -alcohol, smoking, hypercalcemia, hyperlipidemia
- Idiopathic
- Genetic- PRSS1, CFTR, and SPINK1 genes
- Autoimmune- autoimmune chronic pancreatitis, IgG4-related disease
- Recurrent and severe acute pancreatitis
- Obstructive- pancreas divisum, duct obstruction, malignancy

The diagnosis can be made using Rosemont criteria. It is based on endoscopic ultrasonographic (EUS) findings. It is explained in the pearl attached below.

ERCP is the most sensitive radiologic test for the diagnosis of chronic pancreatitis.

The below image shows the X-ray findings:



Other options:

Option B: Mesenteric calcification would present in a different anatomical region and is associated with conditions like ischemia.

Option C: Horseshoe kidney involves the kidneys and does not cause pancreatic calcifications.

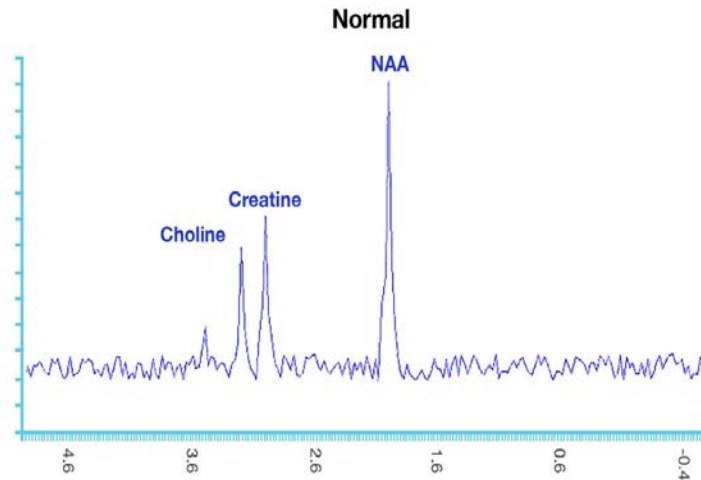
Option D: Jejunal fecolith refers to hardened stool within the small intestine.

Solution to Question 14:

The given clinical history of a child along with an MRI showing a ring-enhancing intraaxial lesion of the brain could likely be a tuberculoma or neurocysticercosis and MR spectroscopy would be the next best investigation for further assessment.

Magnetic Resonance (MR) spectroscopy is a non-invasive test based on the principle of magnetic resonance used to analyse the metabolite distribution within the brain tissue. It contains N-acetyl aspartate (NAA), a neuronal viability signal; choline (a component of cell membranes and a marker of cellular turnover); creatine (present in metabolically active tissues); lipids (released with cell membrane disruption); and lactate (a hallmark of anaerobic metabolism).

The image below shows a normal MR spectroscopy.



In tuberculoma, MR spectroscopy has reduced NAA owing to the destruction of the normal brain parenchyma and increased levels of lactate or lipids. In neurocysticercosis, there is reduced NAA and creatine and elevated lactate, alanine, succinate, and choline levels.

Other options:

Option B- Ocular B- scan is done to rule out diseases in the posterior segment in the presence of media opacities which is not significant for this case.

Option C- CBNAAT of CSF will not be the next investigation as the lumbar puncture analysis was normal.

Option D- PET-CT is done to assess the stage of cancer and is hence not required in this case.